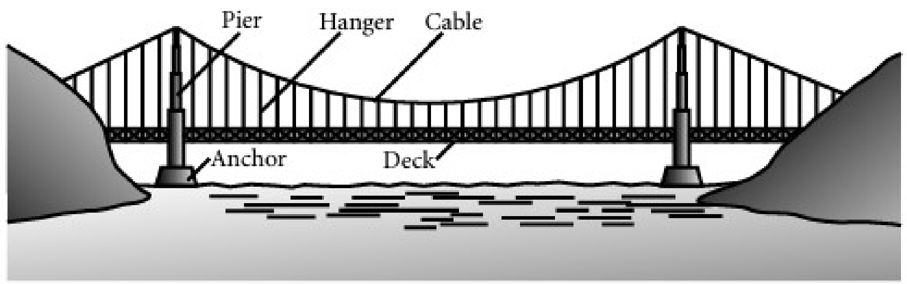
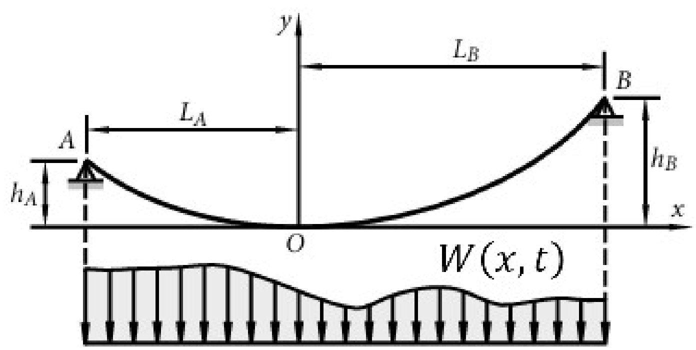
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

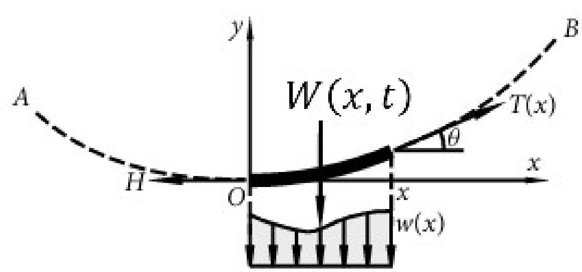
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A SIMPLY SUPPORTED CABLE ELEMENT: AN OPENSEES FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY ANALYSIS

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



A suspension bridge.

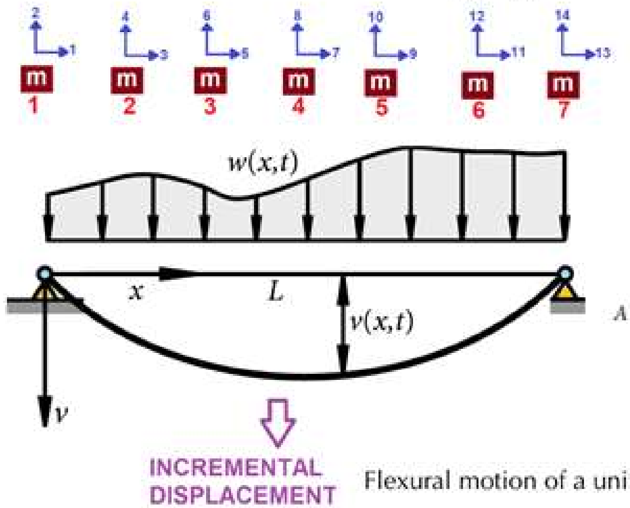




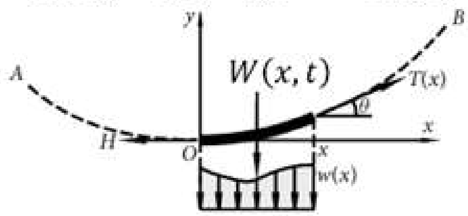
$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

$$m\ddot{u} + c\dot{u} + ku = P(t)$$

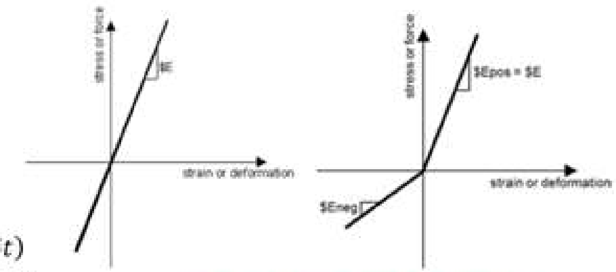
$$Structure\ Ductility\ Damage\ Index = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$



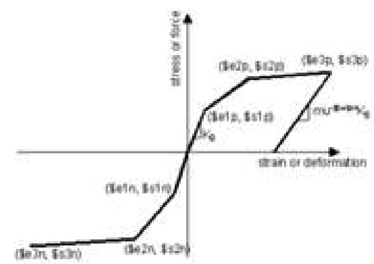
$$W(x,t) = P(t) = W_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$



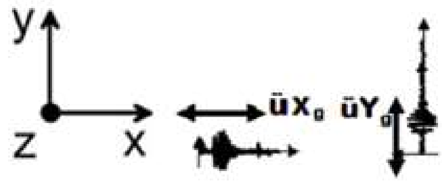
Flexural motion of a uniform elastic beam



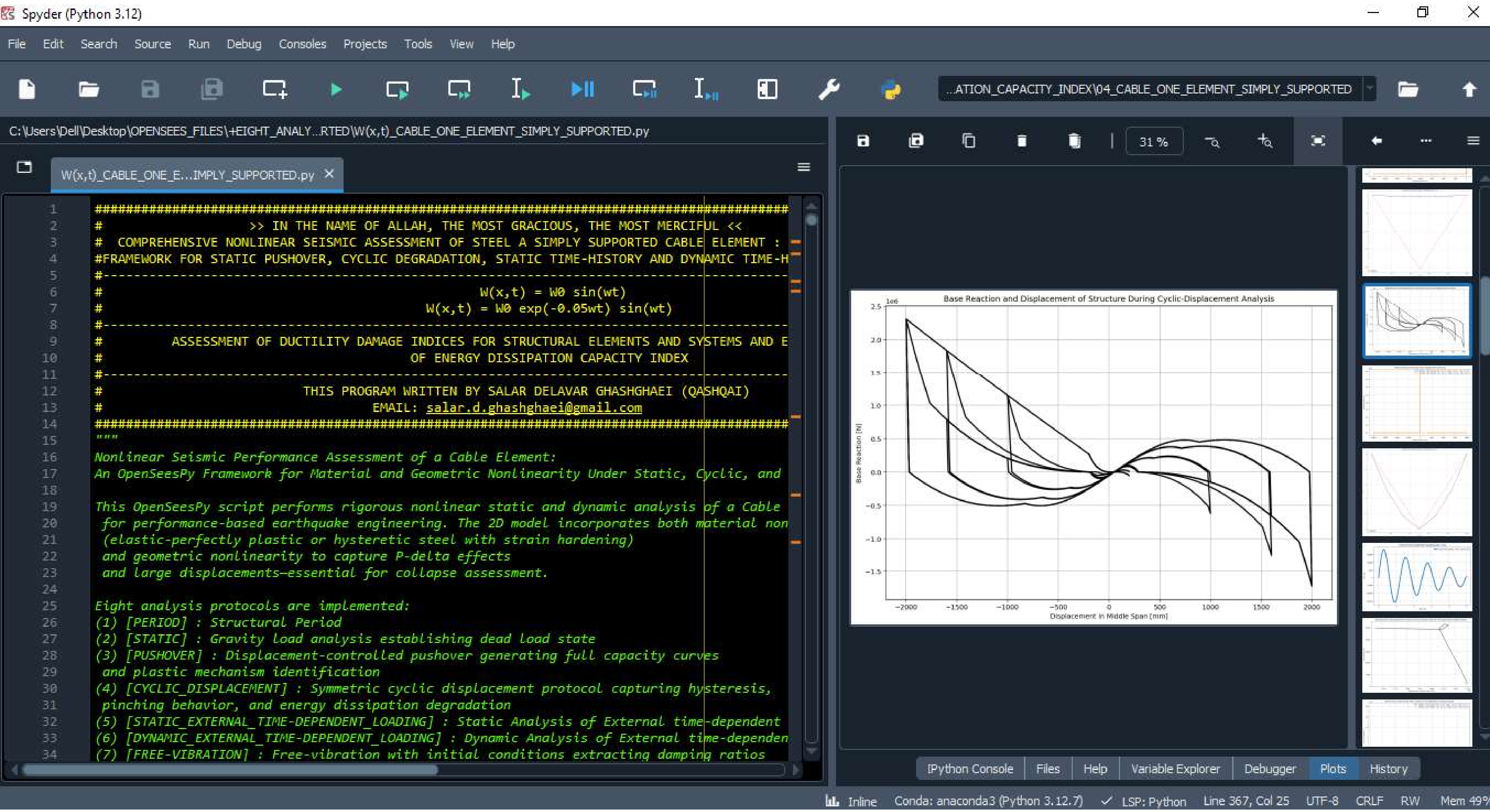
ELEMENT ELASTIC MATERIAL



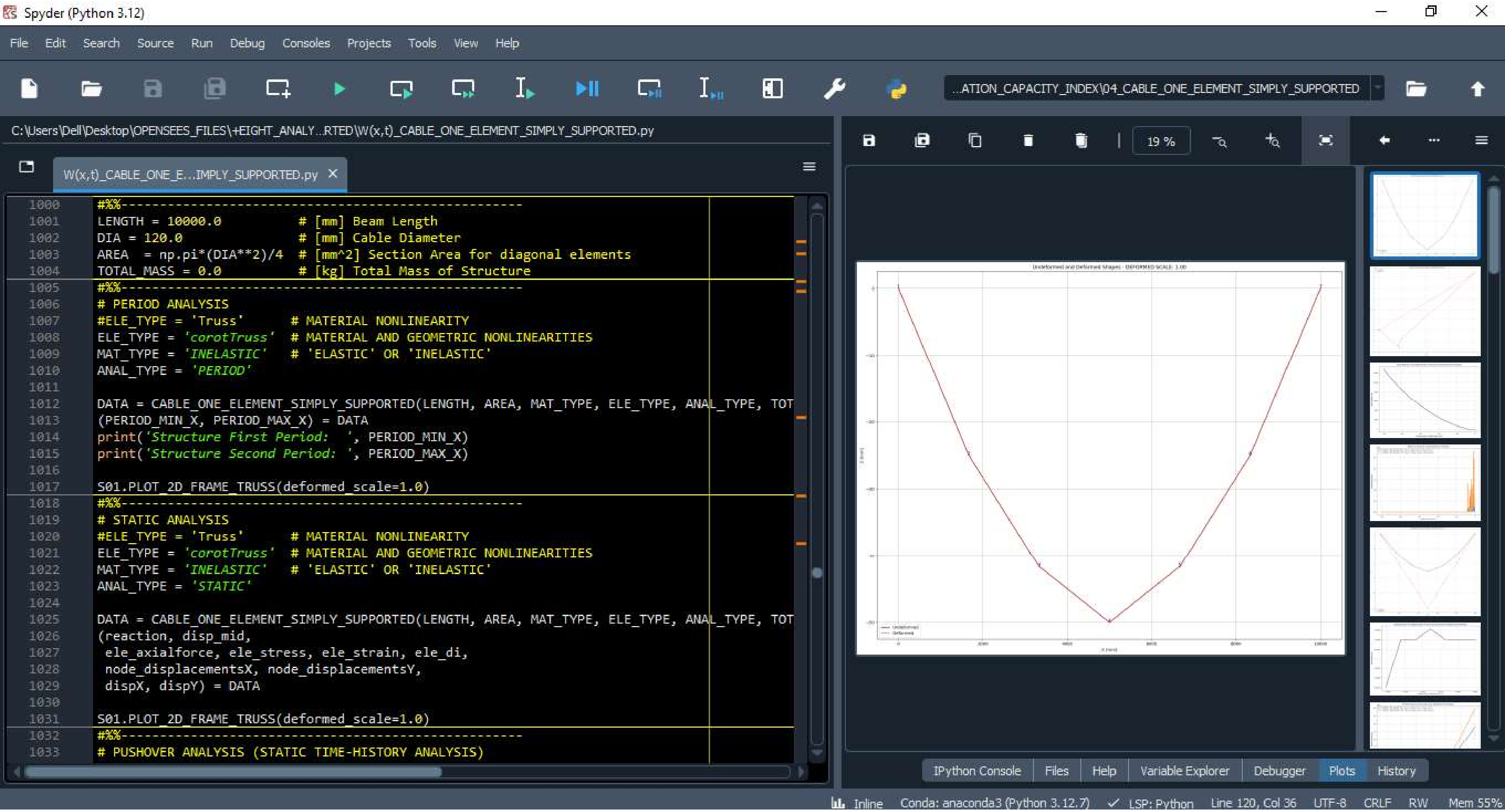
ELEMENT INELASTIC MATERIAL



$$\rho A \frac{\partial^2 v}{\partial t^2} = w(x,t).$$


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...ATION_CAPACITY_INDEX\04_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RTED\W(x,t)_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF STEEL A SIMPLY SUPPORTED CABLE ELEMENT :
#FRAMEWORK FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-H
W(x,t) = W0 sin(wt)
W(x,t) = W0 exp(-0.05wt) sin(wt)
ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND E
OF ENERGY DISSIPATION CAPACITY INDEX
THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
EMAIL: salar.d.ghashghaei@gmail.com
"""
Nonlinear Seismic Performance Assessment of a Cable Element:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and
This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a Cable
for performance-based earthquake engineering. The 2D model incorporates both material non
(elastic-perfectly plastic or hysteretic steel with strain hardening)
and geometric nonlinearity to capture P-delta effects
and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
and plastic mechanism identification
(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturing hysteresis,
pinching behavior, and energy dissipation degradation
(5) [STATIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Static Analysis of External time-dependent
(6) [DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING] : Dynamic Analysis of External time-dependen
(7) [FREE-VIBRATION] : Free-vibration with initial conditions extracting damping ratios
31 %
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 367, Col 25 UTF-8 CRLF RW Mem 49%

STATIC ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...ATION_CAPACITY_INDEX\04_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED
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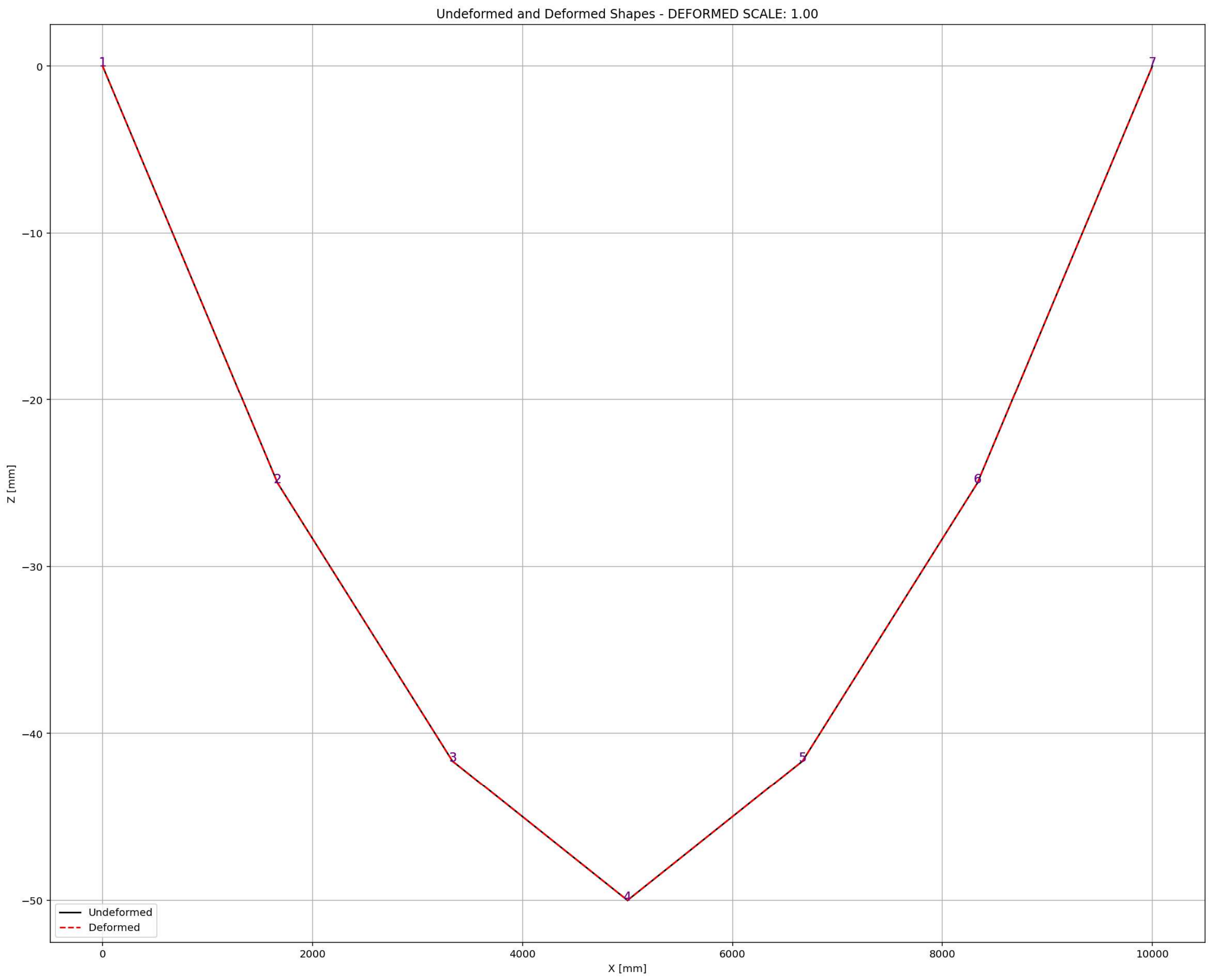
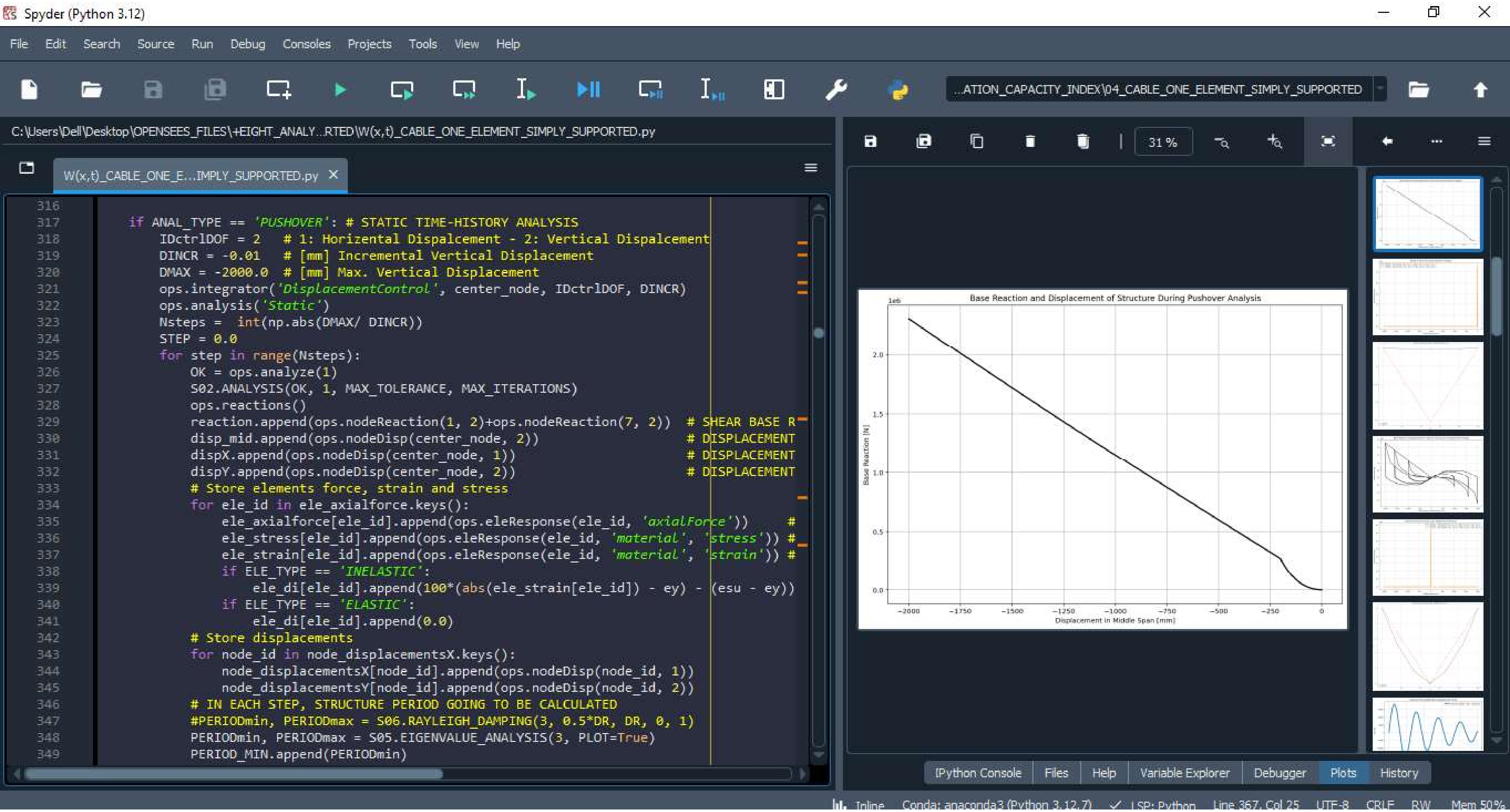
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
#%%---
LENGTH = 10000.0   # [mm] Beam Length
DIA = 120.0   # [mm] Cable Diameter
AREA = np.pi*(DIA**2)/4 # [mm^2] Section Area for diagonal elements
TOTAL_MASS = 0.0   # [kg] Total Mass of Structure
#%%---
PERIOD ANALYSIS
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'PERIOD'
DATA = CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOT
(PERIOD_MIN_X, PERIOD_MAX_X) = DATA
print('Structure First Period: ', PERIOD_MIN_X)
print('Structure Second Period: ', PERIOD_MAX_X)
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%---
STATIC ANALYSIS
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss'   # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC'
DATA = CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOT
(reaction, disp_mid,
 ele_axialforce, ele_stress, ele_strain, ele_di,
 node_displacementsX, node_displacementsY,
 dispX, dispY) = DATA
S01.PLOT_2D_FRAME_TRUSS(deformed_scale=1.0)
#%%---
PUSHOVER ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
19 %
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 120, Col 36 UTF-8 CRLF RW Mem 55%



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
0
−10
−20
−30
−40
−50
Z [mm]
0
2000
4000
6000
8000
10000
X [mm]
1
2
3
4
5
6
7
Undeformed
Deformed

PUSHOVER ANALYSIS


Spyder (Python 3.12)
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...ATION_CAPACITY_INDEX\04_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED
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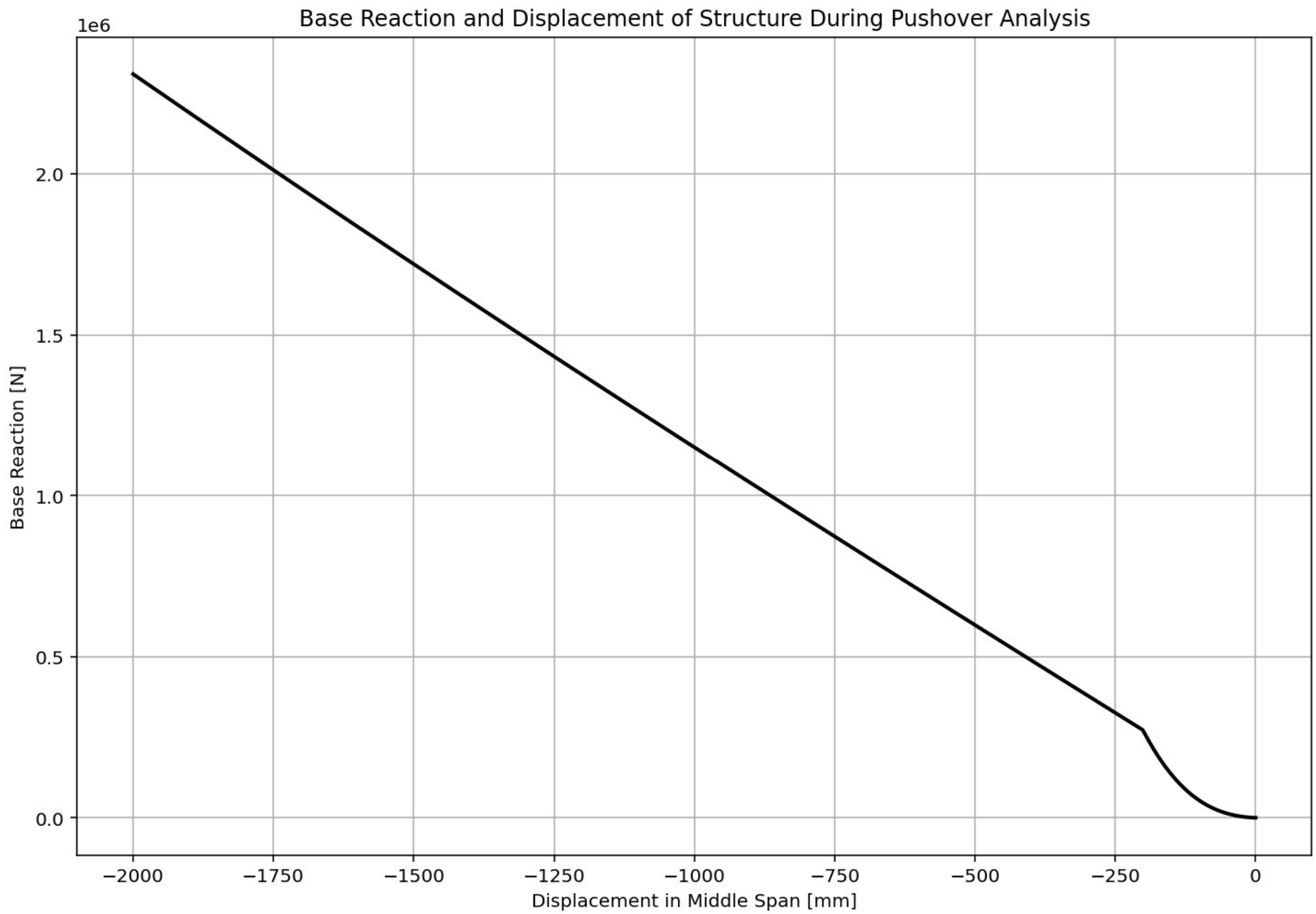
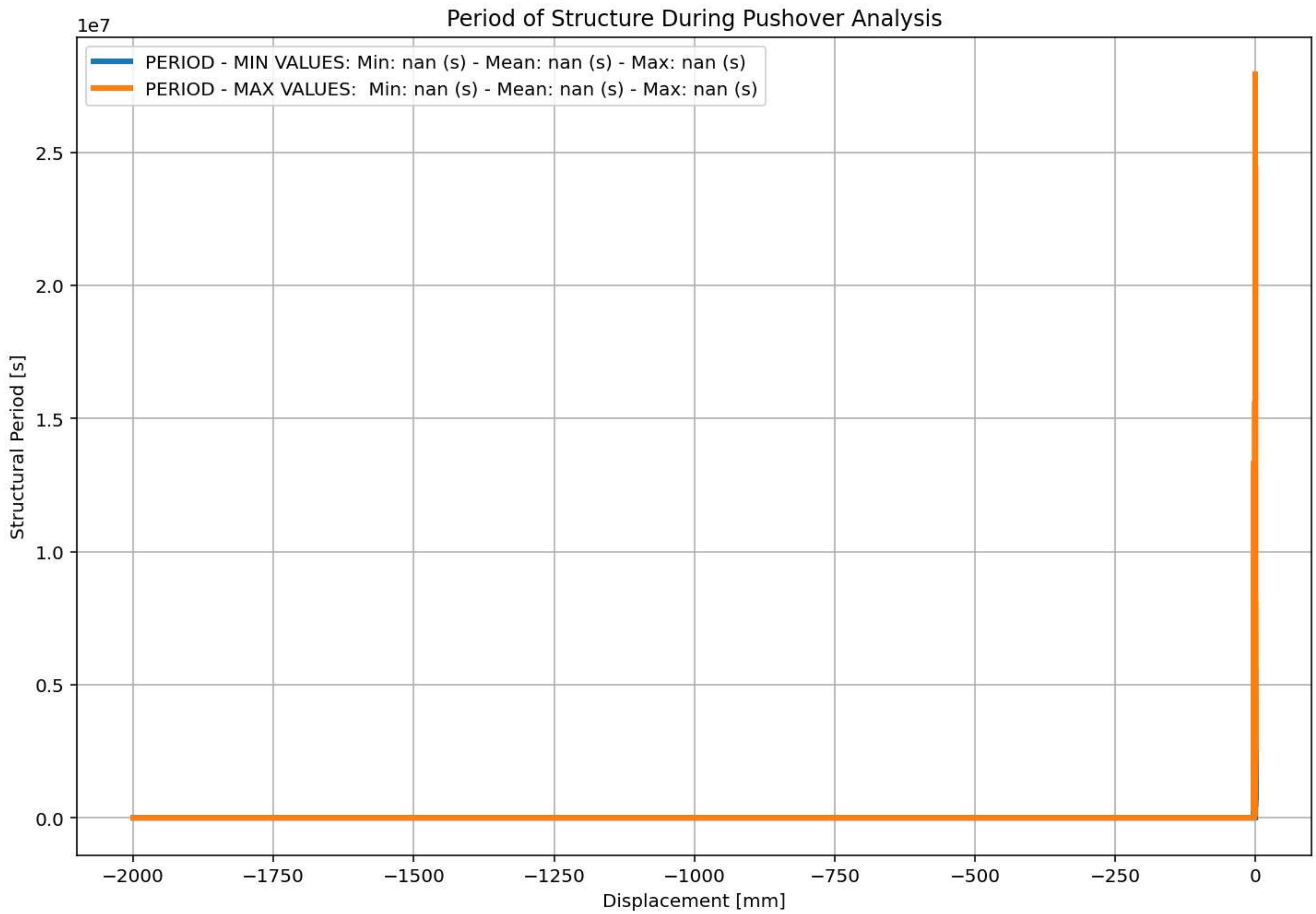
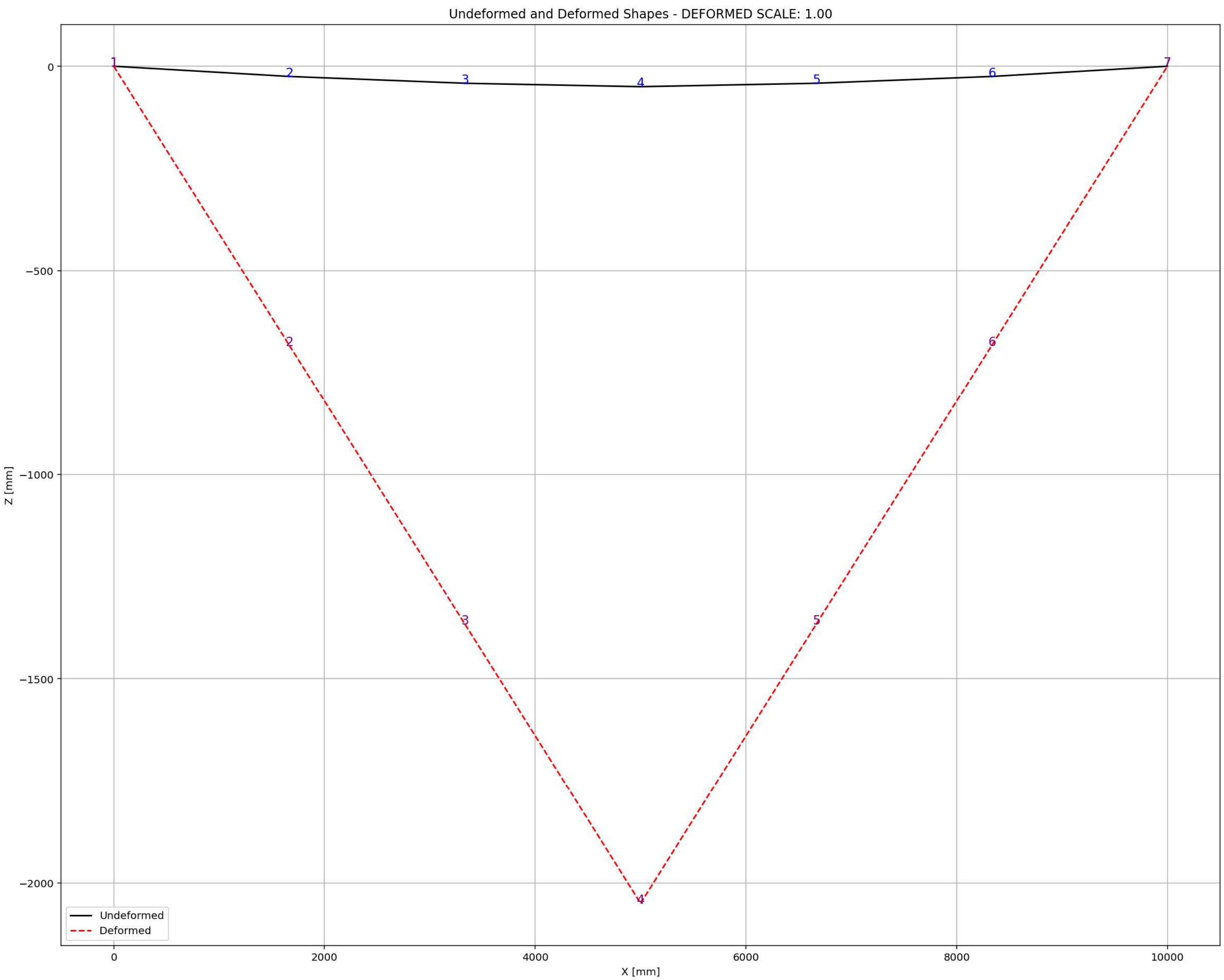
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
    if ANAL_TYPE == 'PUSHOVER': # STATIC TIME-HISTORY ANALYSIS
        IDctrlDOF = 2   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
        DINCR = -0.01   # [mm] Incremental Vertical Displacement
        DMAX = -2000.0 # [mm] Max. Vertical Displacement
        ops.integrator('DisplacementControl', center_node, IDctrlDOF, DINCR)
        ops.analysis('Static')
        Nsteps = int(np.abs(DMAX/ DINCR))
        STEP = 0.0
        for step in range(Nsteps):
            OK = ops.analyze(1)
            S02.ANALYSIS(OK, 1, MAX_TOLERANCE, MAX_ITERATIONS)
            ops.reactions()
            reaction.append(ops.nodeReaction(1, 2)+ops.nodeReaction(7, 2)) # SHEAR BASE R
            disp_mid.append(ops.nodeDisp(center_node, 2))   # DISPLACEMENT
            dispX.append(ops.nodeDisp(center_node, 1))   # DISPLACEMENT
            dispY.append(ops.nodeDisp(center_node, 2))   # DISPLACEMENT
            # Store elements force, strain and stress
            for ele_id in ele_axialforce.keys():
                ele_axialforce[ele_id].append(ops.eleResponse(ele_id, 'axialForce'))   #
                ele_stress[ele_id].append(ops.eleResponse(ele_id, 'material', 'stress')) #
                ele_strain[ele_id].append(ops.eleResponse(ele_id, 'material', 'strain')) #
                if ELE_TYPE == 'INELASTIC':
                    ele_di[ele_id].append(100*(abs(ele_strain[ele_id]) - ey) - (esu - ey))
                if ELE_TYPE == 'ELASTIC':
                    ele_di[ele_id].append(0.0)
            # Store displacements
            for node_id in node_displacementsX.keys():
                node_displacementsX[node_id].append(ops.nodeDisp(node_id, 1))
                node_displacementsY[node_id].append(ops.nodeDisp(node_id, 2))
            # IN EACH STEP, STRUCTURE PERIOD GOING TO BE CALCULATED
            #PERIODmin, PERIODmax = S06.RAYLEIGH_DAMPING(3, 0.5*DR, DR, 0, 1)
            PERIODmin, PERIODmax = S05.EIGENVALUE_ANALYSIS(3, PLOT=True)
            PERIOD_MIN.append(PERIODmin)
31 %
Base Reaction and Displacement of Structure During Pushover Analysis
1e6
2.0
1.5
1.0
0.5
0.0
Base Reaction [N]
−2000 −1750 −1500 −1250 −1000 −750 −500 −250 0
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 367, Col 25 UTF-8 CRLF RW Mem 50%

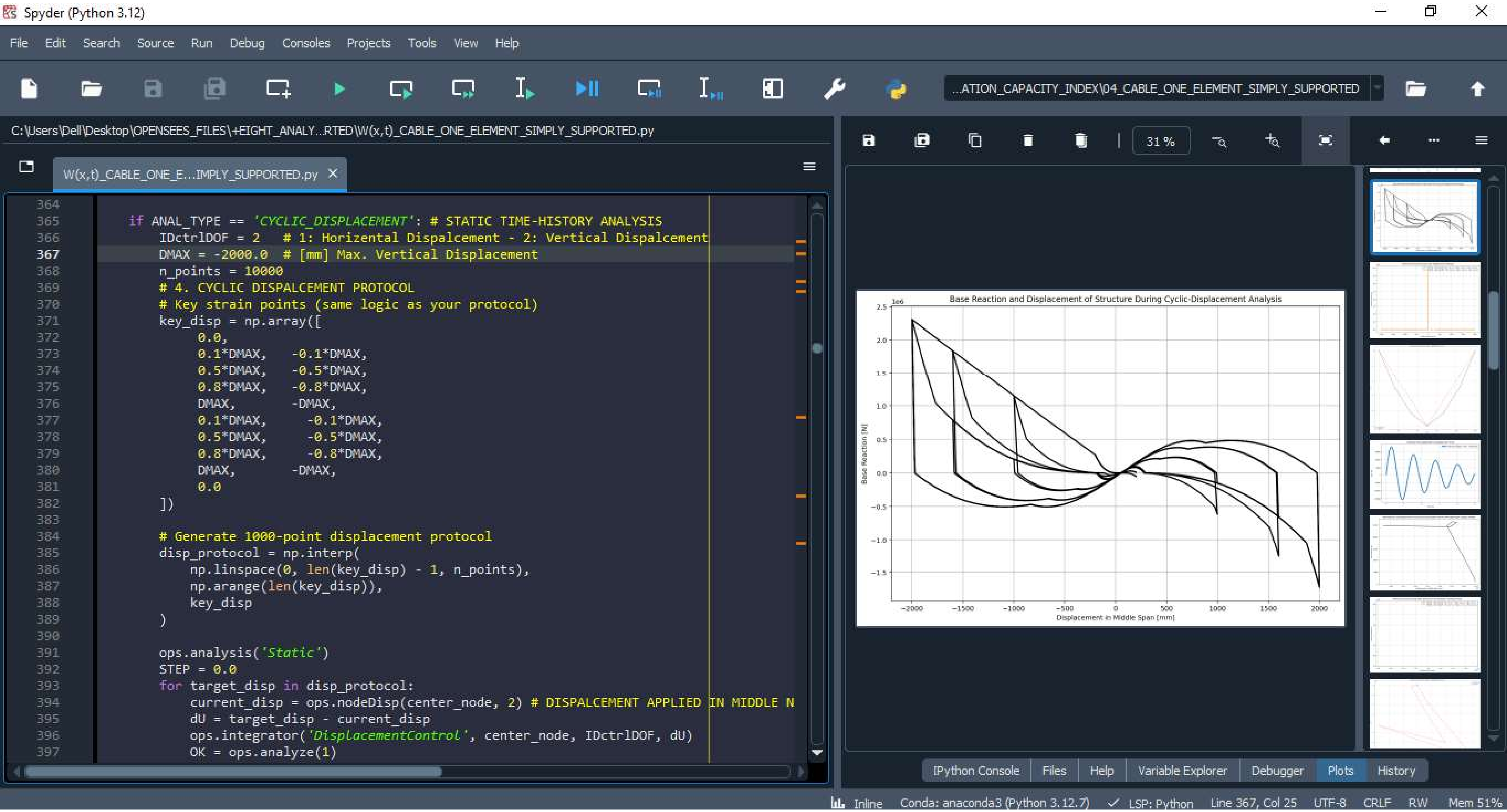


Base Reaction and Displacement of Structure During Pushover Analysis
1e6
2.0
1.5
1.0
0.5
0.0
Base Reaction [N]
−2000
−1750
−1500
−1250
−1000
−750
−500
−250
0
Displacement in Middle Span [mm]


Period of Structure During Pushover Analysis
1e7
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
2.5
2.0
1.5
1.0
0.5
0.0
Structural Period [s]
−2000
−1750
−1500
−1250
−1000
−750
−500
−250
0
Displacement [mm]


Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
Undeformed
Deformed
0
−500
−1000
−1500
−2000
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
7

CYCLIC DISPLACEMENT ANALYSIS


Spyder (Python 3.12)
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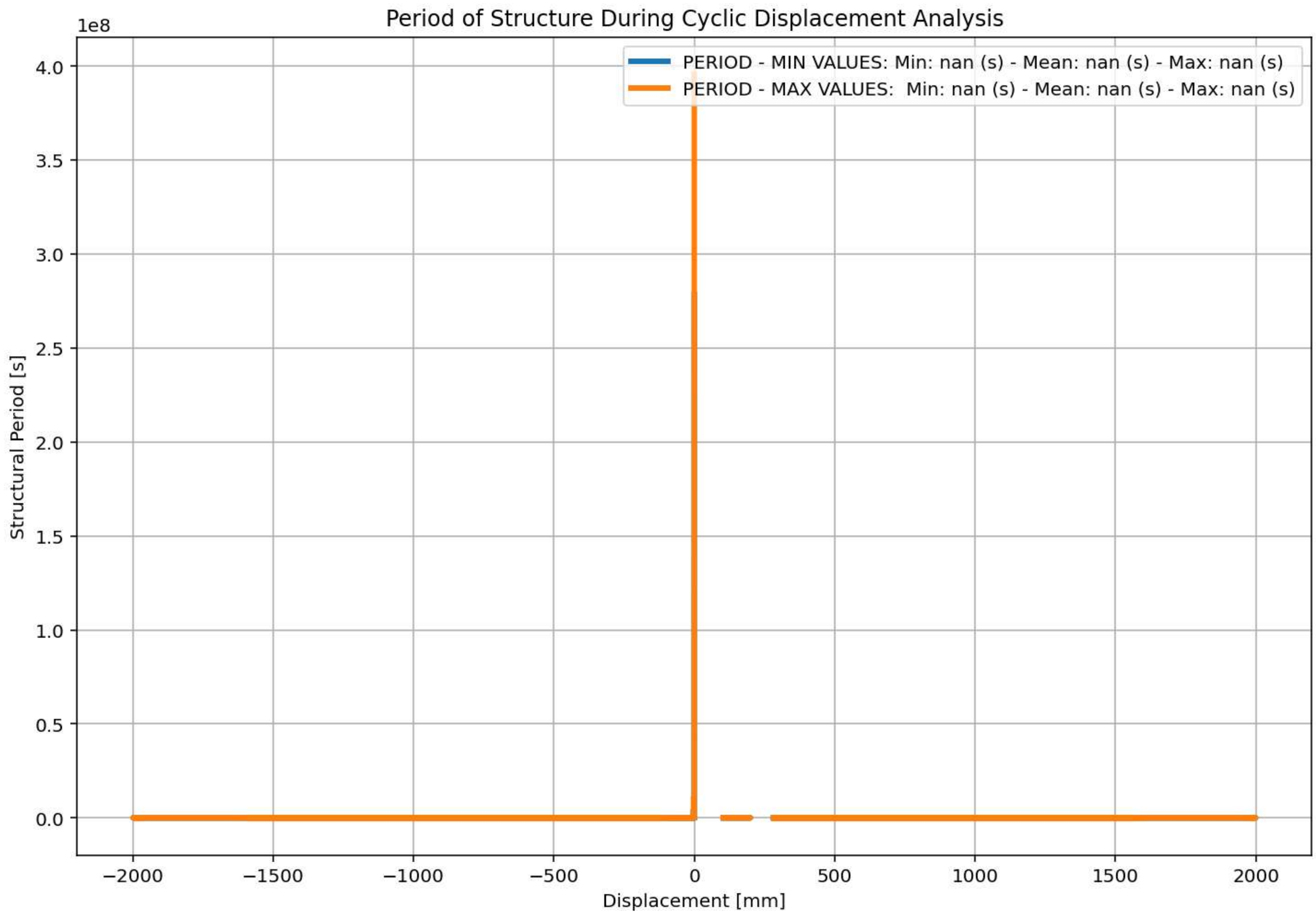
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
    if ANAL_TYPE == 'CYCLIC_DISPLACEMENT': # STATIC TIME-HISTORY ANALYSIS
        IDctrlDOF = 2   # 1: Horizental Dispalcement - 2: Vertical Dispalcement
        DMAX = -2000.0 # [mm] Max. Vertical Displacement
        n_points = 10000
        # 4. CYCLIC DISPALCEMENT PROTOCOL
        # Key strain points (same logic as your protocol)
        key_disp = np.array([
             0.0,
             0.1*DMAX,   -0.1*DMAX,
             0.5*DMAX,   -0.5*DMAX,
             0.8*DMAX,   -0.8*DMAX,
             DMAX,   -DMAX,
             0.1*DMAX,   -0.1*DMAX,
             0.5*DMAX,   -0.5*DMAX,
             0.8*DMAX,   -0.8*DMAX,
             DMAX,   -DMAX,
             0.0
        ])

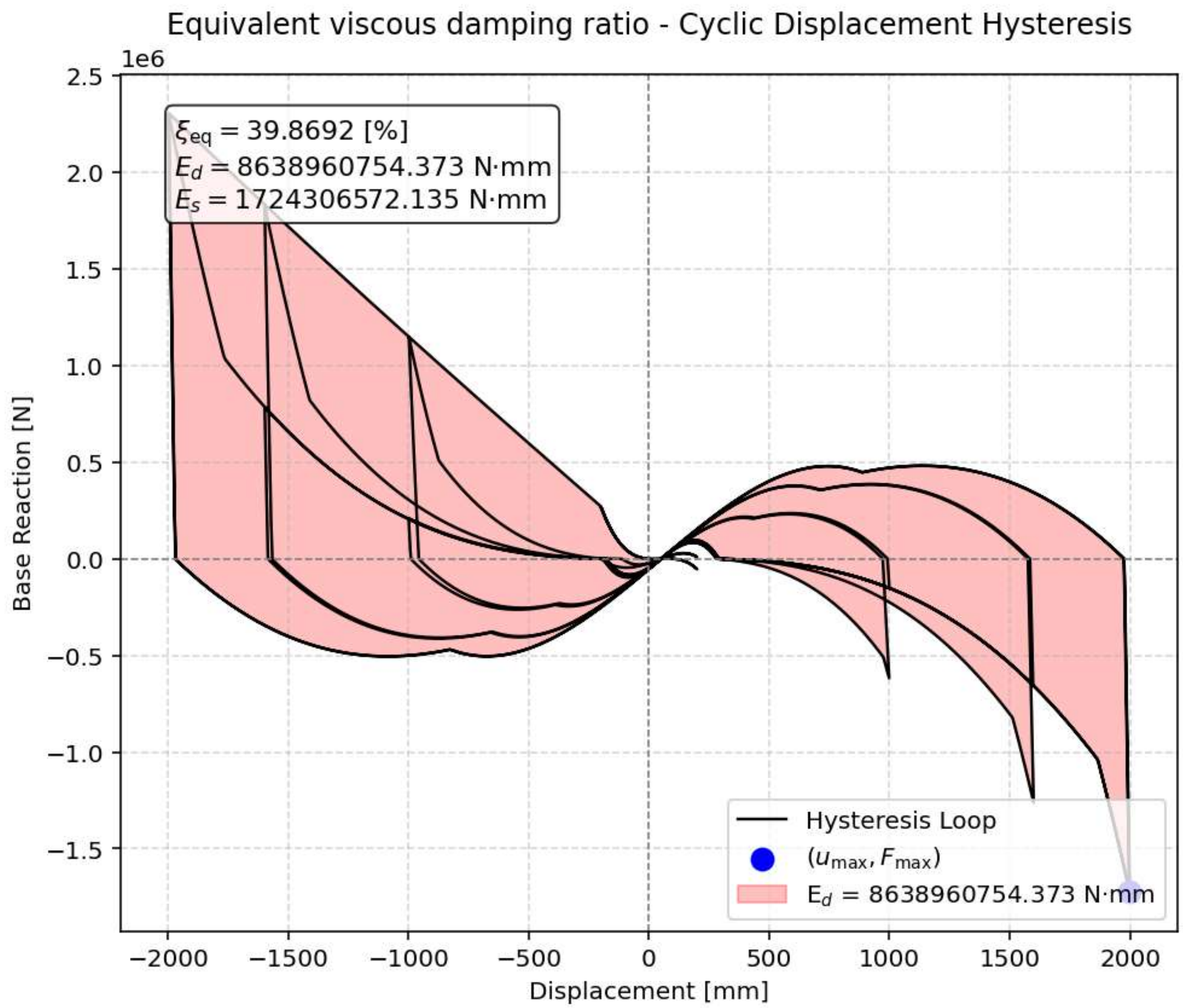
        # Generate 1000-point displacement protocol
        disp_protocol = np.interp(
            np.linspace(0, len(key_disp) - 1, n_points),
            np.arange(len(key_disp)),
            key_disp
)

        ops.analysis('Static')
        STEP = 0.0
        for target_disp in disp_protocol:
            current_disp = ops.nodeDisp(center_node, 2) # DISPALCEMENT APPLIED IN MIDDLE N
            dU = target_disp - current_disp
            ops.integrator('DisplacementControl', center_node, IDctrlDOF, dU)
            OK = ops.analyze(1)
31 %
Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 367, Col 25 UTF-8 CRLF RW Mem 51%

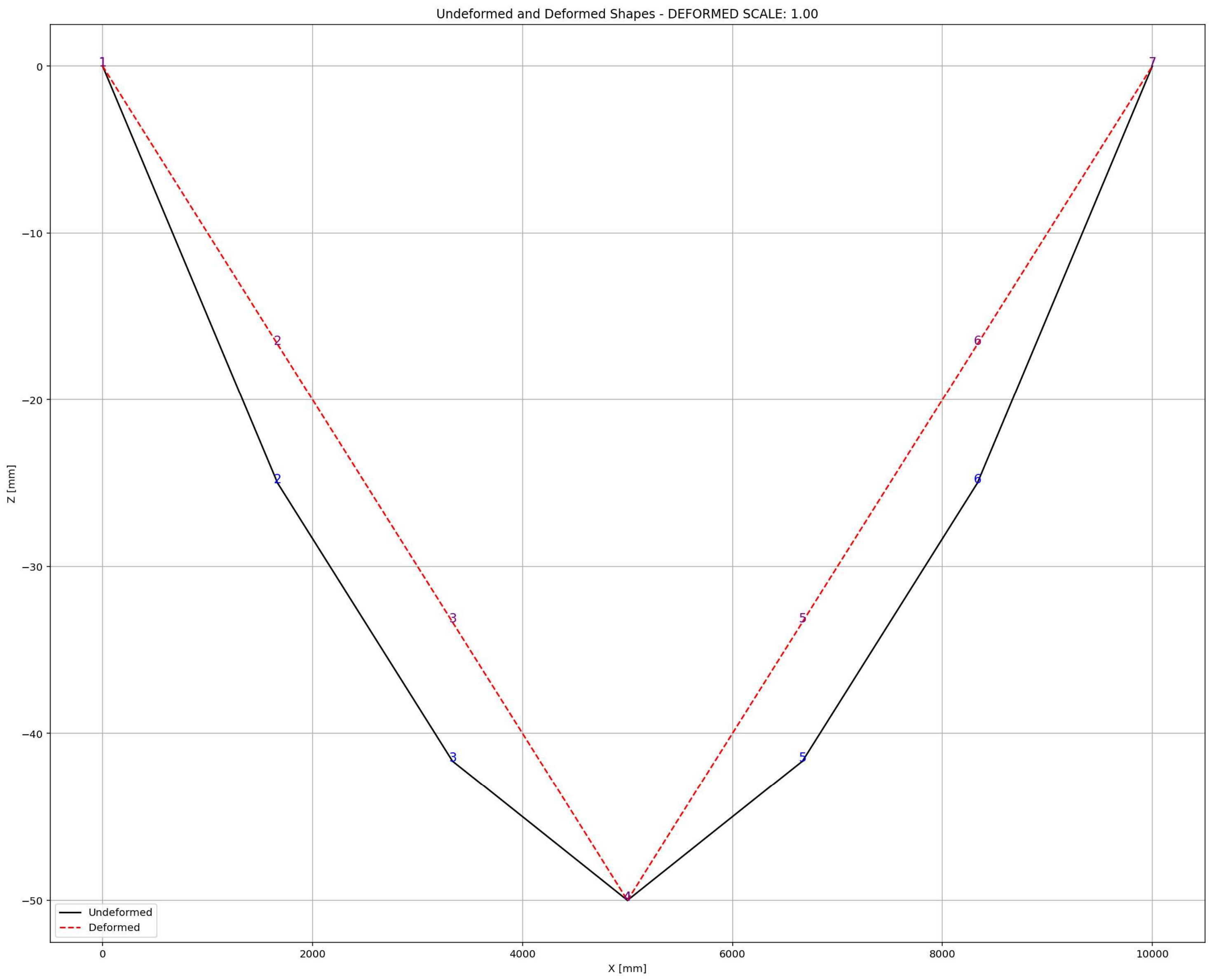

Base Reaction and Displacement of Structure During Cyclic-Displacement Analysis
1e6
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Base Reaction [N]
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Displacement in Middle Span [mm]



Period of Structure During Cyclic Displacement Analysis
1e8
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−2000
−1500
−1000
−500
0
500
1000
1500
2000

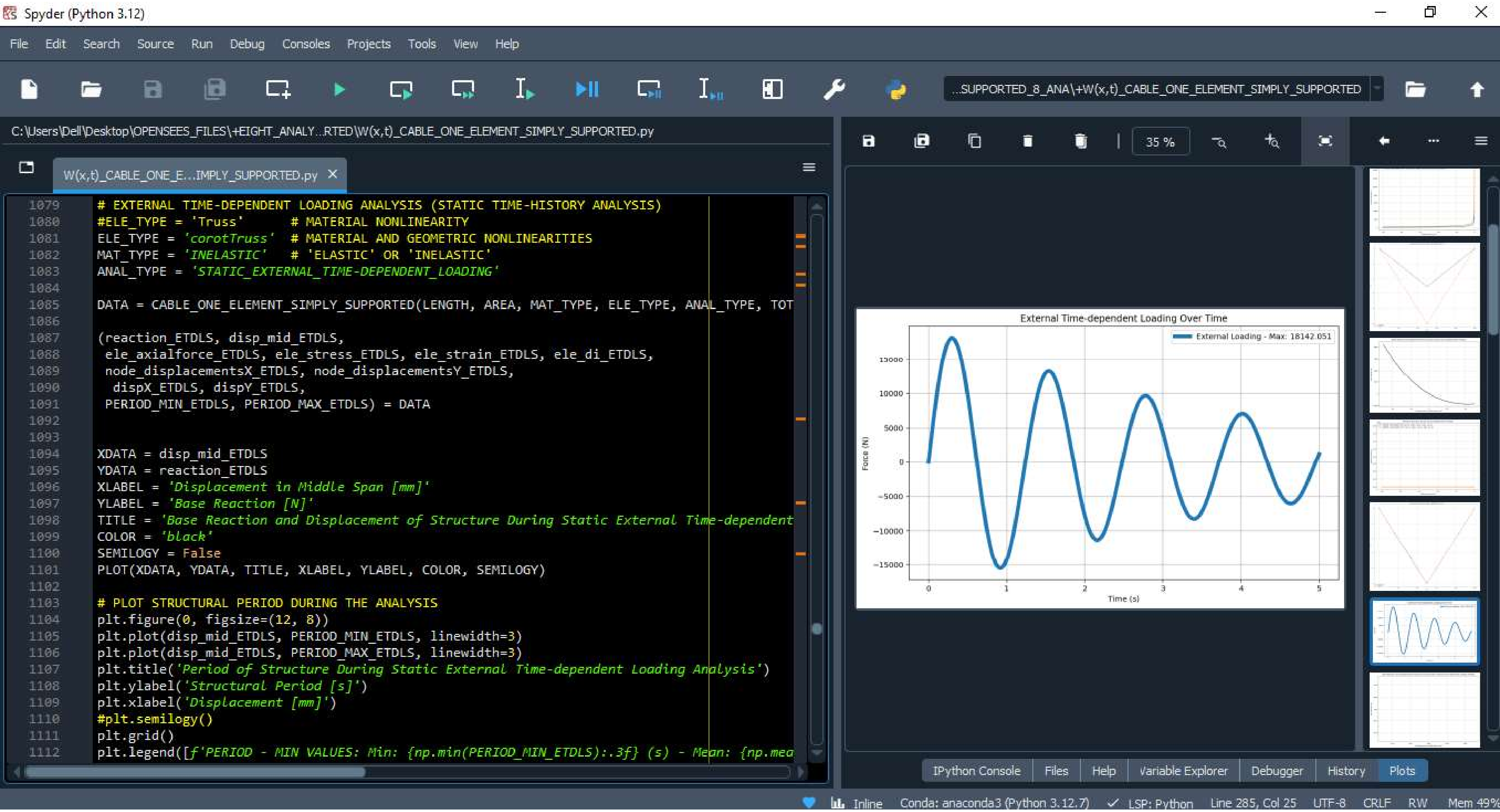


Equivalent viscous damping ratio - Cyclic Displacement Hysteresis
1e6
$\xi_{eq} = 39.8692$ [%]
$E_d = 8638960754.373$ N·mm
$E_s = 1724306572.135$ N·mm
Hysteresis Loop
(u_{max}, F_{max})
E_d = 8638960754.373 N·mm
Base Reaction [N]
Displacement [mm]
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2000
−1500
−1000
−500
0
500
1000
1500
2000



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
−10
−20
−30
−40
−50
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
7
Undeformed
Deformed

STATIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


Spyder (Python 3.12)
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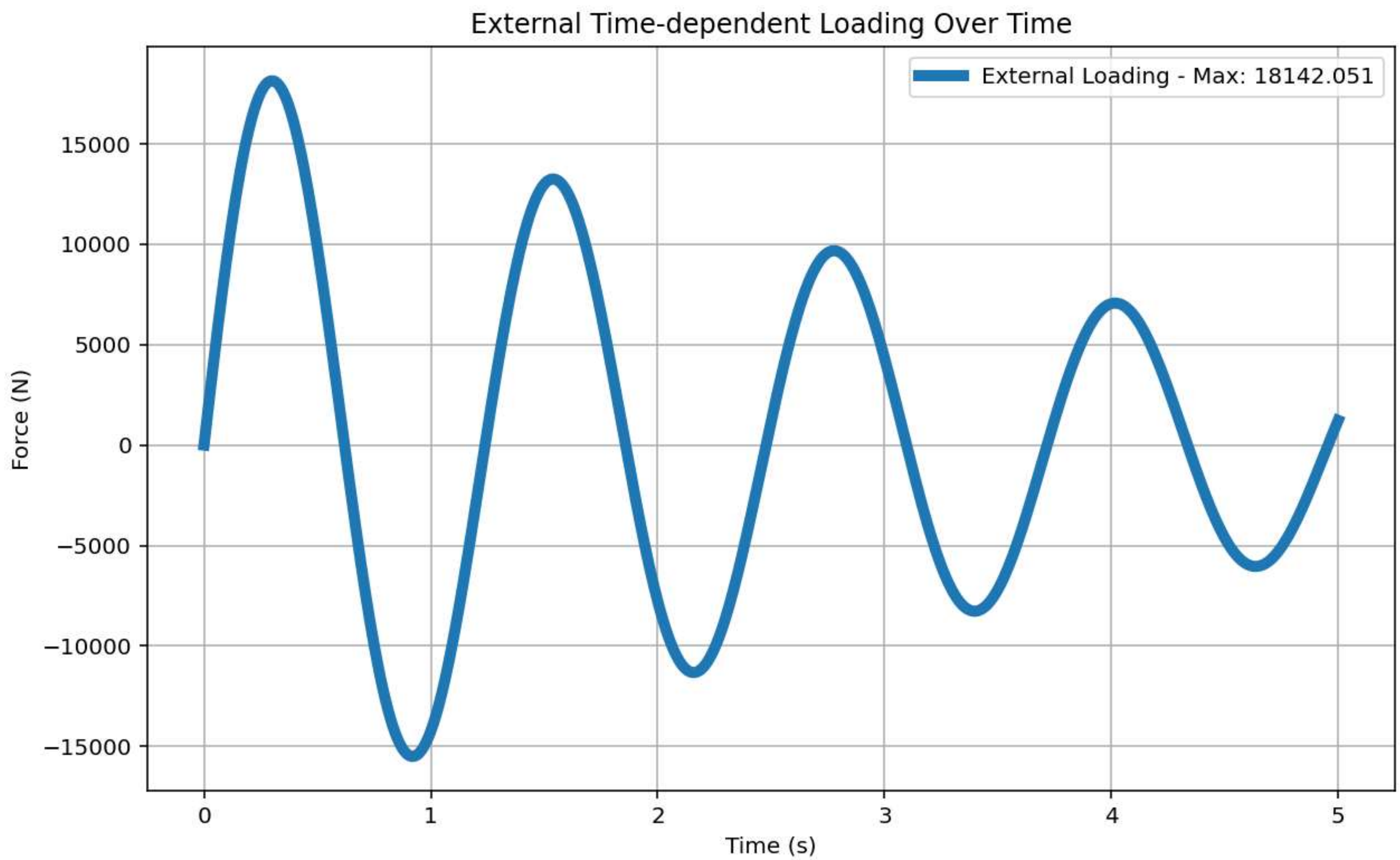
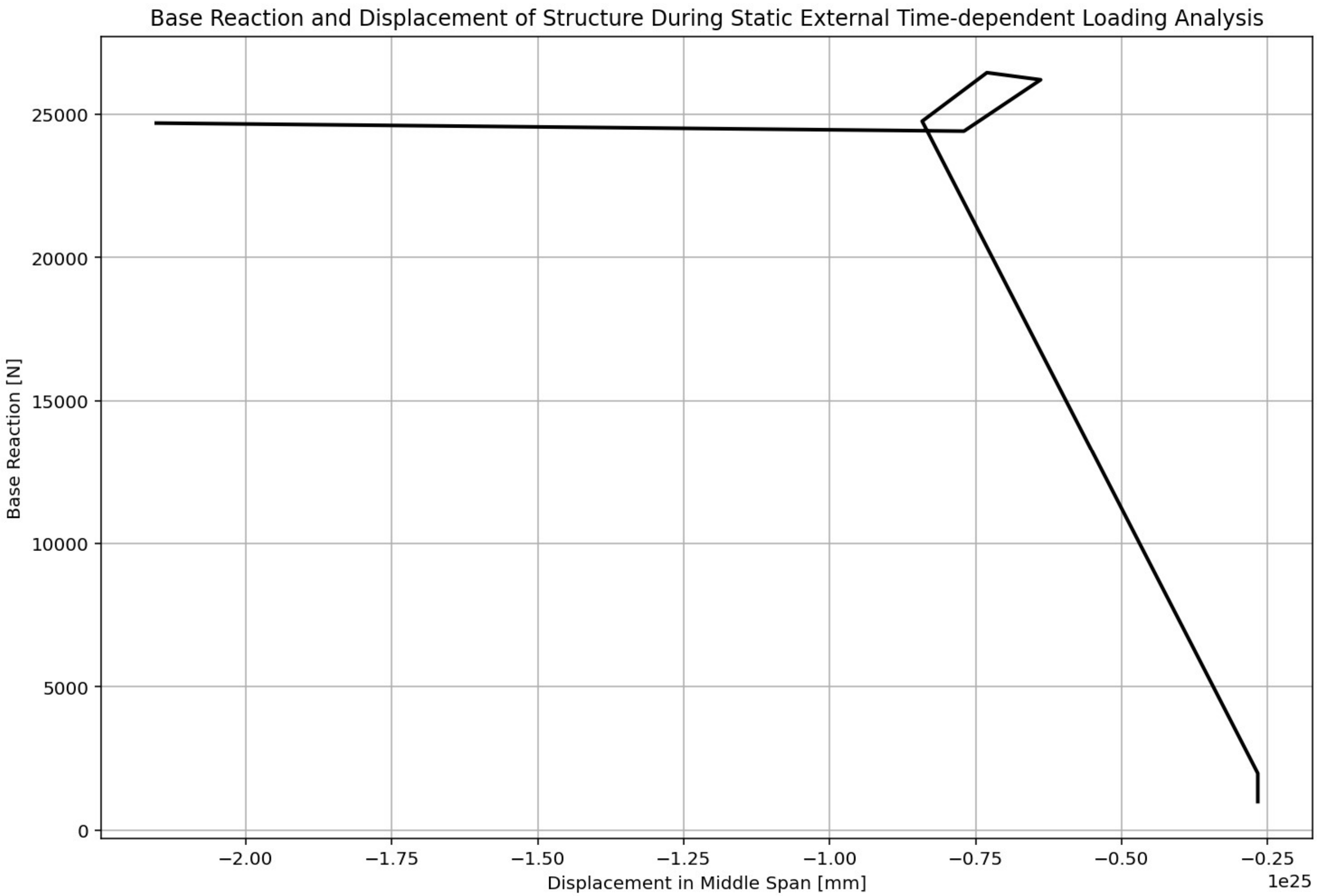
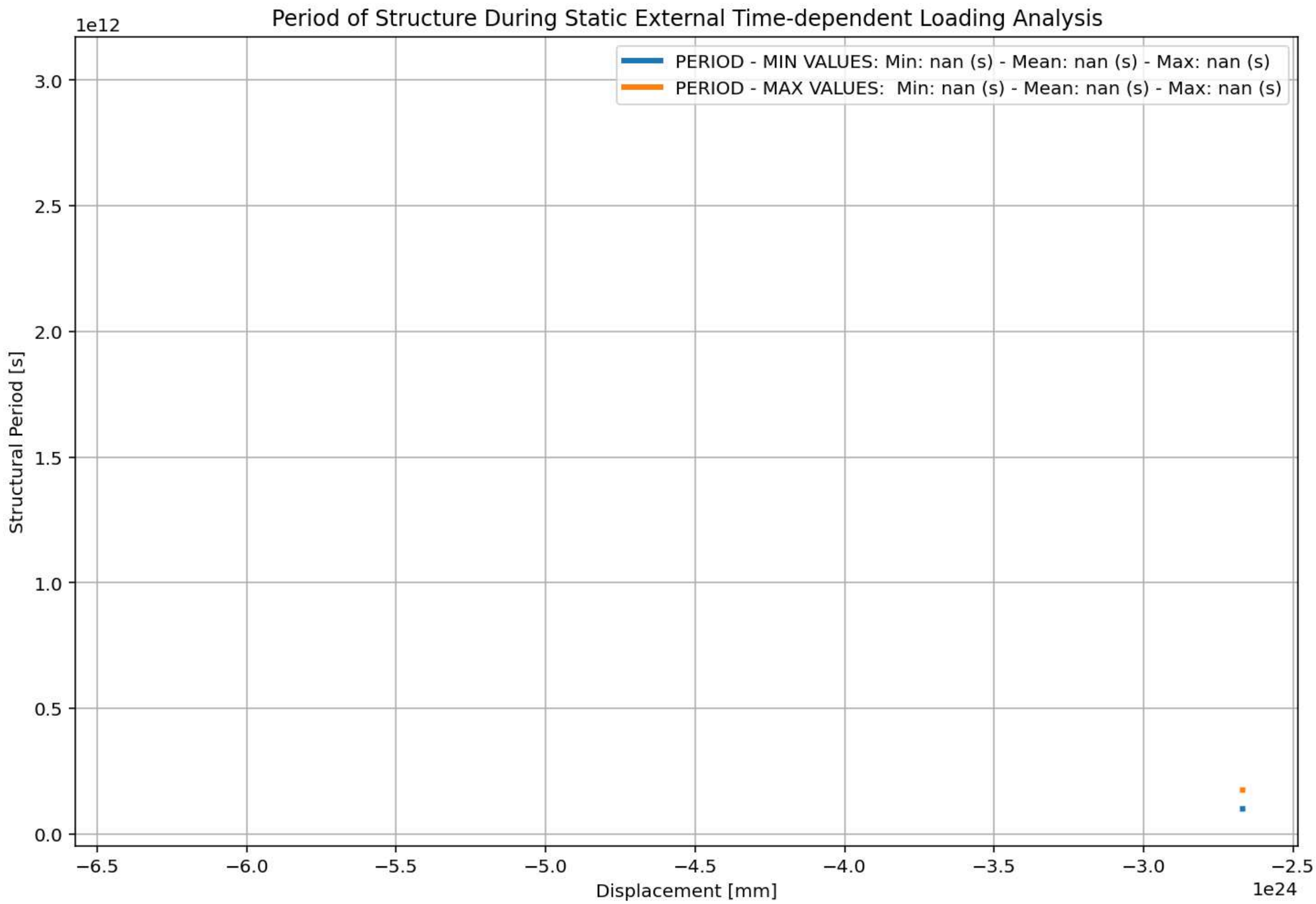
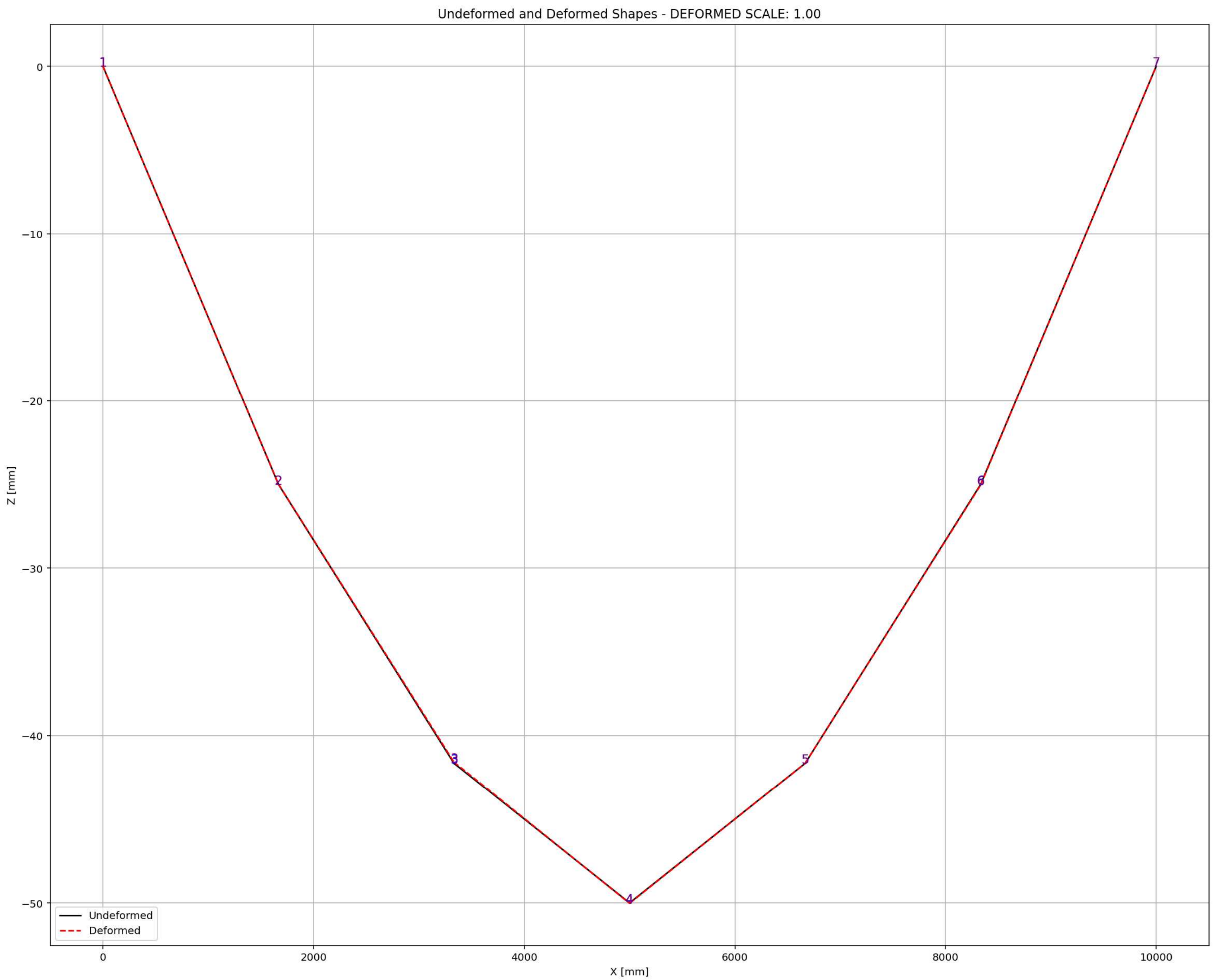
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RTED\W(x,t)_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (STATIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'STATIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOT
(reaction_ETDLS, disp_mid_ETDLS,
ele_axialforce_ETDLS, ele_stress_ETDLS, ele_strain_ETDLS, ele_di_ETDLS,
node_displacementsX_ETDLS, node_displacementsY_ETDLS,
dispX_ETDLS, dispY_ETDLS,
PERIOD_MIN_ETDLS, PERIOD_MAX_ETDLS) = DATA
XDATA = disp_mid_ETDLS
YDATA = reaction_ETDLS
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Static External Time-dependent
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
# PLOT STRUCTURAL PERIOD DURING THE ANALYSIS
plt.figure(0, figsize=(12, 8))
plt.plot(disp_mid_ETDLS, PERIOD_MIN_ETDLS, linewidth=3)
plt.plot(disp_mid_ETDLS, PERIOD_MAX_ETDLS, linewidth=3)
plt.title('Period of Structure During Static External Time-dependent Loading Analysis')
plt.ylabel('Structural Period [s]')
plt.xlabel('Displacement [mm]')
#plt.semilogy()
plt.grid()
plt.legend([f'PERIOD - MIN VALUES: Min: {np.min(PERIOD_MIN_ETDLS):.3f} (s) - Mean: {np.mea
35 %
External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
Time (s)
IPython Console Files Help Variable Explorer Debugger History Plots
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 285, Col 25 UTF-8 CRLF RW Mem 49%


External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
Time (s)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5


Base Reaction and Displacement of Structure During Static External Time-dependent Loading Analysis
Base Reaction [N]
25000
20000
15000
10000
5000
0
−2.00
−1.75
−1.50
−1.25
−1.00
−0.75
−0.50
−0.25
Displacement in Middle Span [mm]
1e25

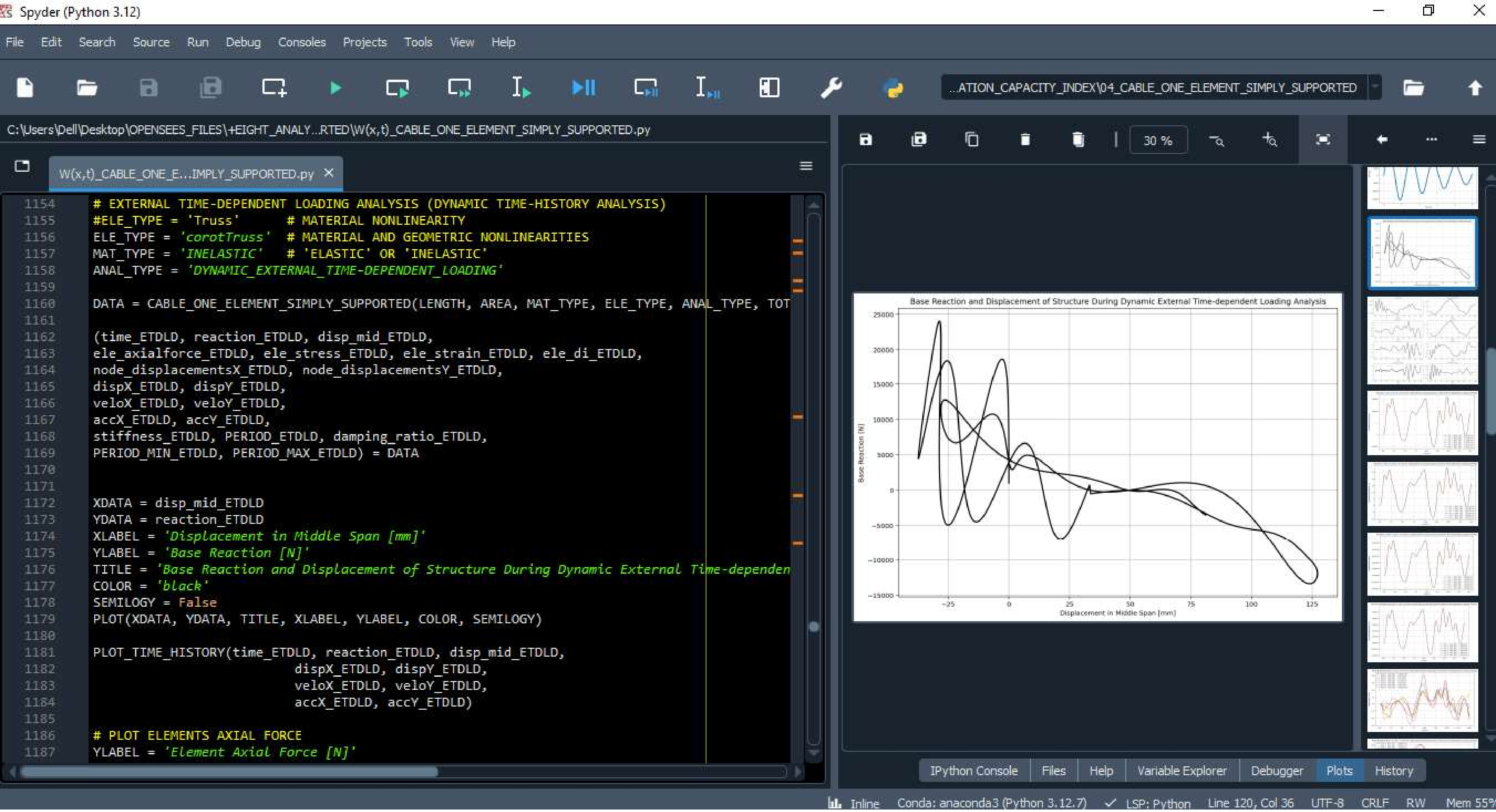


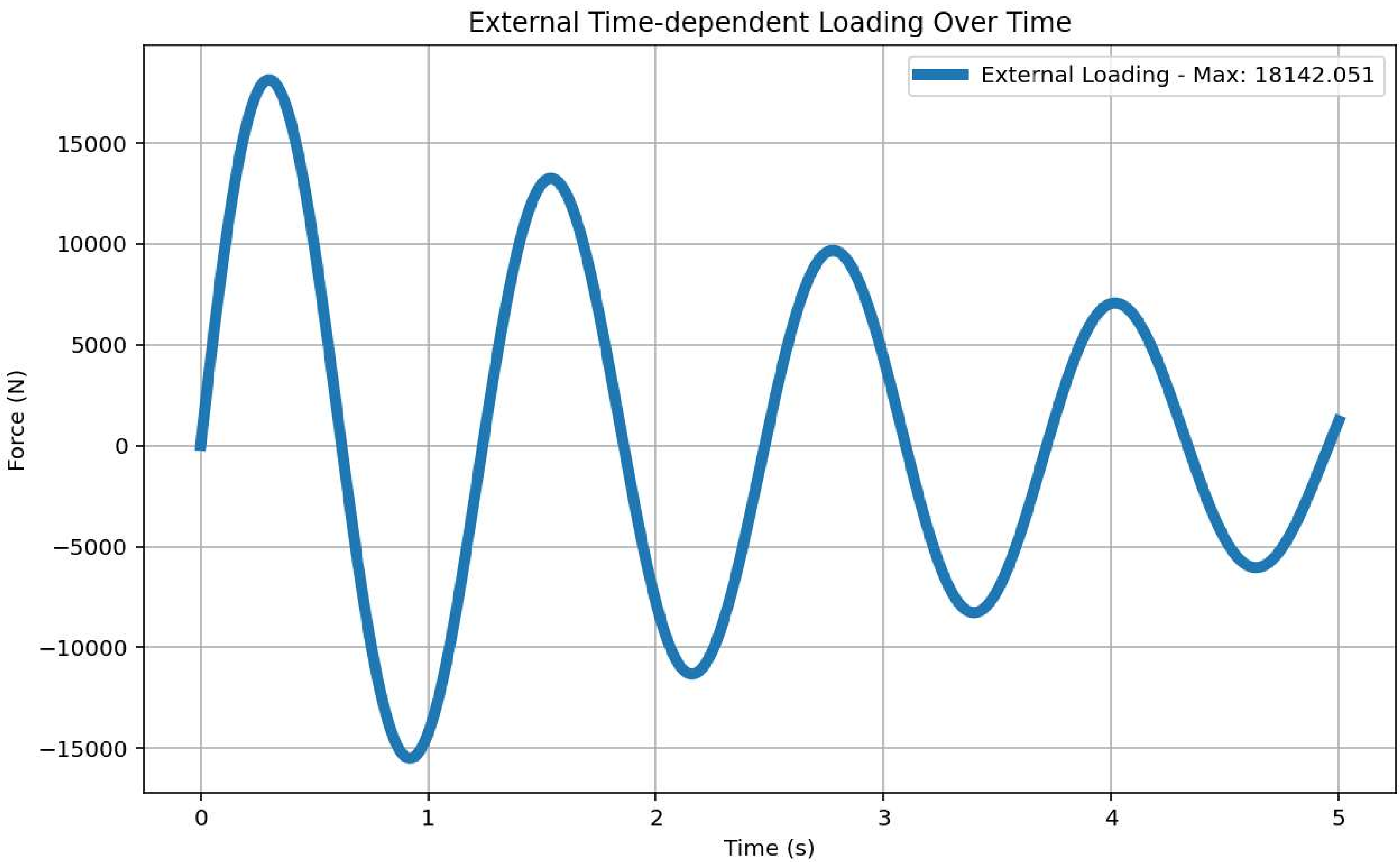
Period of Structure During Static External Time-dependent Loading Analysis
1e12
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
Structural Period [s]
−6.5
−6.0
−5.5
−5.0
−4.5
−4.0
−3.5
−3.0
−2.5
1e24
Displacement [mm]

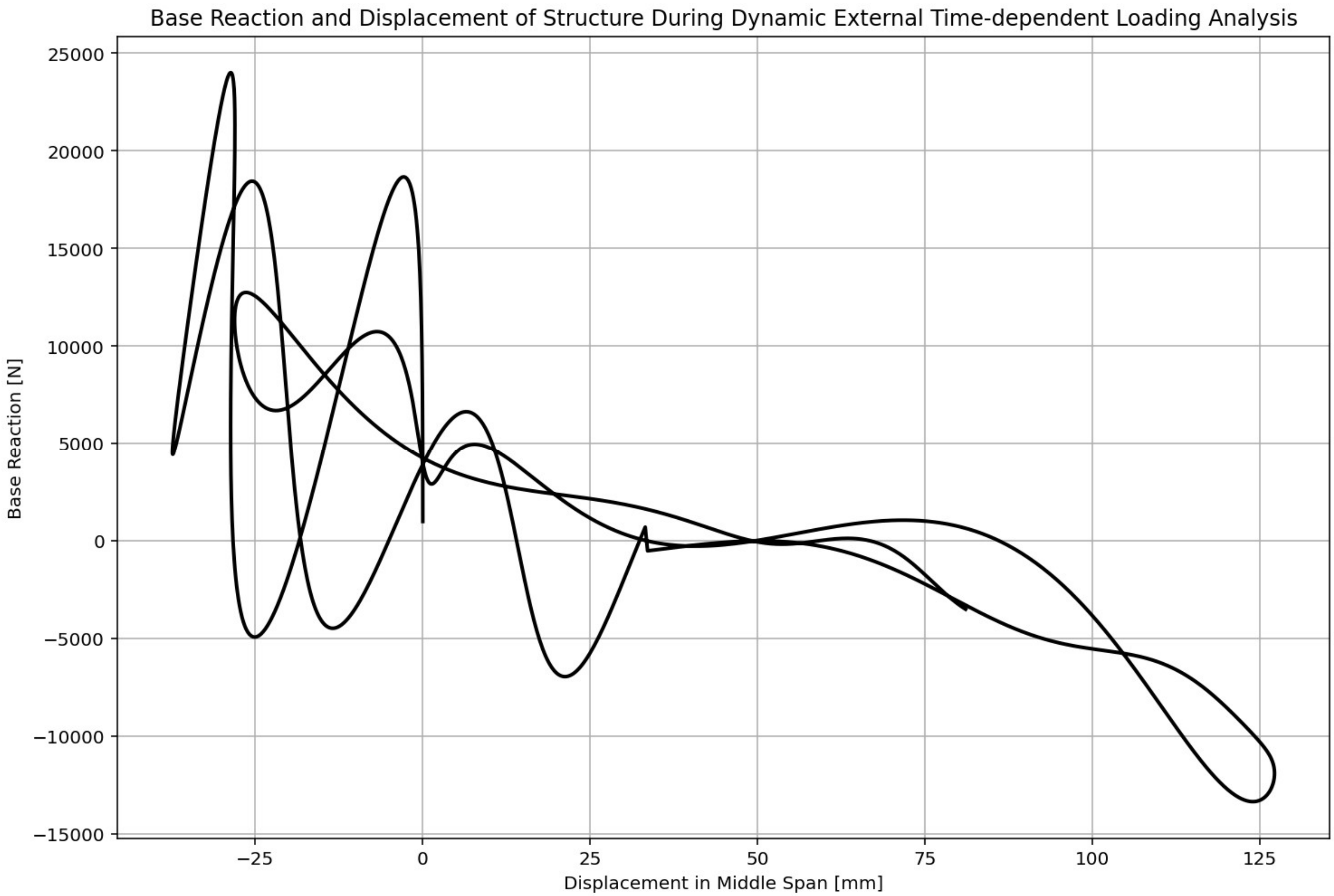


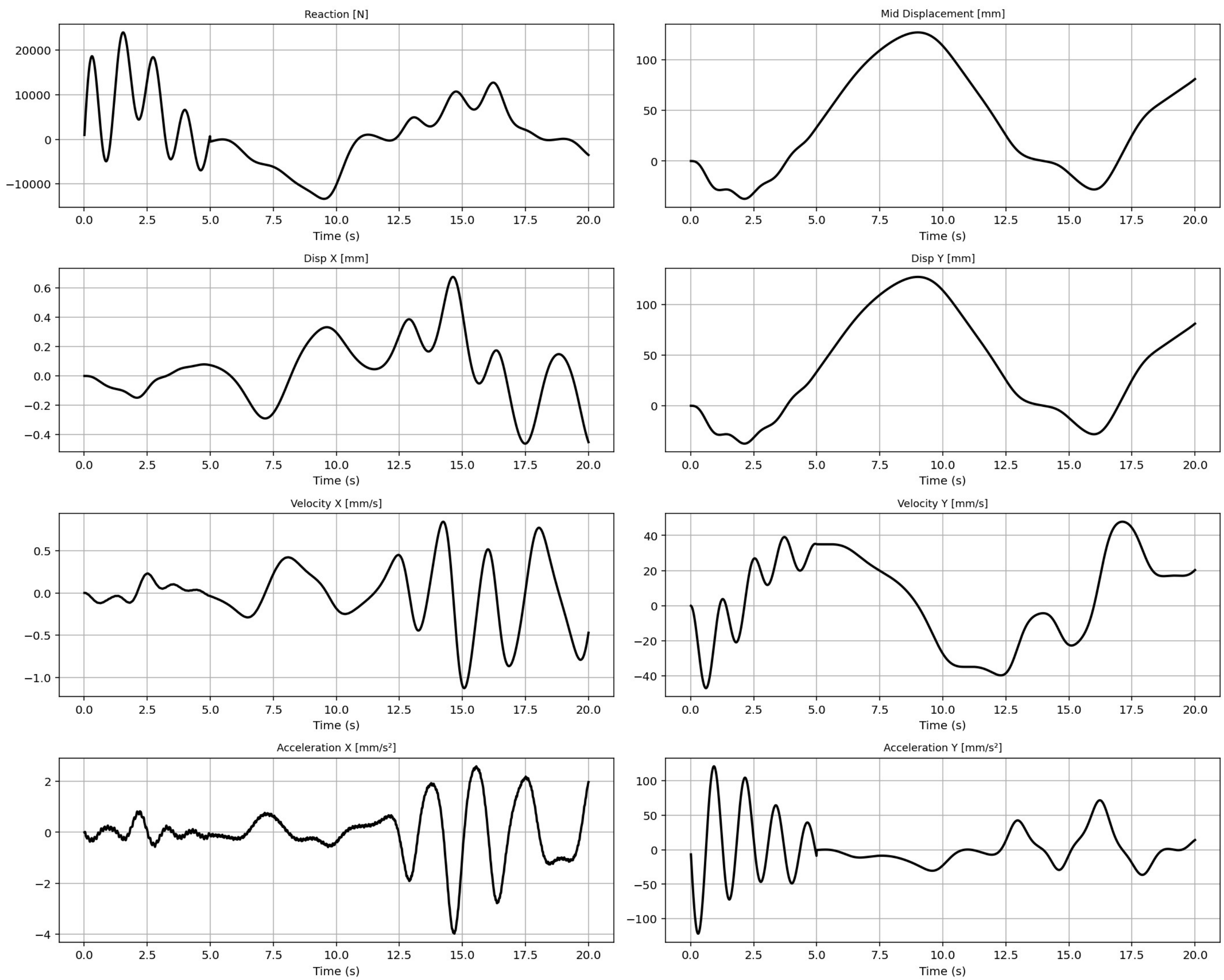
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
−10
−20
−30
−40
−50
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
7
Undeformed
Deformed

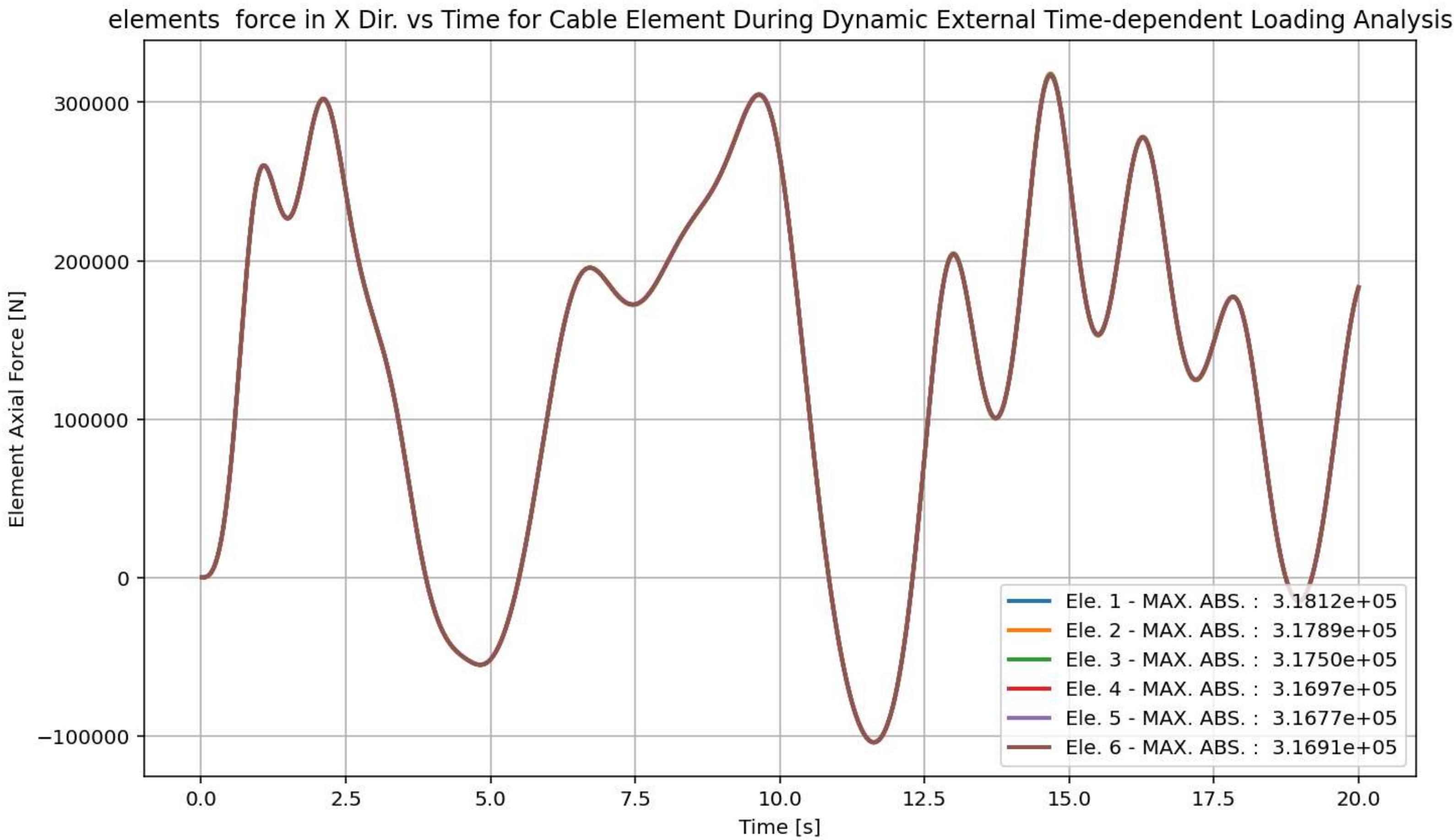
DYNAMIC TIME-HISTORY WITH EXTERNAL TIME-DEPENDENT LOADING ANALYSIS


Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...ATION_CAPACITY_INDEX\04_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RTED\W(x,t)_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
EXTERNAL TIME-DEPENDENT LOADING ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'DYNAMIC_EXTERNAL_TIME-DEPENDENT_LOADING'
DATA = CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOT
(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
ele_axialforce_ETDLD, ele_stress_ETDLD, ele_strain_ETDLD, ele_di_ETDLD,
node_displacementsX_ETDLD, node_displacementsY_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD,
stiffness_ETDLD, PERIOD_ETDLD, damping_ratio_ETDLD,
PERIOD_MIN_ETDLD, PERIOD_MAX_ETDLD) = DATA
XDATA = disp_mid_ETDLD
YDATA = reaction_ETDLD
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Displacement of Structure During Dynamic External Time-dependen
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_ETDLD, reaction_ETDLD, disp_mid_ETDLD,
dispX_ETDLD, dispY_ETDLD,
veloX_ETDLD, veloY_ETDLD,
accX_ETDLD, accY_ETDLD)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
30 %
Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 120, Col 36 UTF-8 CRLF RW Mem 55%

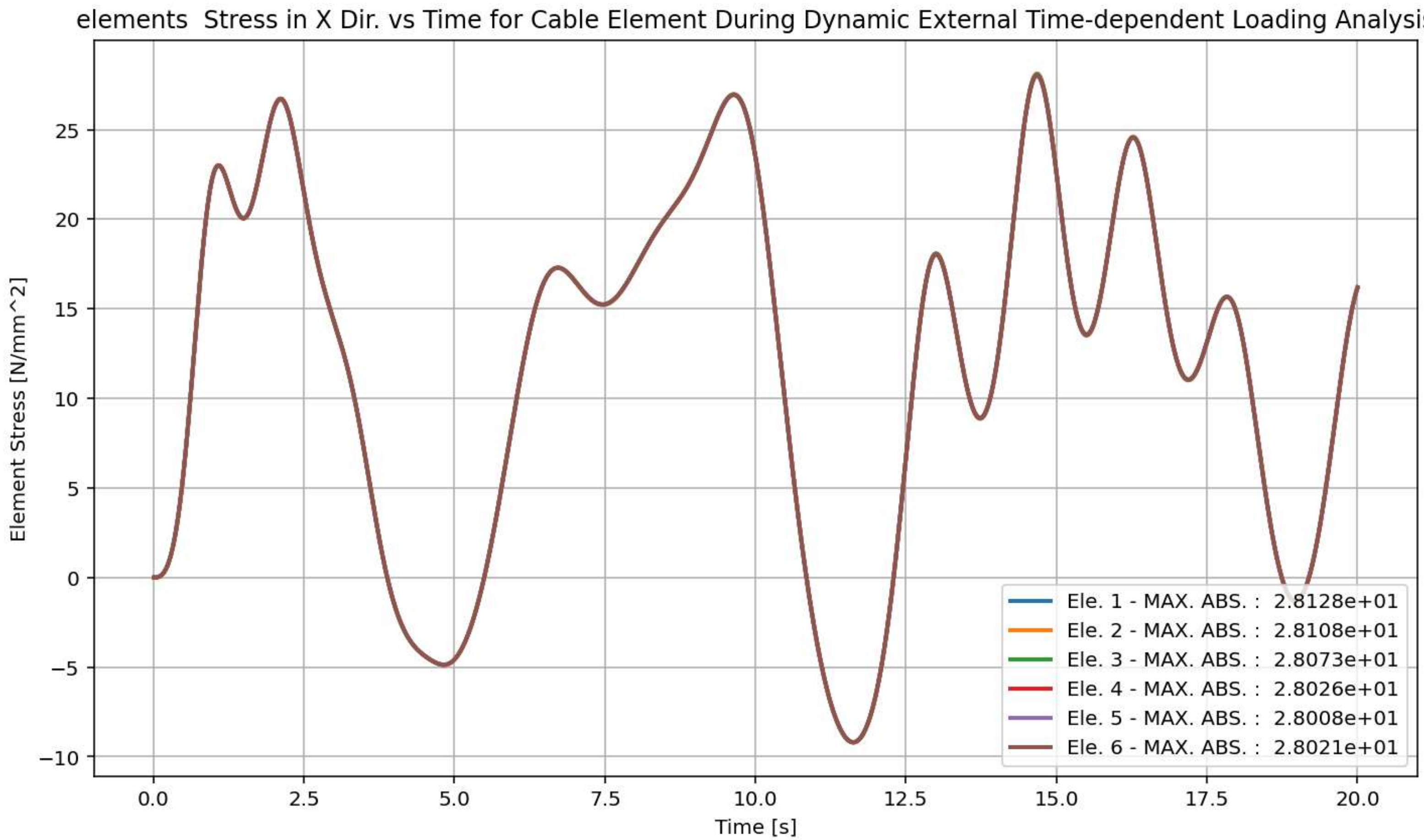

External Time-dependent Loading Over Time
External Loading - Max: 18142.051
Force (N)
Time (s)
15000
10000
5000
0
−5000
−10000
−15000
0
1
2
3
4
5

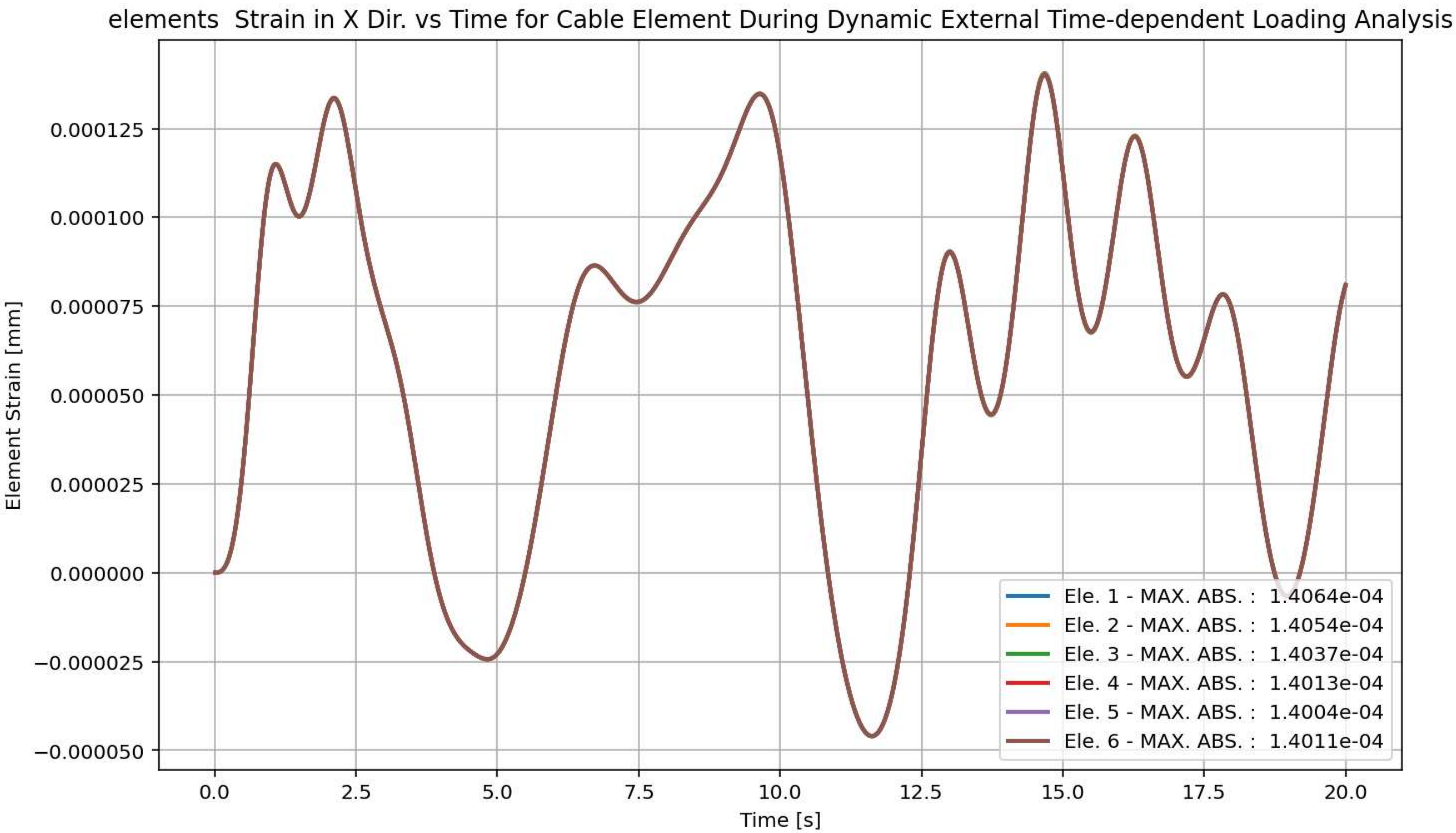

Base Reaction and Displacement of Structure During Dynamic External Time-dependent Loading Analysis
25000
20000
15000
10000
5000
0
−5000
−10000
−15000
Base Reaction [N]
−25
0
25
50
75
100
125
Displacement in Middle Span [mm]


Reaction [N]
Time (s)
Mid Displacement [mm]
Time (s)
Disp X [mm]
Time (s)
Disp Y [mm]
Time (s)
Velocity X [mm/s]
Time (s)
Velocity Y [mm/s]
Time (s)
Acceleration X [mm/s²]
Time (s)
Acceleration Y [mm/s²]
Time (s)

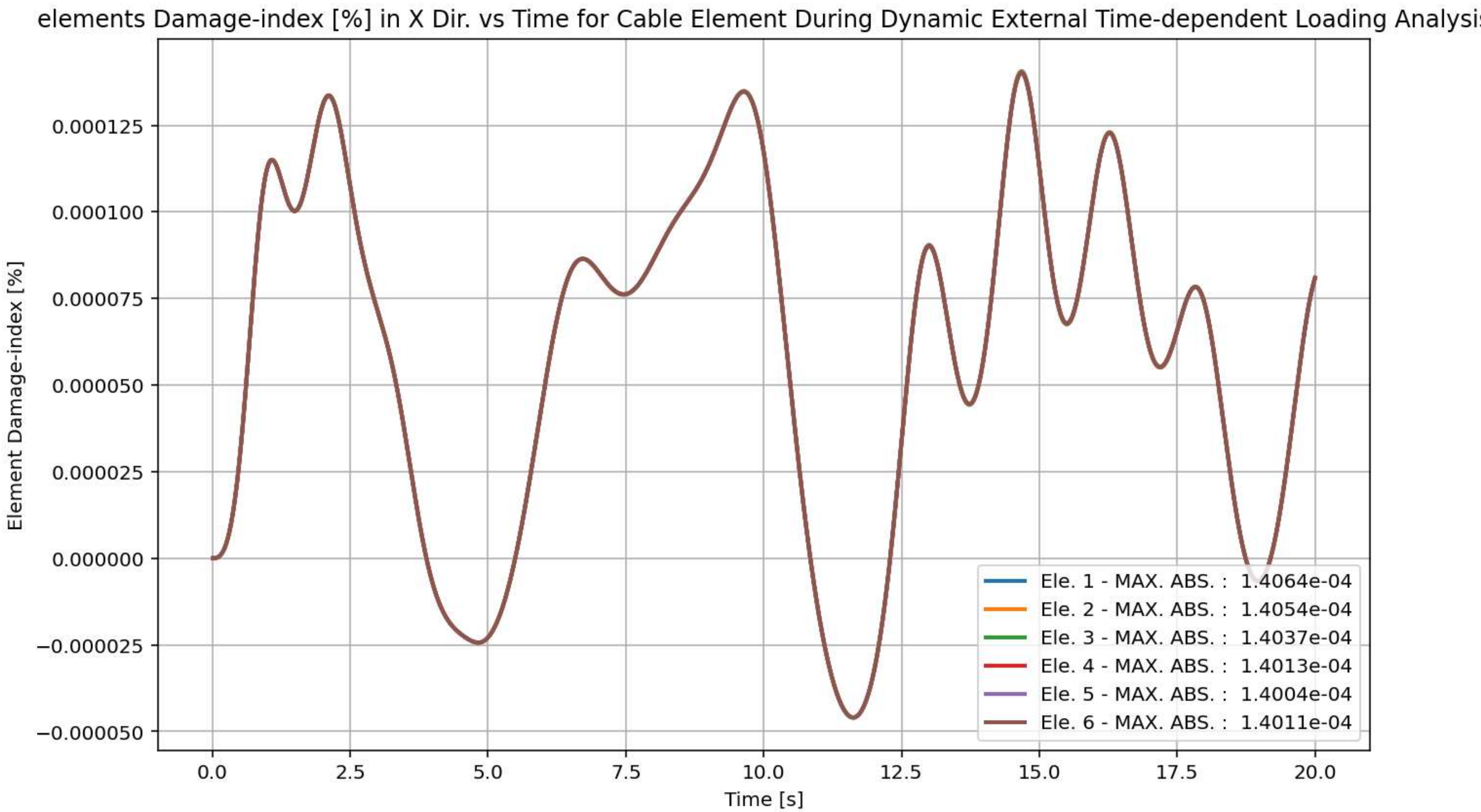


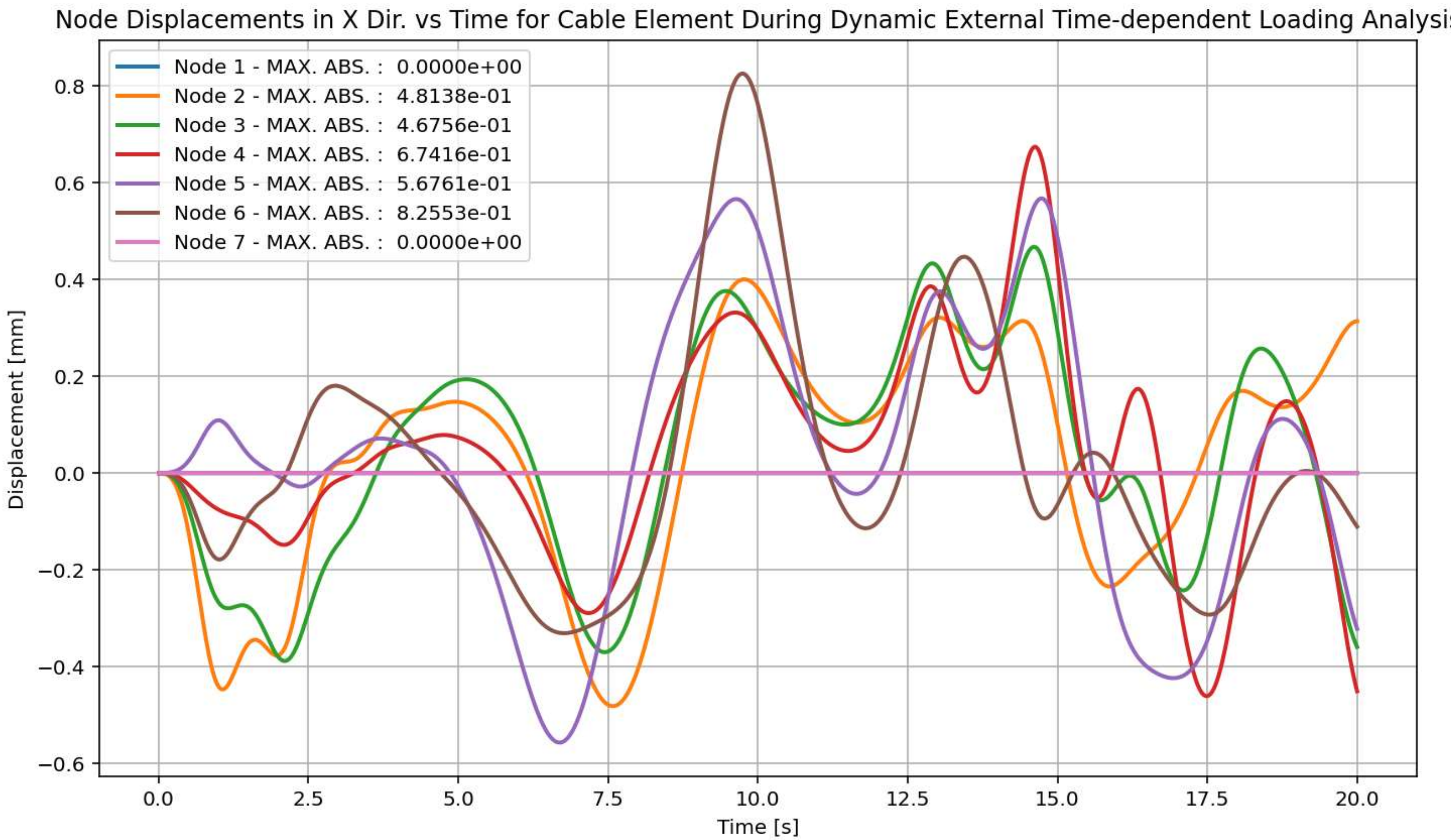
elements force in X Dir. vs Time for Cable Element During Dynamic External Time-dependent Loading Analysis
Element Axial Force [N]
300000
200000
100000
0
−100000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 3.1812e+05
Ele. 2 - MAX. ABS. : 3.1789e+05
Ele. 3 - MAX. ABS. : 3.1750e+05
Ele. 4 - MAX. ABS. : 3.1697e+05
Ele. 5 - MAX. ABS. : 3.1677e+05
Ele. 6 - MAX. ABS. : 3.1691e+05

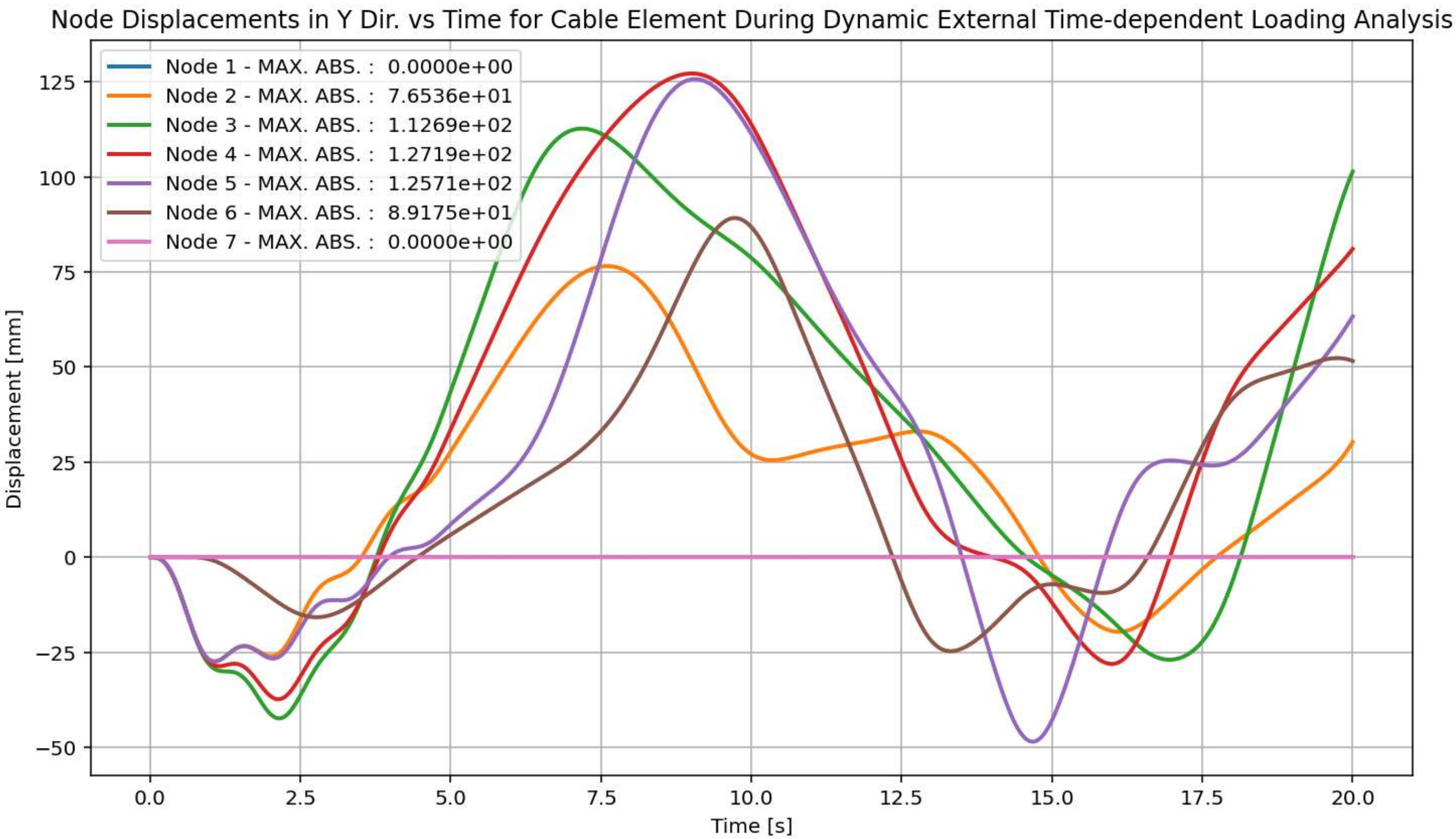

elements Stress in X Dir. vs Time for Cable Element During Dynamic External Time-dependent Loading Analysis
Element Stress [N/mm^2]
Time [s]
Ele. 1 - MAX. ABS. : 2.8128e+01
Ele. 2 - MAX. ABS. : 2.8108e+01
Ele. 3 - MAX. ABS. : 2.8073e+01
Ele. 4 - MAX. ABS. : 2.8026e+01
Ele. 5 - MAX. ABS. : 2.8008e+01
Ele. 6 - MAX. ABS. : 2.8021e+01



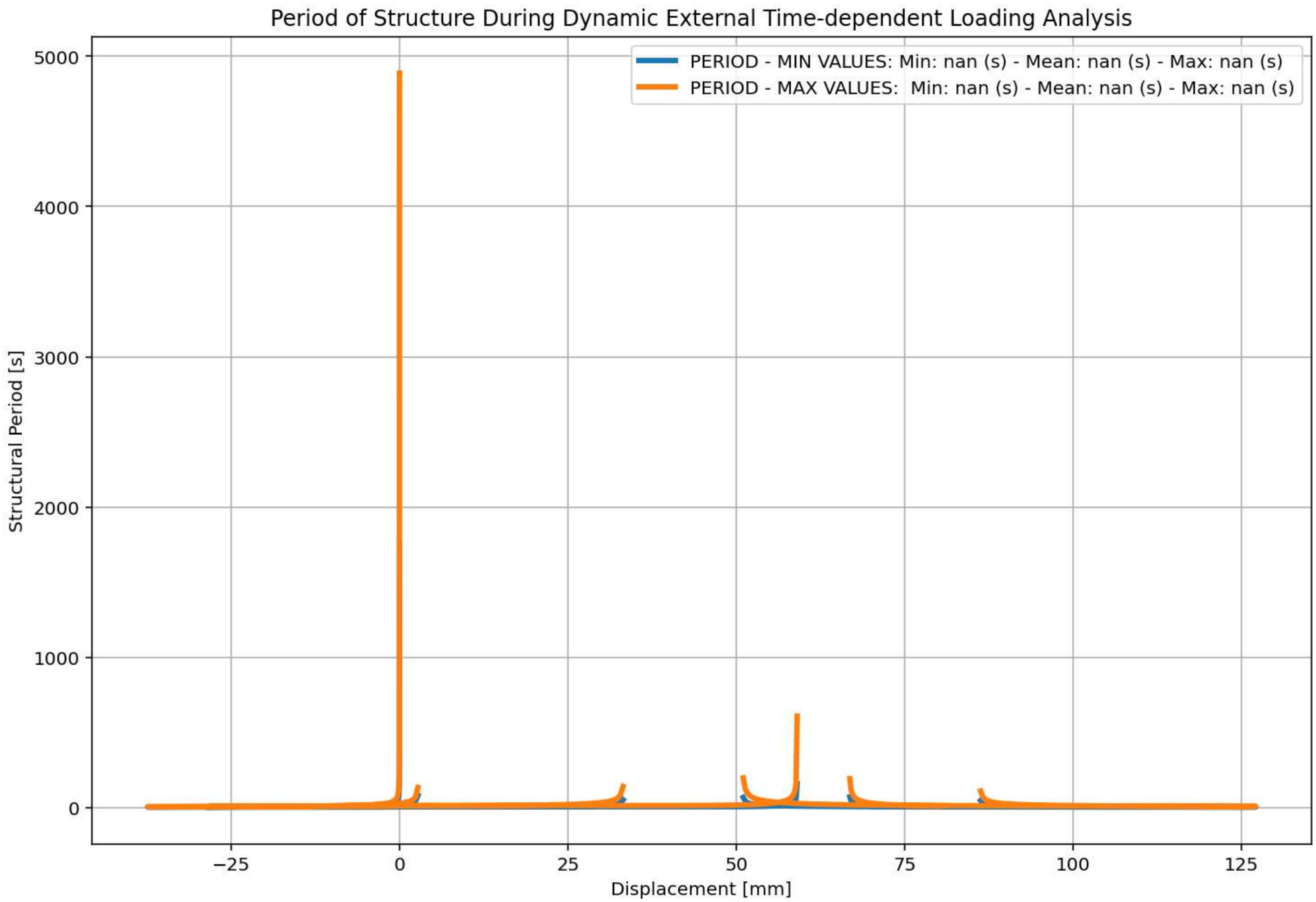
elements Strain in X Dir. vs Time for Cable Element During Dynamic External Time-dependent Loading Analysis
Element Strain [mm]
0.000125
0.000100
0.000075
0.000050
0.000025
0.000000
−0.000025
−0.000050
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 1.4064e-04
Ele. 2 - MAX. ABS. : 1.4054e-04
Ele. 3 - MAX. ABS. : 1.4037e-04
Ele. 4 - MAX. ABS. : 1.4013e-04
Ele. 5 - MAX. ABS. : 1.4004e-04
Ele. 6 - MAX. ABS. : 1.4011e-04

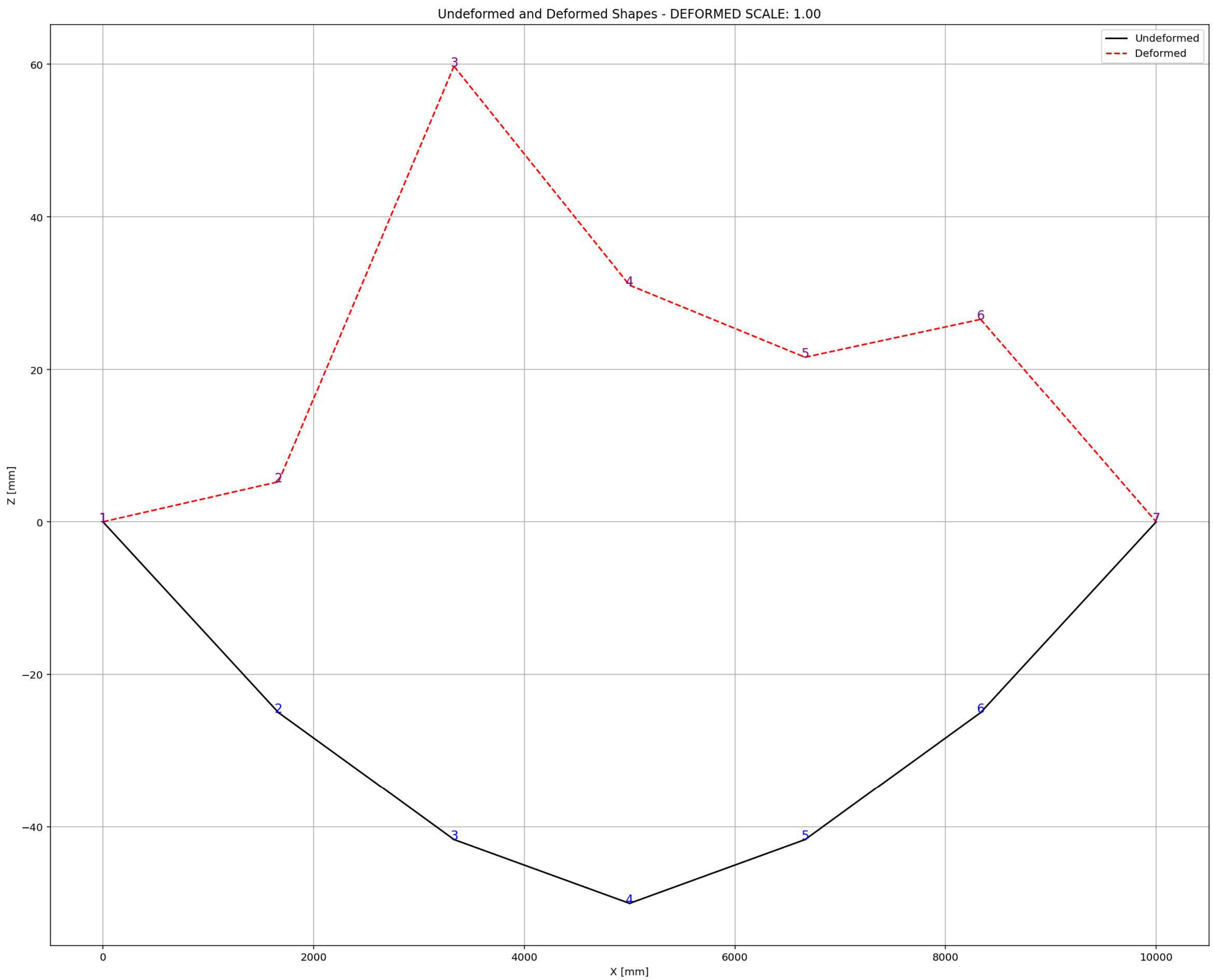

elements Damage-index [%] in X Dir. vs Time for Cable Element During Dynamic External Time-dependent Loading Analysis
Element Damage-index [%]
Time [s]
Ele. 1 - MAX. ABS. : 1.4064e-04
Ele. 2 - MAX. ABS. : 1.4054e-04
Ele. 3 - MAX. ABS. : 1.4037e-04
Ele. 4 - MAX. ABS. : 1.4013e-04
Ele. 5 - MAX. ABS. : 1.4004e-04
Ele. 6 - MAX. ABS. : 1.4011e-04


Node Displacements in X Dir. vs Time for Cable Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 4.8138e-01
Node 3 - MAX. ABS. : 4.6756e-01
Node 4 - MAX. ABS. : 6.7416e-01
Node 5 - MAX. ABS. : 5.6761e-01
Node 6 - MAX. ABS. : 8.2553e-01
Node 7 - MAX. ABS. : 0.0000e+00
Displacement [mm]
Time [s]
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



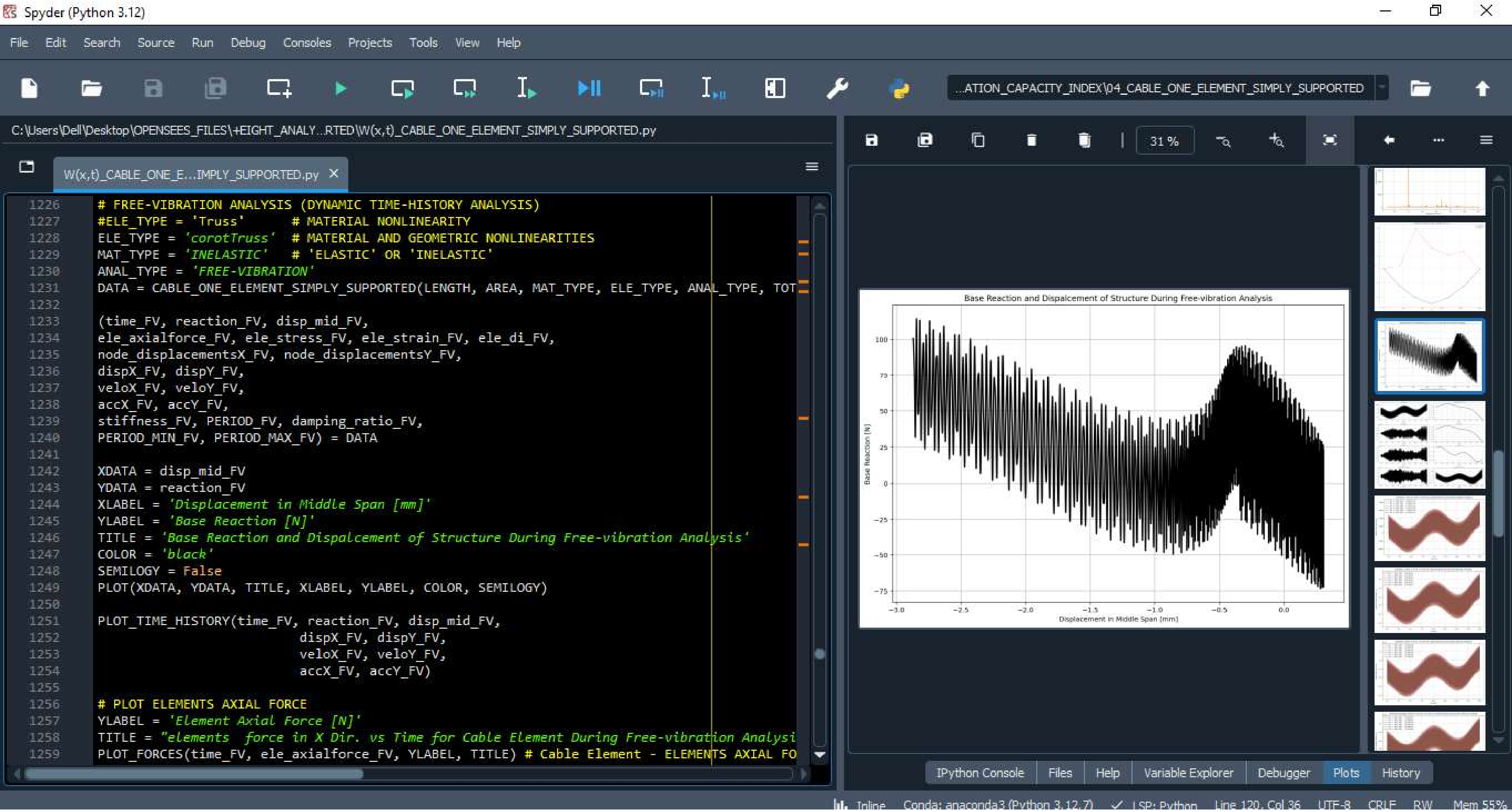
Node Displacements in Y Dir. vs Time for Cable Element During Dynamic External Time-dependent Loading Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 7.6536e+01
Node 3 - MAX. ABS. : 1.1269e+02
Node 4 - MAX. ABS. : 1.2719e+02
Node 5 - MAX. ABS. : 1.2571e+02
Node 6 - MAX. ABS. : 8.9175e+01
Node 7 - MAX. ABS. : 0.0000e+00
Displacement [mm]
125
100
75
50
25
0
−25
−50
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

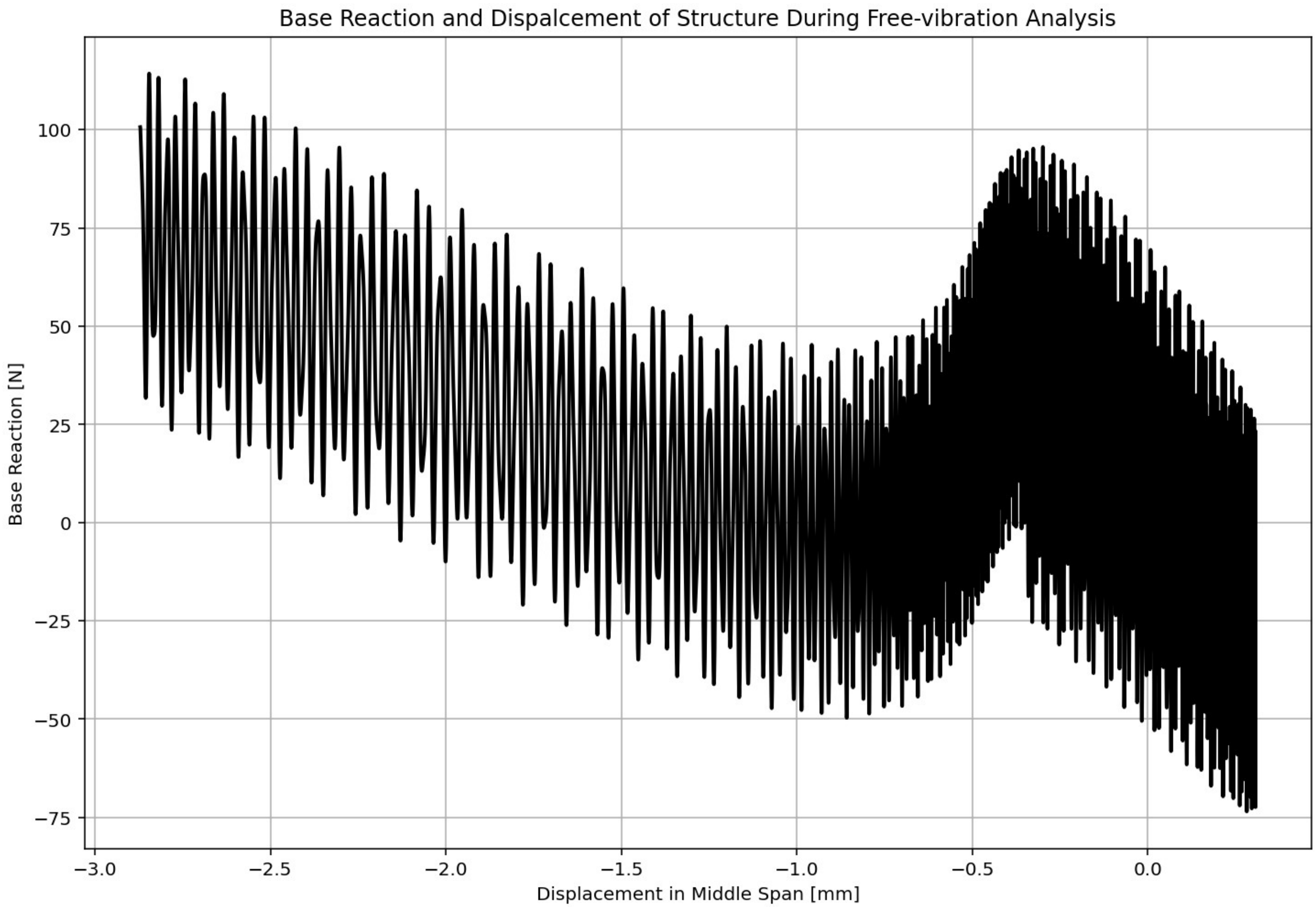

Period of Structure During Dynamic External Time-dependent Loading Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Displacement [mm]
5000
4000
3000
2000
1000
0
−25
0
25
50
75
100
125

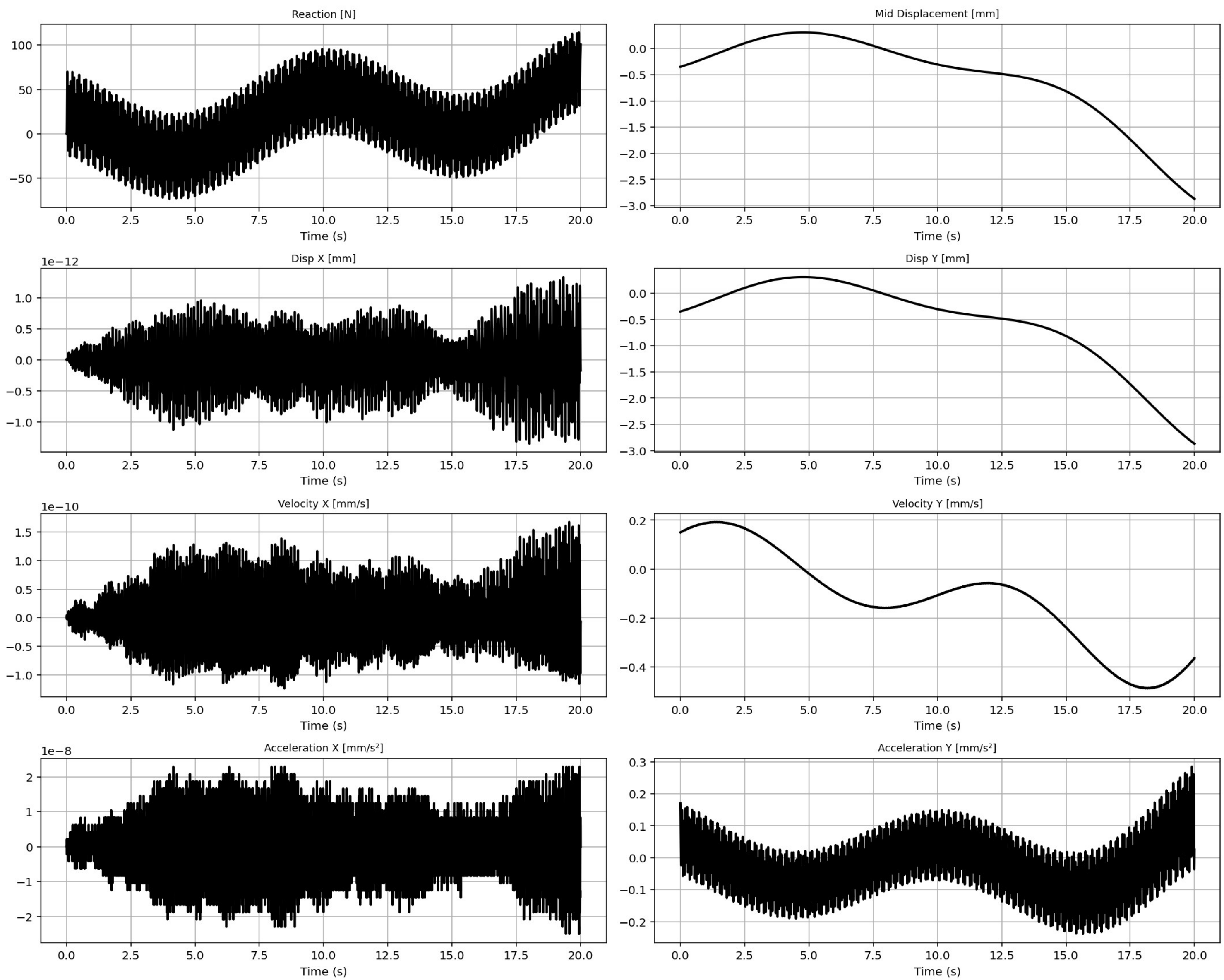


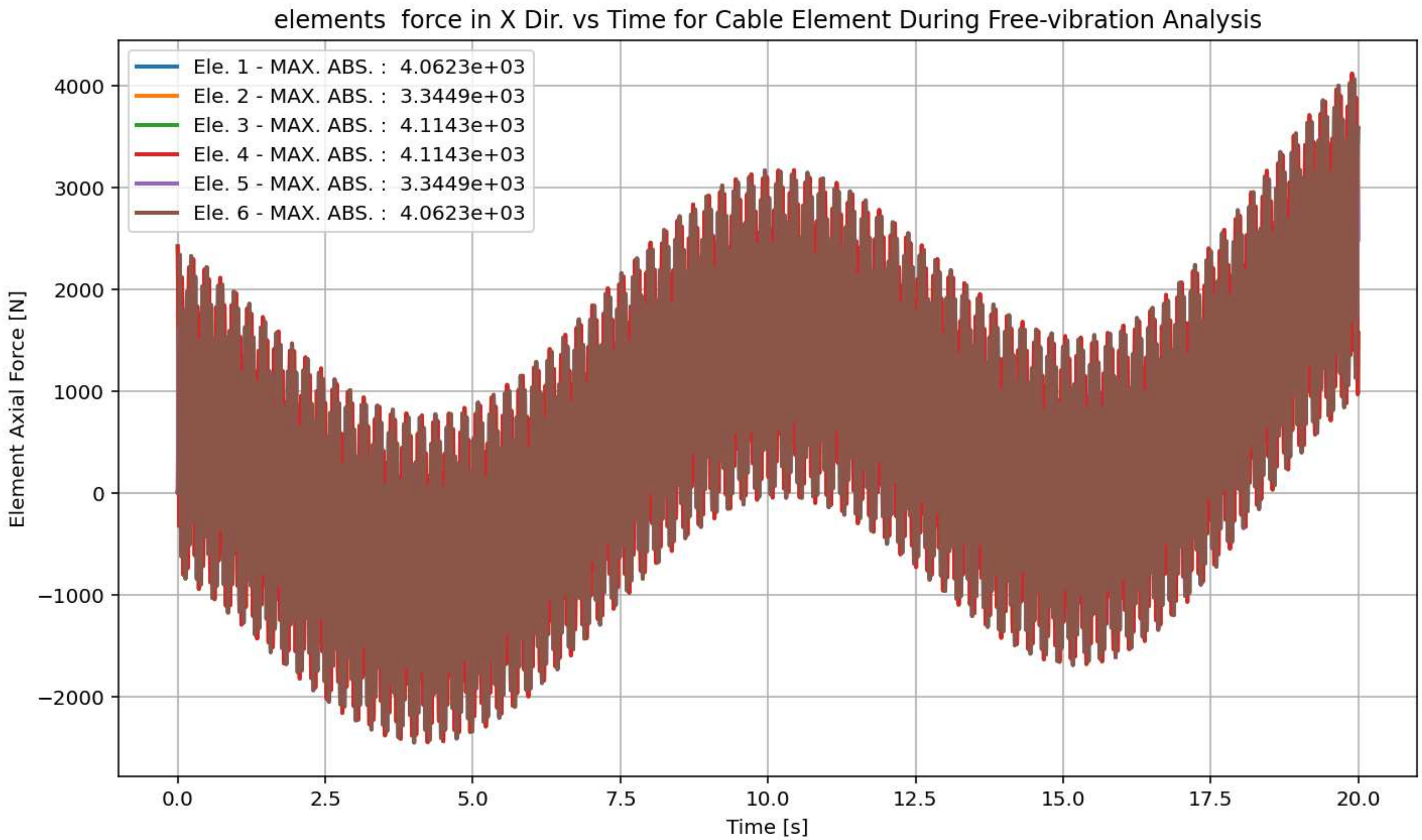
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
60
40
20
0
−20
−40
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
7

FREE-VIBRATION ANALYSIS

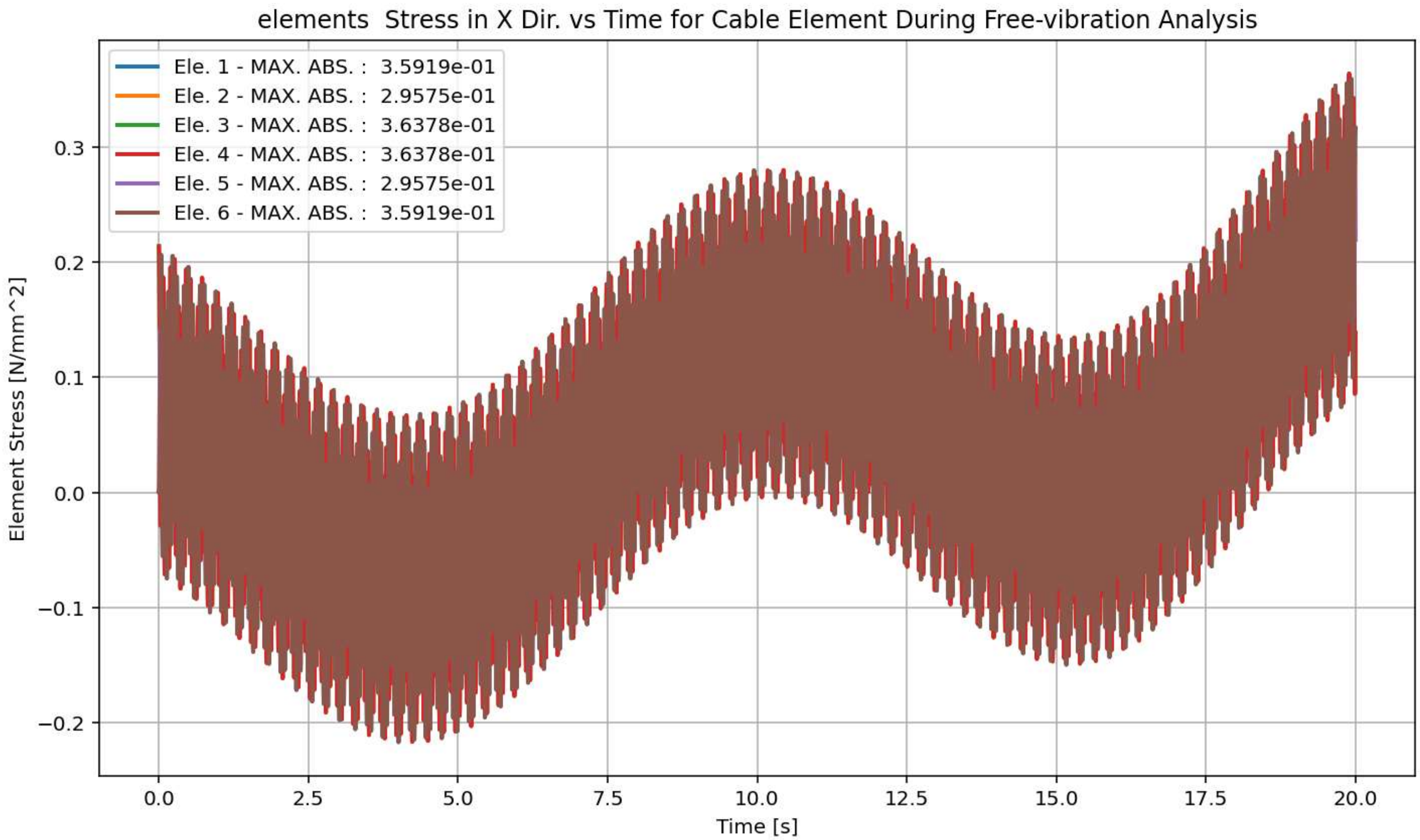

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...ATION_CAPACITY_INDEX\04_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RTED\W(x,t)_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
FREE-VIBRATION ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'FREE-VIBRATION'
DATA = CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOT
(time_FV, reaction_FV, disp_mid_FV,
ele_axialforce_FV, ele_stress_FV, ele_strain_FV, ele_di_FV,
node_displacementsX_FV, node_displacementsY_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV,
stiffness_FV, PERIOD_FV, damping_ratio_FV,
PERIOD_MIN_FV, PERIOD_MAX_FV) = DATA
XDATA = disp_mid_FV
YDATA = reaction_FV
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Free-vibration Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_FV, reaction_FV, disp_mid_FV,
dispX_FV, dispY_FV,
veloX_FV, veloY_FV,
accX_FV, accY_FV)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
TITLE = "elements force in X Dir. vs Time for Cable Element During Free-vibration Analysi
PLOT_FORCES(time_FV, ele_axialforce_FV, YLABEL, TITLE) # Cable Element - ELEMENTS AXIAL FO
31 %
Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 120, Col 36 UTF-8 CRLF RW Mem 55%

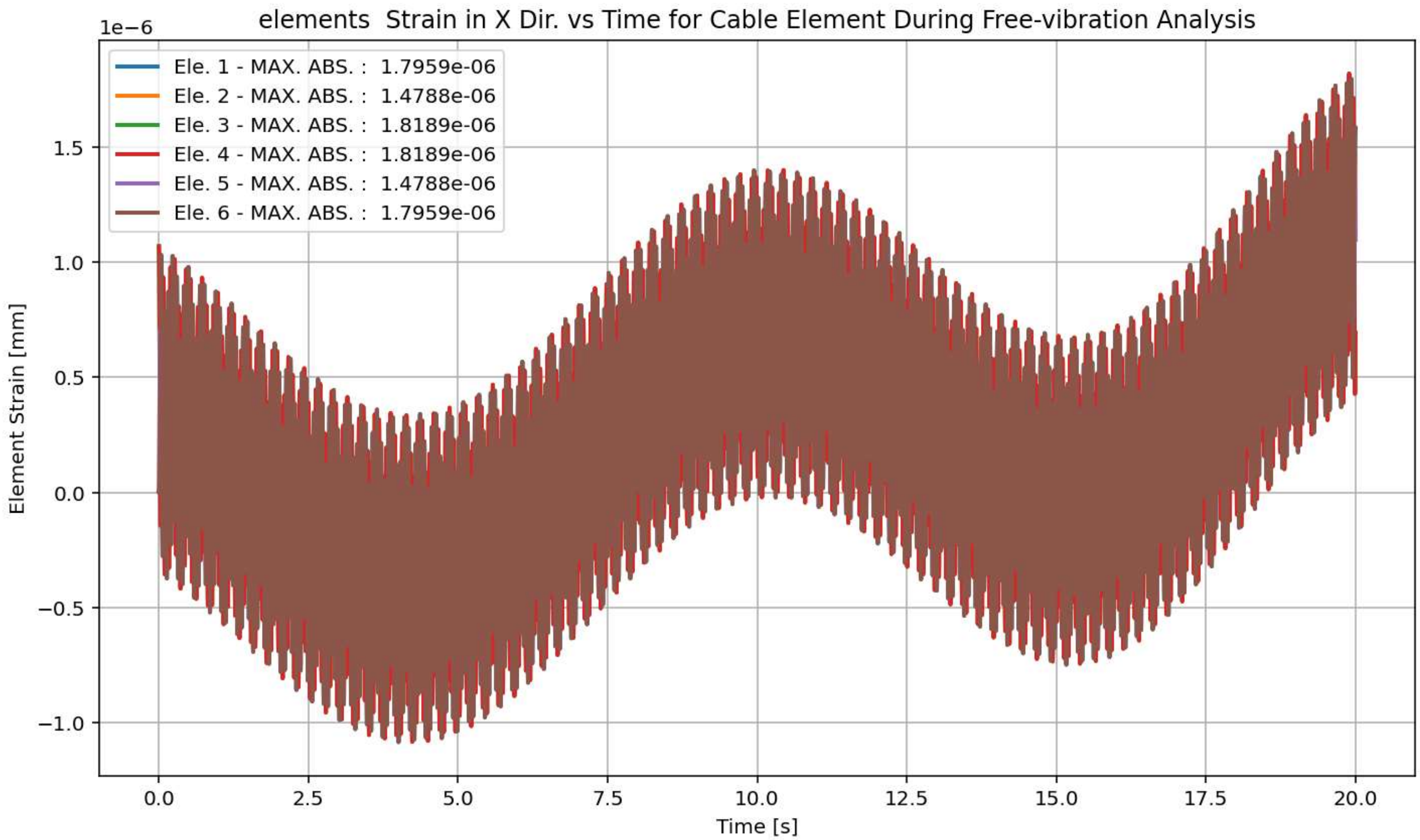

Base Reaction and Dispalcement of Structure During Free-vibration Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
100
75
50
25
0
−25
−50
−75
−3.0
−2.5
−2.0
−1.5
−1.0
−0.5
0.0


Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)
1e−12
1e−10
1e−8

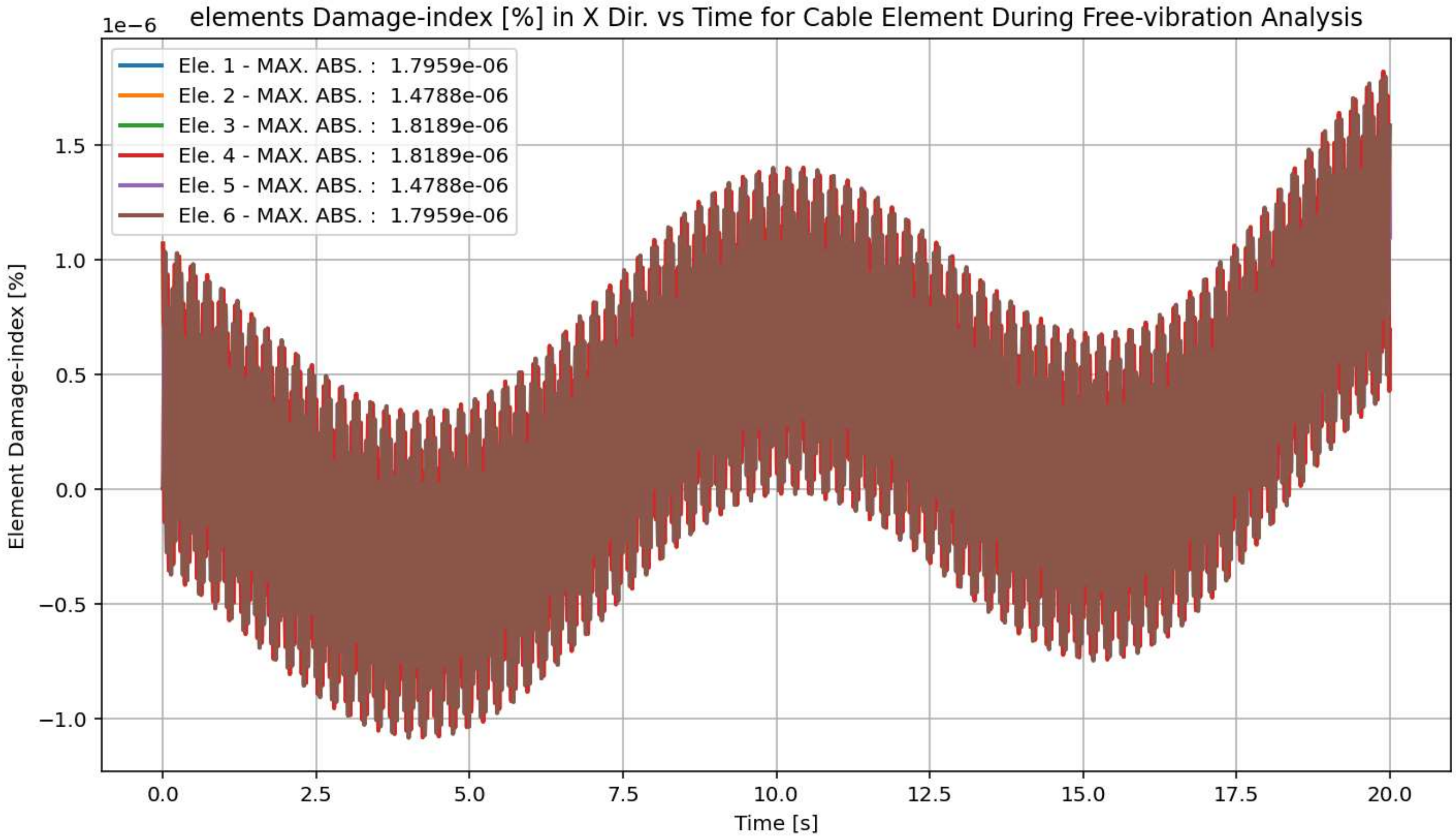


elements force in X Dir. vs Time for Cable Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 4.0623e+03
Ele. 2 - MAX. ABS. : 3.3449e+03
Ele. 3 - MAX. ABS. : 4.1143e+03
Ele. 4 - MAX. ABS. : 4.1143e+03
Ele. 5 - MAX. ABS. : 3.3449e+03
Ele. 6 - MAX. ABS. : 4.0623e+03
Element Axial Force [N]
Time [s]
4000
3000
2000
1000
0
−1000
−2000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

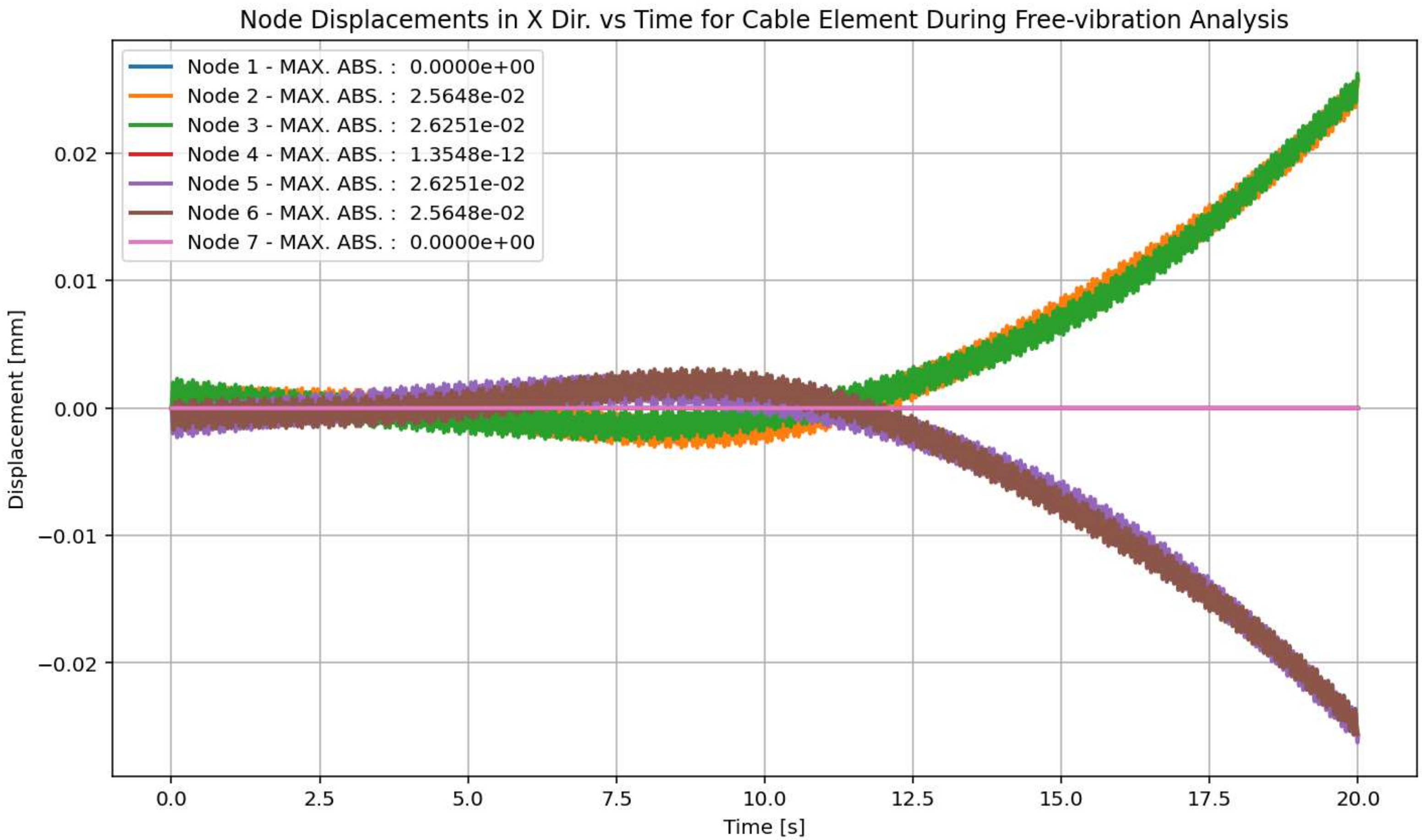

elements Stress in X Dir. vs Time for Cable Element During Free-vibration Analysis
Ele. 1 - MAX. ABS. : 3.5919e-01
Ele. 2 - MAX. ABS. : 2.9575e-01
Ele. 3 - MAX. ABS. : 3.6378e-01
Ele. 4 - MAX. ABS. : 3.6378e-01
Ele. 5 - MAX. ABS. : 2.9575e-01
Ele. 6 - MAX. ABS. : 3.5919e-01
Element Stress [N/mm^2]
Time [s]
0.3
0.2
0.1
0.0
−0.1
−0.2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



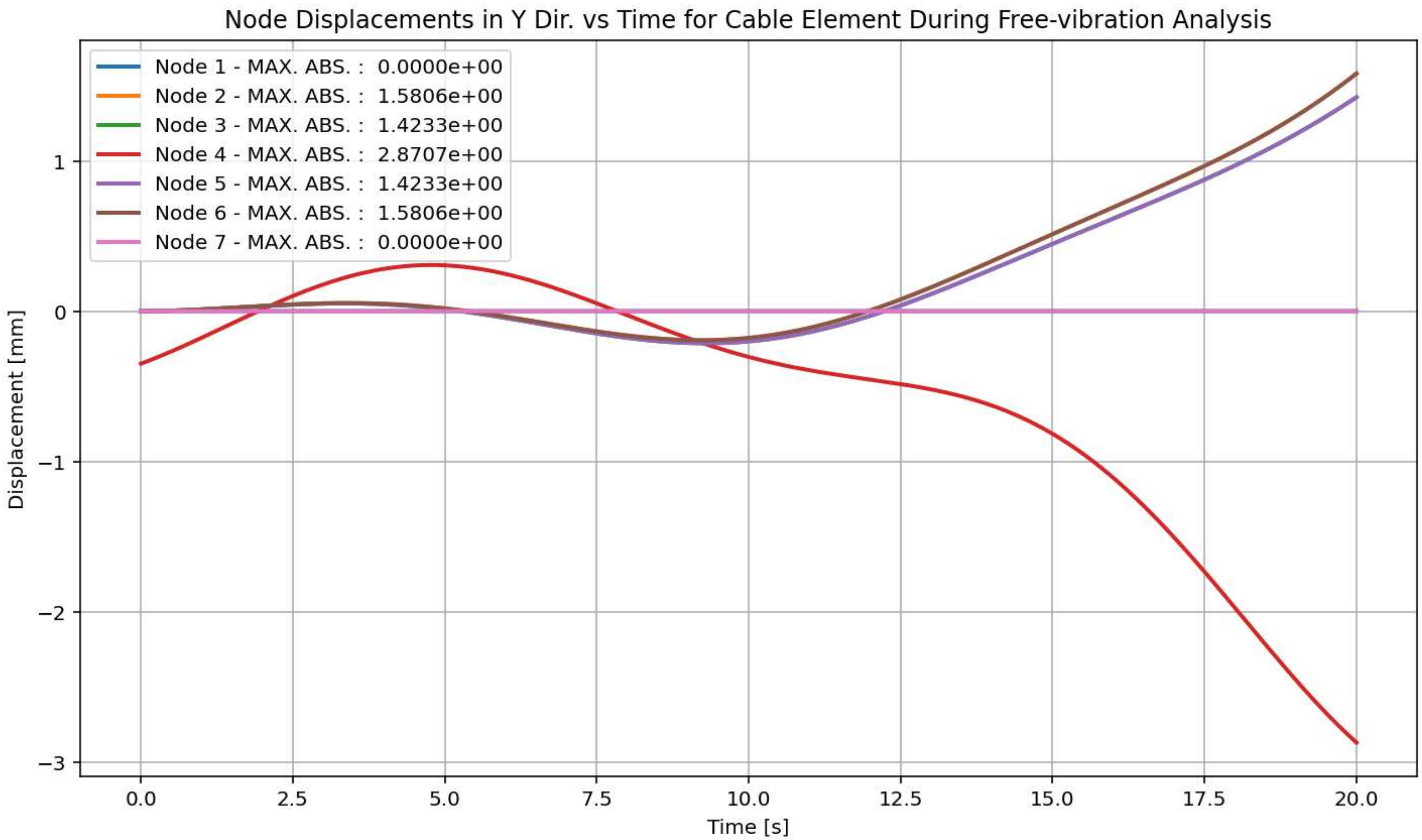
elements Strain in X Dir. vs Time for Cable Element During Free-vibration Analysis
1e−6
Ele. 1 - MAX. ABS. : 1.7959e-06
Ele. 2 - MAX. ABS. : 1.4788e-06
Ele. 3 - MAX. ABS. : 1.8189e-06
Ele. 4 - MAX. ABS. : 1.8189e-06
Ele. 5 - MAX. ABS. : 1.4788e-06
Ele. 6 - MAX. ABS. : 1.7959e-06
1.5
1.0
0.5
0.0
−0.5
−1.0
Element Strain [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

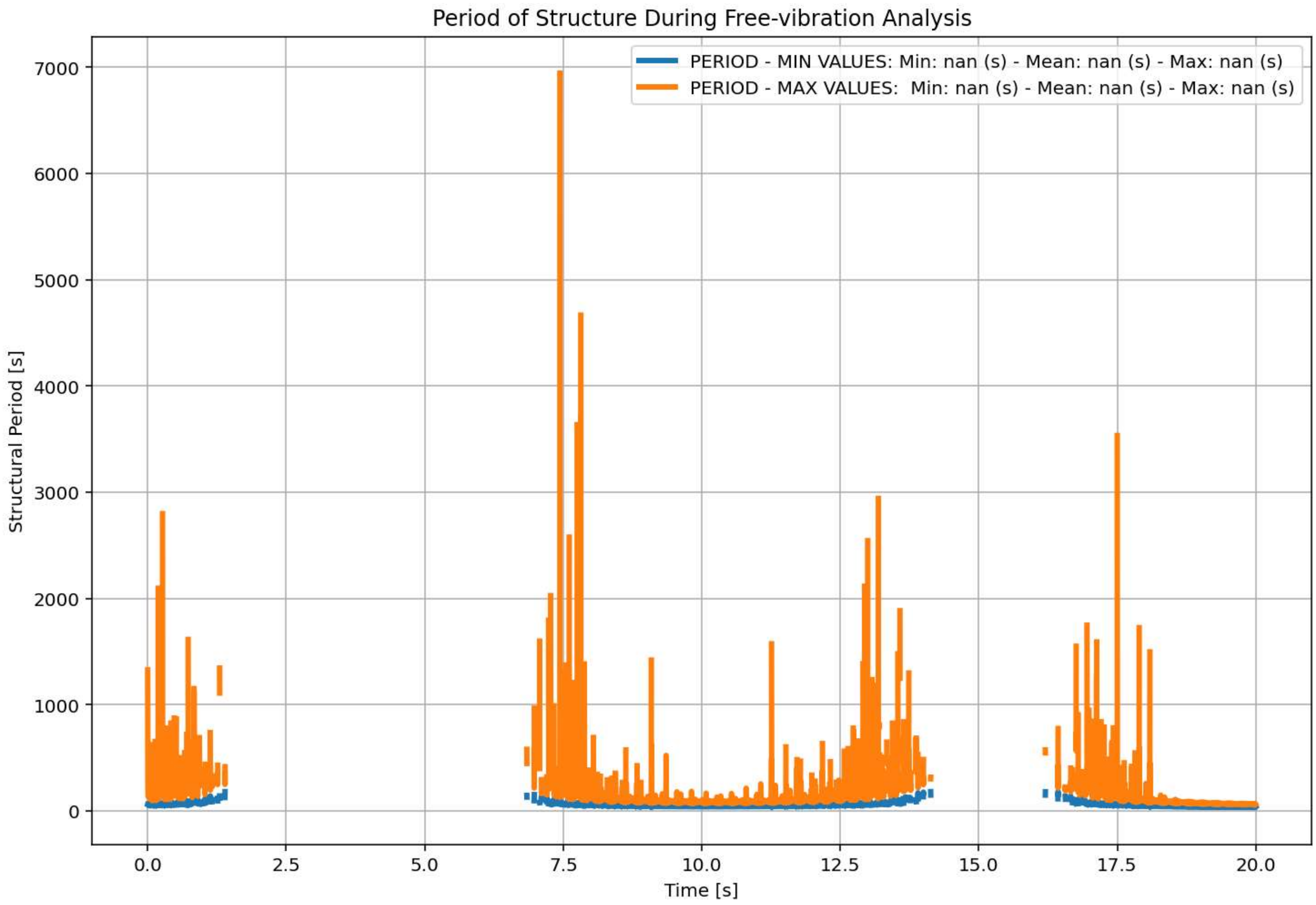


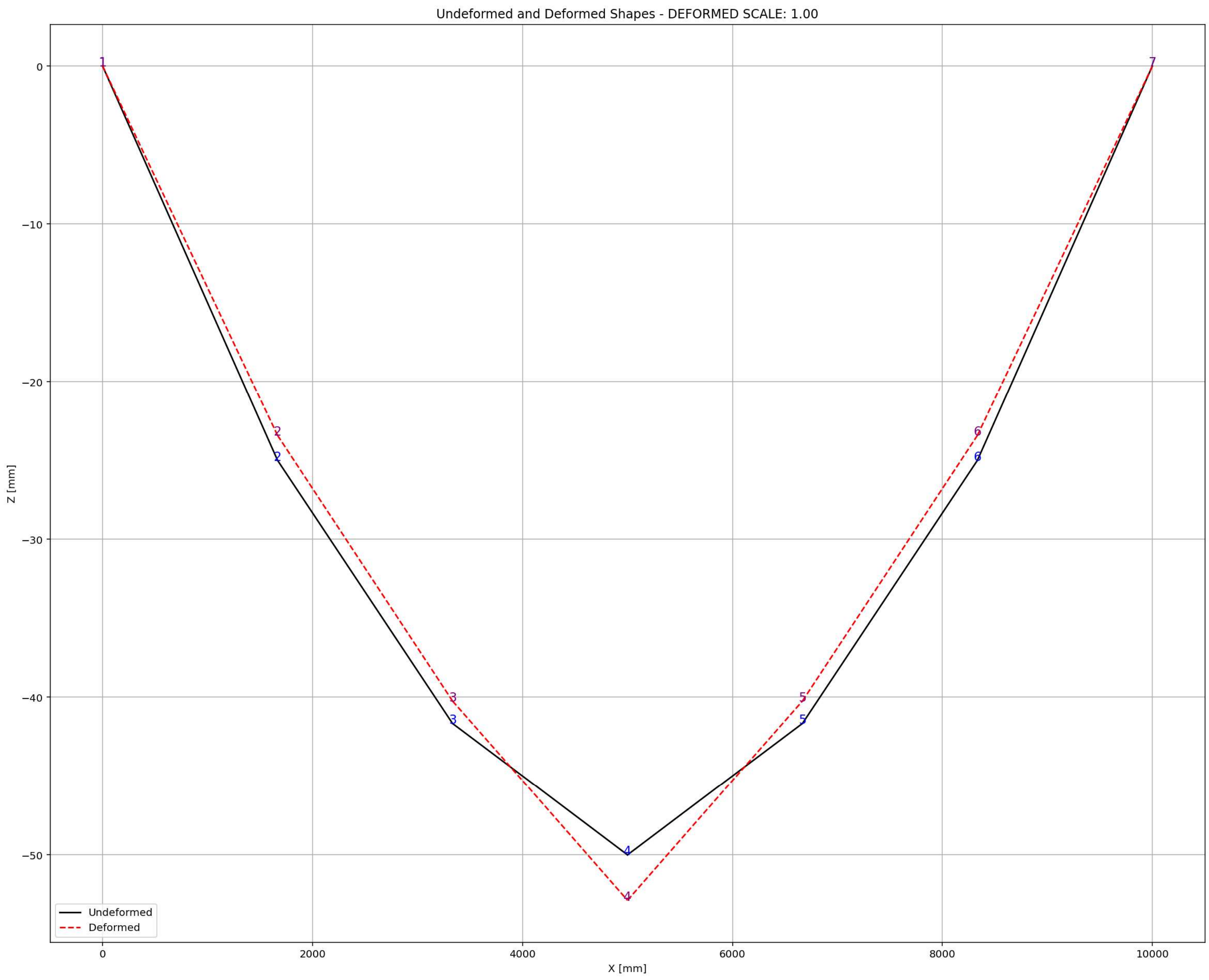
elements Damage-index [%] in X Dir. vs Time for Cable Element During Free-vibration Analysis
1e−6
Ele. 1 - MAX. ABS. : 1.7959e-06
Ele. 2 - MAX. ABS. : 1.4788e-06
Ele. 3 - MAX. ABS. : 1.8189e-06
Ele. 4 - MAX. ABS. : 1.8189e-06
Ele. 5 - MAX. ABS. : 1.4788e-06
Ele. 6 - MAX. ABS. : 1.7959e-06
Element Damage-index [%]
Time [s]



Node Displacements in X Dir. vs Time for Cable Element During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 2.5648e-02
Node 3 - MAX. ABS. : 2.6251e-02
Node 4 - MAX. ABS. : 1.3548e-12
Node 5 - MAX. ABS. : 2.6251e-02
Node 6 - MAX. ABS. : 2.5648e-02
Node 7 - MAX. ABS. : 0.0000e+00
0.02
0.01
0.00
−0.01
−0.02
Displacement [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]

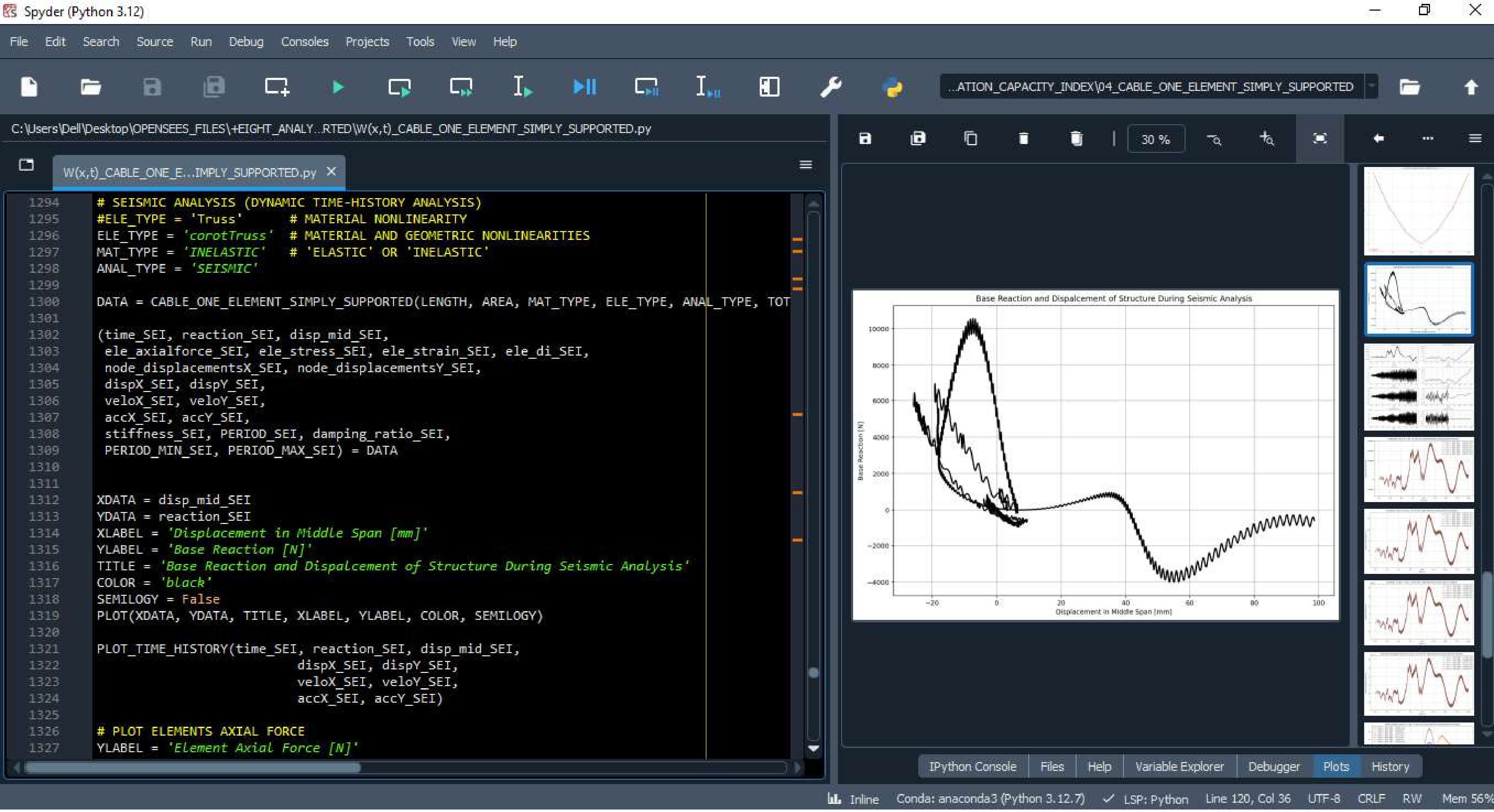

Node Displacements in Y Dir. vs Time for Cable Element During Free-vibration Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 1.5806e+00
Node 3 - MAX. ABS. : 1.4233e+00
Node 4 - MAX. ABS. : 2.8707e+00
Node 5 - MAX. ABS. : 1.4233e+00
Node 6 - MAX. ABS. : 1.5806e+00
Node 7 - MAX. ABS. : 0.0000e+00
Displacement [mm]
Time [s]
1
0
−1
−2
−3
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

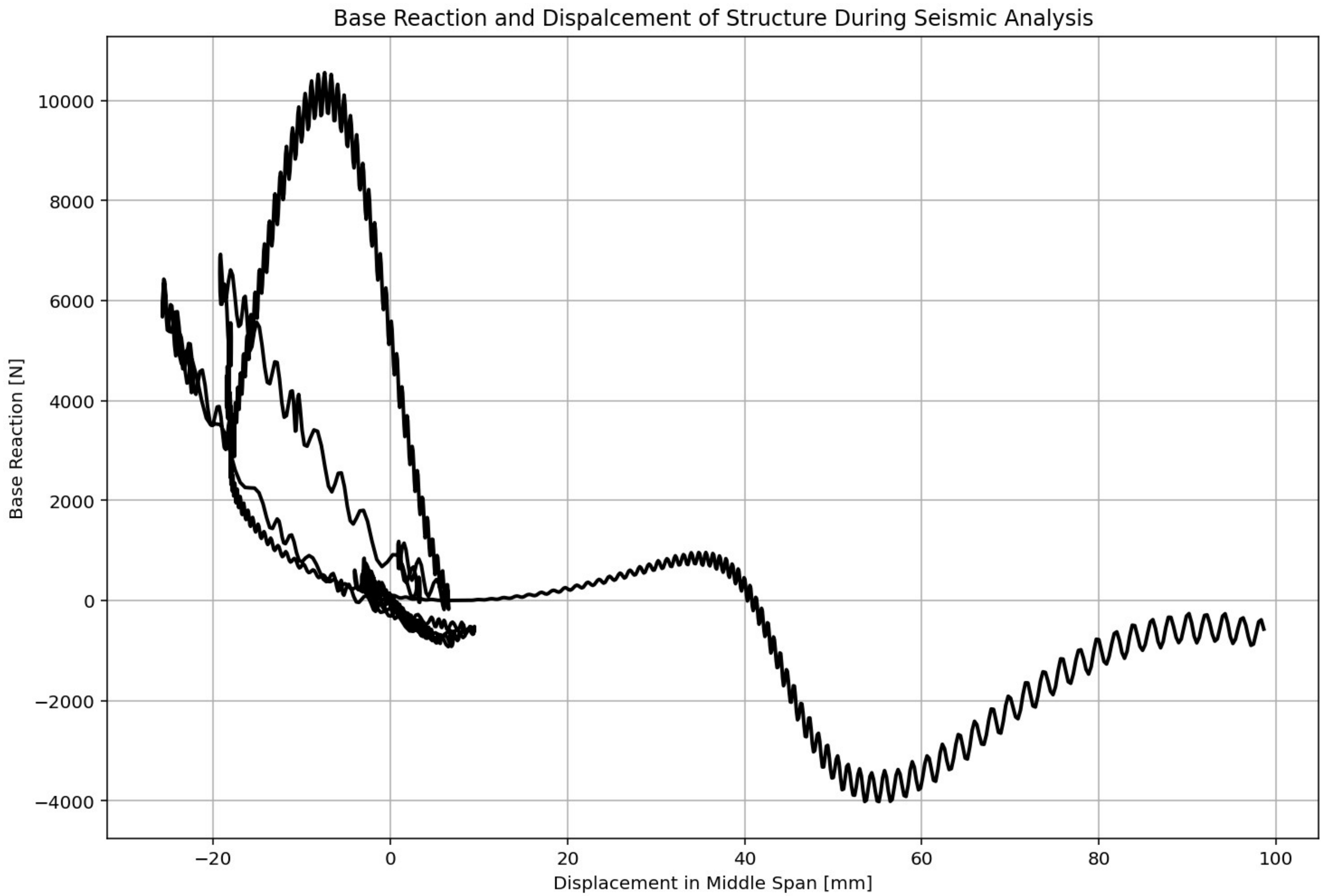

Period of Structure During Free-vibration Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Time [s]
0
1000
2000
3000
4000
5000
6000
7000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



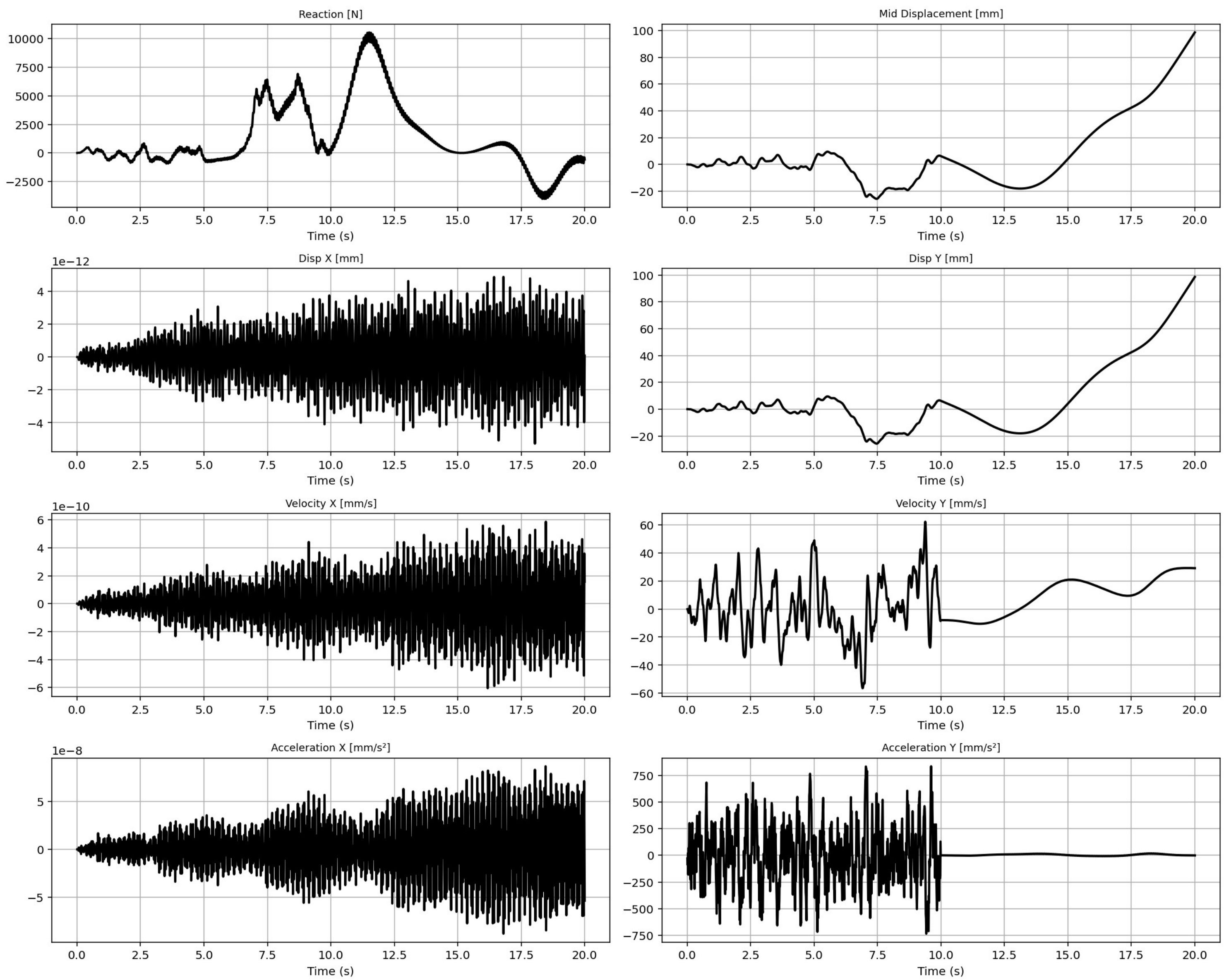
Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Z [mm]
X [mm]
0
−10
−20
−30
−40
−50
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
7
Undeformed
Deformed

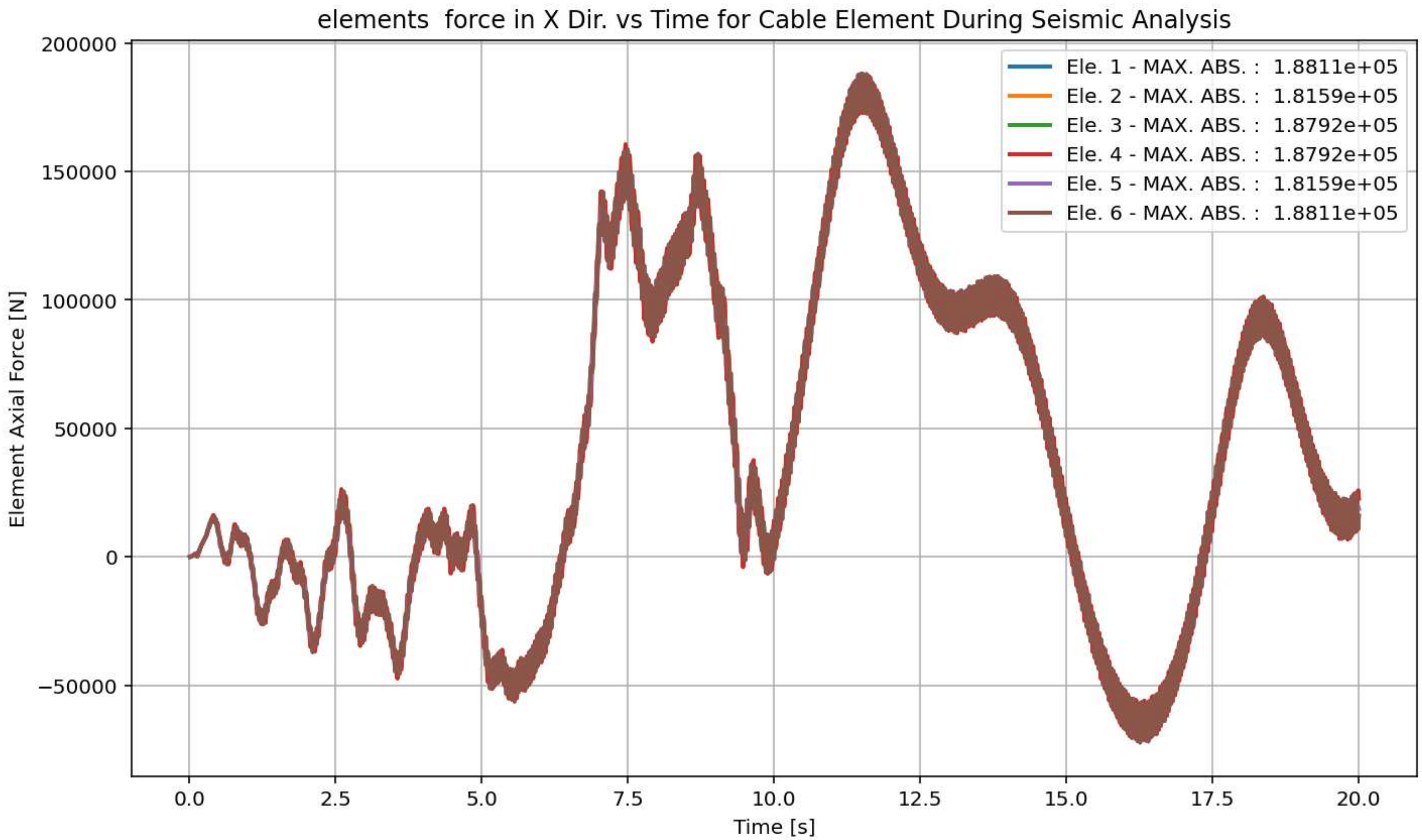
SEISMIC ANALYSIS

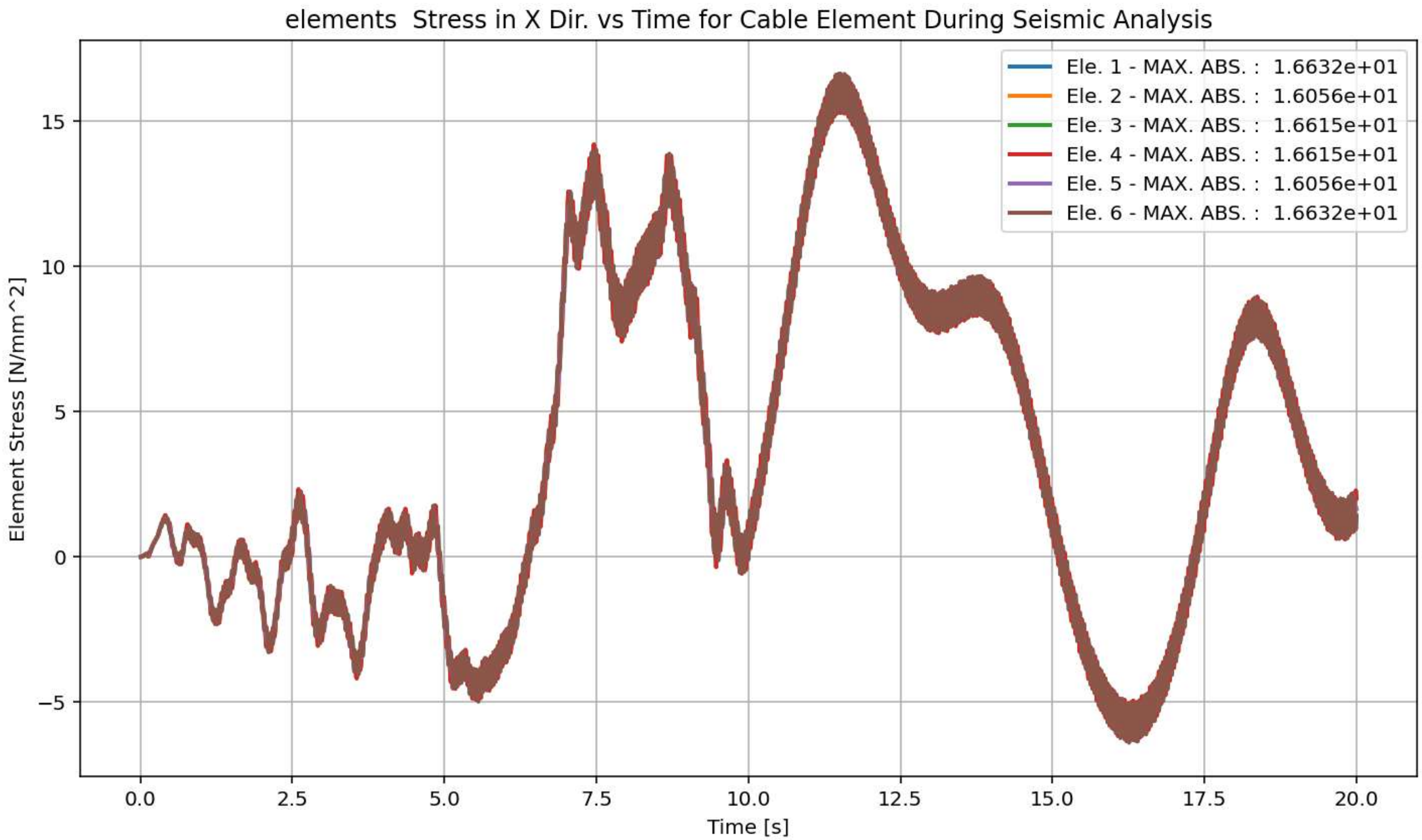

Spyder (Python 3.12)
File Edit Search Source Run Debug Consoles Projects Tools View Help
...ATION_CAPACITY_INDEX\04_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED
C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...RTED\W(x,t)_CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED.py
W(x,t)_CABLE_ONE_E...IMPLY_SUPPORTED.py
SEISMIC ANALYSIS (DYNAMIC TIME-HISTORY ANALYSIS)
#ELE_TYPE = 'Truss'   # MATERIAL NONLINEARITY
ELE_TYPE = 'corotTruss' # MATERIAL AND GEOMETRIC NONLINEARITIES
MAT_TYPE = 'INELASTIC'   # 'ELASTIC' OR 'INELASTIC'
ANAL_TYPE = 'SEISMIC'
DATA = CABLE_ONE_ELEMENT_SIMPLY_SUPPORTED(LENGTH, AREA, MAT_TYPE, ELE_TYPE, ANAL_TYPE, TOT
(time_SEI, reaction_SEI, disp_mid_SEI,
 ele_axialforce_SEI, ele_stress_SEI, ele_strain_SEI, ele_di_SEI,
 node_displacementsX_SEI, node_displacementsY_SEI,
 dispX_SEI, dispY_SEI,
 veloX_SEI, veloY_SEI,
 accX_SEI, accY_SEI,
 stiffness_SEI, PERIOD_SEI, damping_ratio_SEI,
 PERIOD_MIN_SEI, PERIOD_MAX_SEI) = DATA
XDATA = disp_mid_SEI
YDATA = reaction_SEI
XLABEL = 'Displacement in Middle Span [mm]'
YLABEL = 'Base Reaction [N]'
TITLE = 'Base Reaction and Dispalcement of Structure During Seismic Analysis'
COLOR = 'black'
SEMILOGY = False
PLOT(XDATA, YDATA, TITLE, XLABEL, YLABEL, COLOR, SEMILOGY)
PLOT_TIME_HISTORY(time_SEI, reaction_SEI, disp_mid_SEI,
                  dispX_SEI, dispY_SEI,
                  veloX_SEI, veloY_SEI,
                  accX_SEI, accY_SEI)
# PLOT ELEMENTS AXIAL FORCE
YLABEL = 'Element Axial Force [N]'
30 %
Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
Displacement in Middle Span [mm]
IPython Console Files Help Variable Explorer Debugger Plots History
Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 120, Col 36 UTF-8 CRLF RW Mem 56%



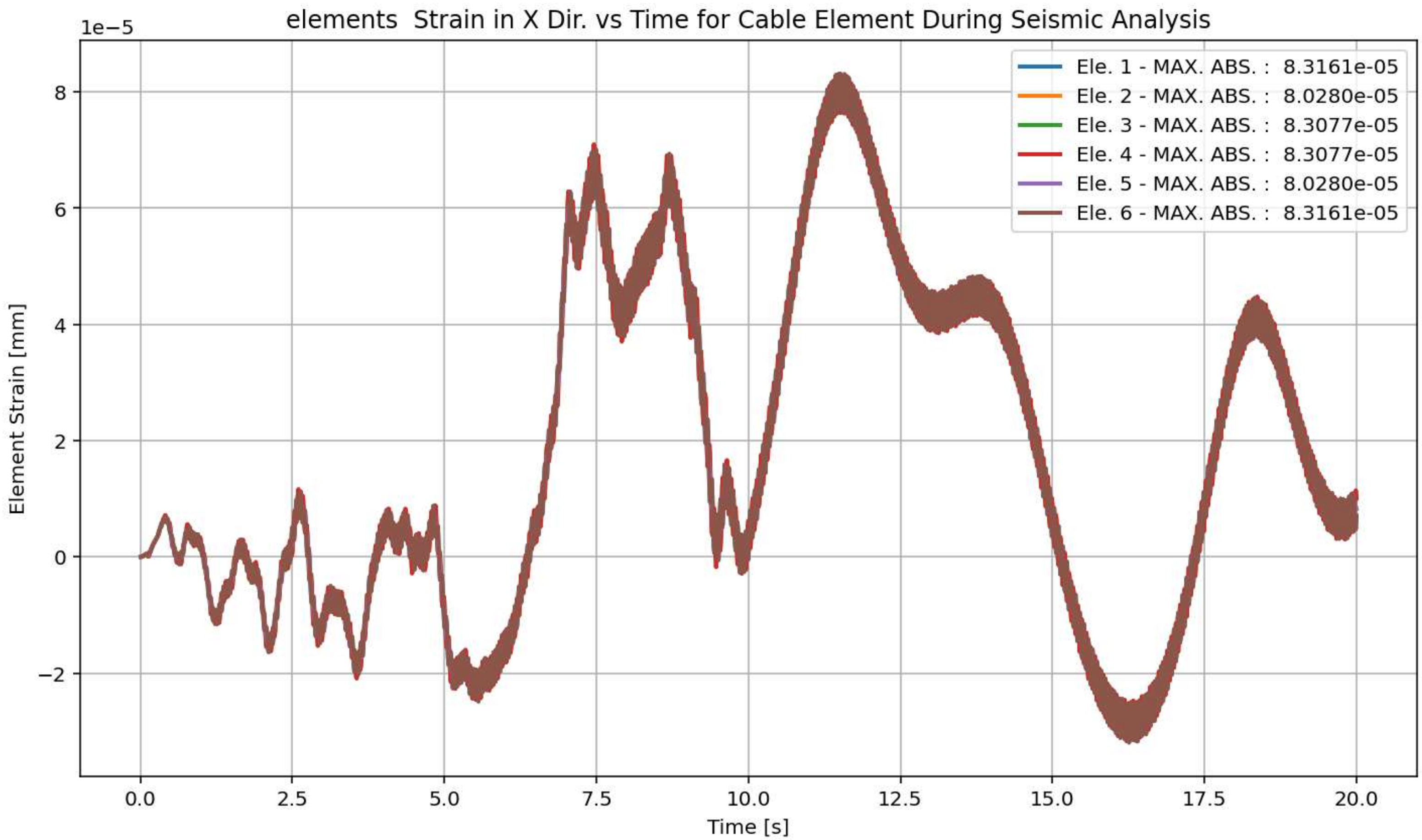
Base Reaction and Dispalcement of Structure During Seismic Analysis
Base Reaction [N]
10000
8000
6000
4000
2000
0
−2000
−4000
−20
0
20
40
60
80
100
Displacement in Middle Span [mm]

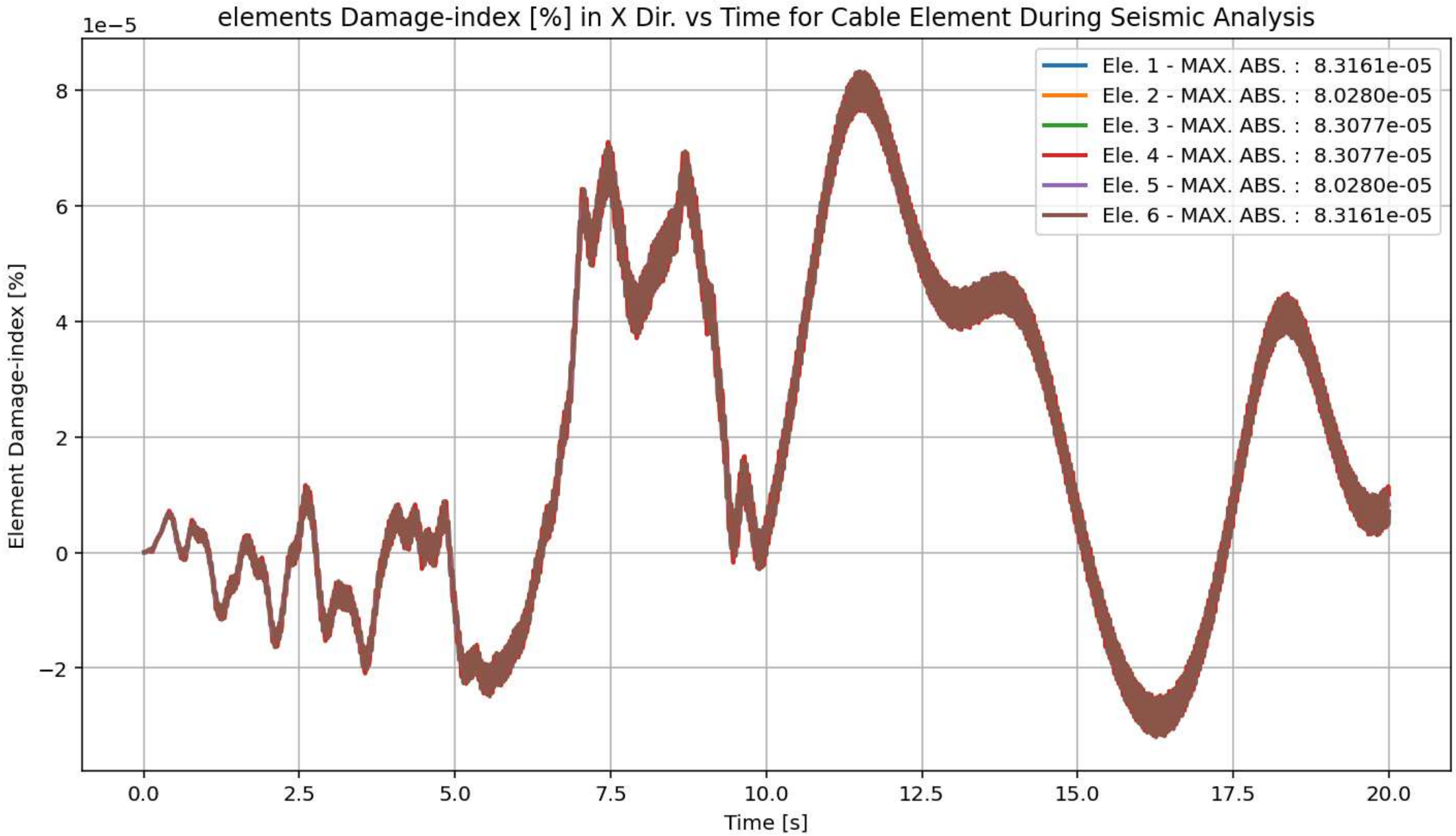

Reaction [N]
Mid Displacement [mm]
Disp X [mm]
Disp Y [mm]
Velocity X [mm/s]
Velocity Y [mm/s]
Acceleration X [mm/s²]
Acceleration Y [mm/s²]
Time (s)
1e−12
1e−10
1e−8

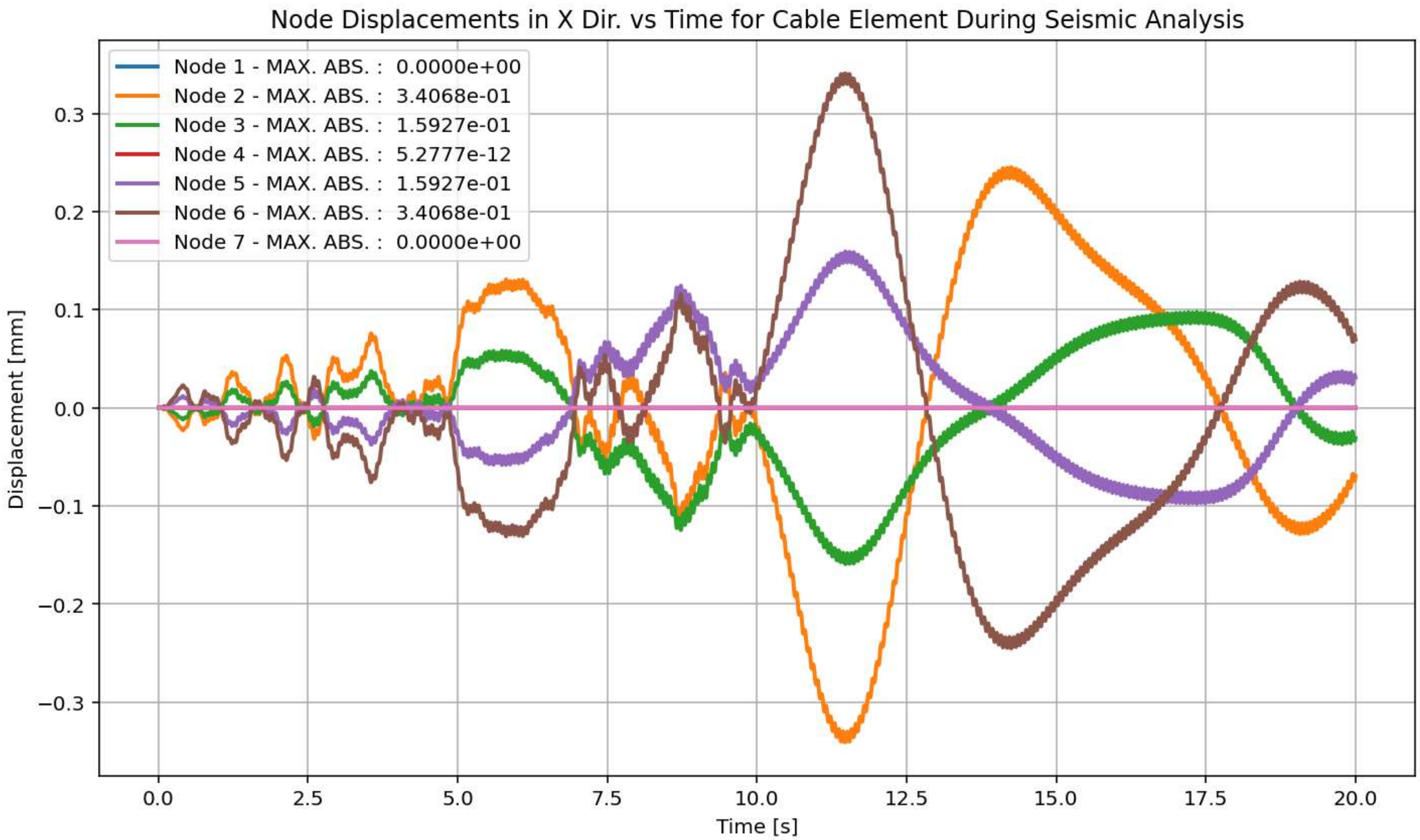

elements force in X Dir. vs Time for Cable Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.8811e+05
Ele. 2 - MAX. ABS. : 1.8159e+05
Ele. 3 - MAX. ABS. : 1.8792e+05
Ele. 4 - MAX. ABS. : 1.8792e+05
Ele. 5 - MAX. ABS. : 1.8159e+05
Ele. 6 - MAX. ABS. : 1.8811e+05
Element Axial Force [N]
200000
150000
100000
50000
0
−50000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]



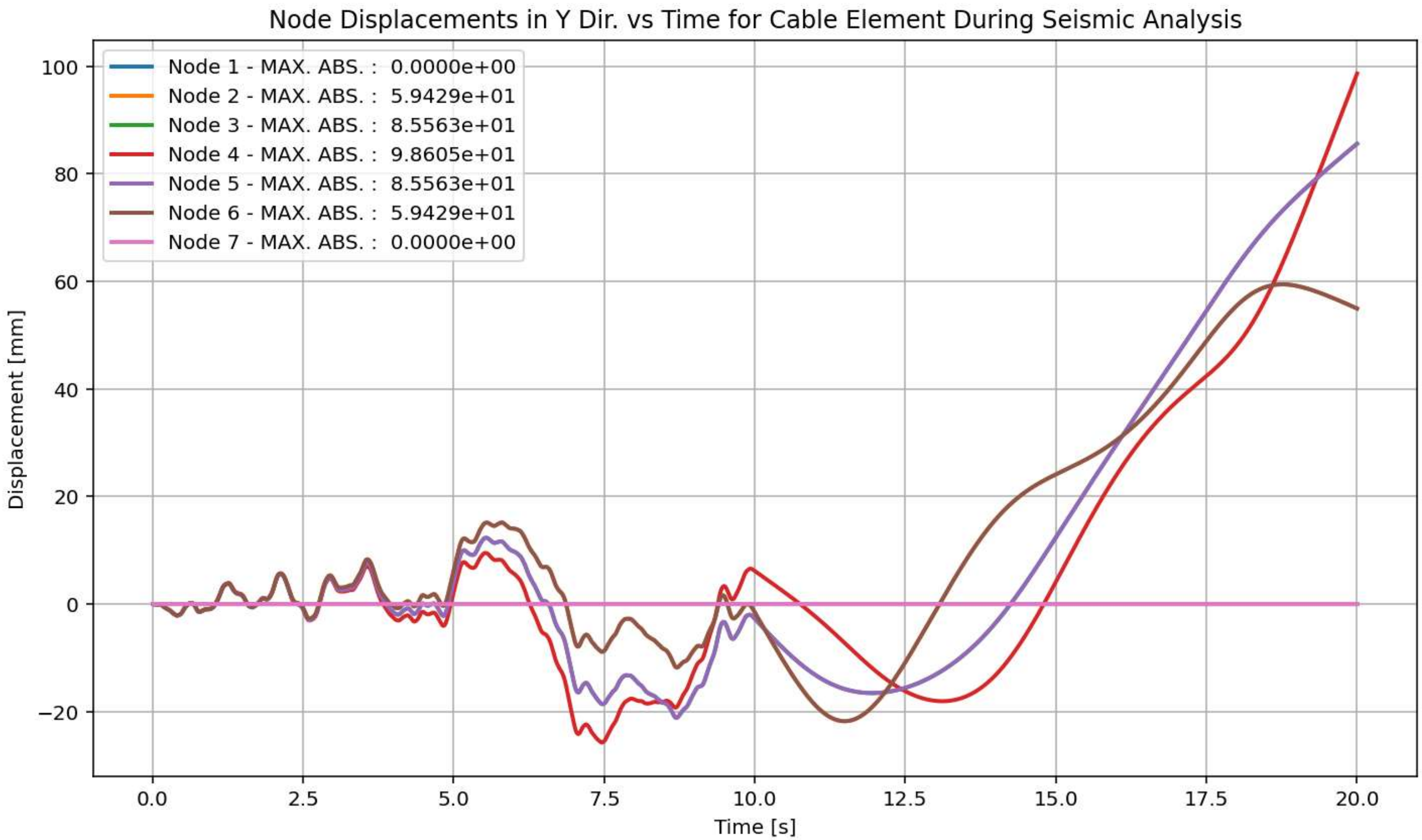
elements Stress in X Dir. vs Time for Cable Element During Seismic Analysis
Ele. 1 - MAX. ABS. : 1.6632e+01
Ele. 2 - MAX. ABS. : 1.6056e+01
Ele. 3 - MAX. ABS. : 1.6615e+01
Ele. 4 - MAX. ABS. : 1.6615e+01
Ele. 5 - MAX. ABS. : 1.6056e+01
Ele. 6 - MAX. ABS. : 1.6632e+01
Element Stress [N/mm^2]
Time [s]
15
10
5
0
−5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

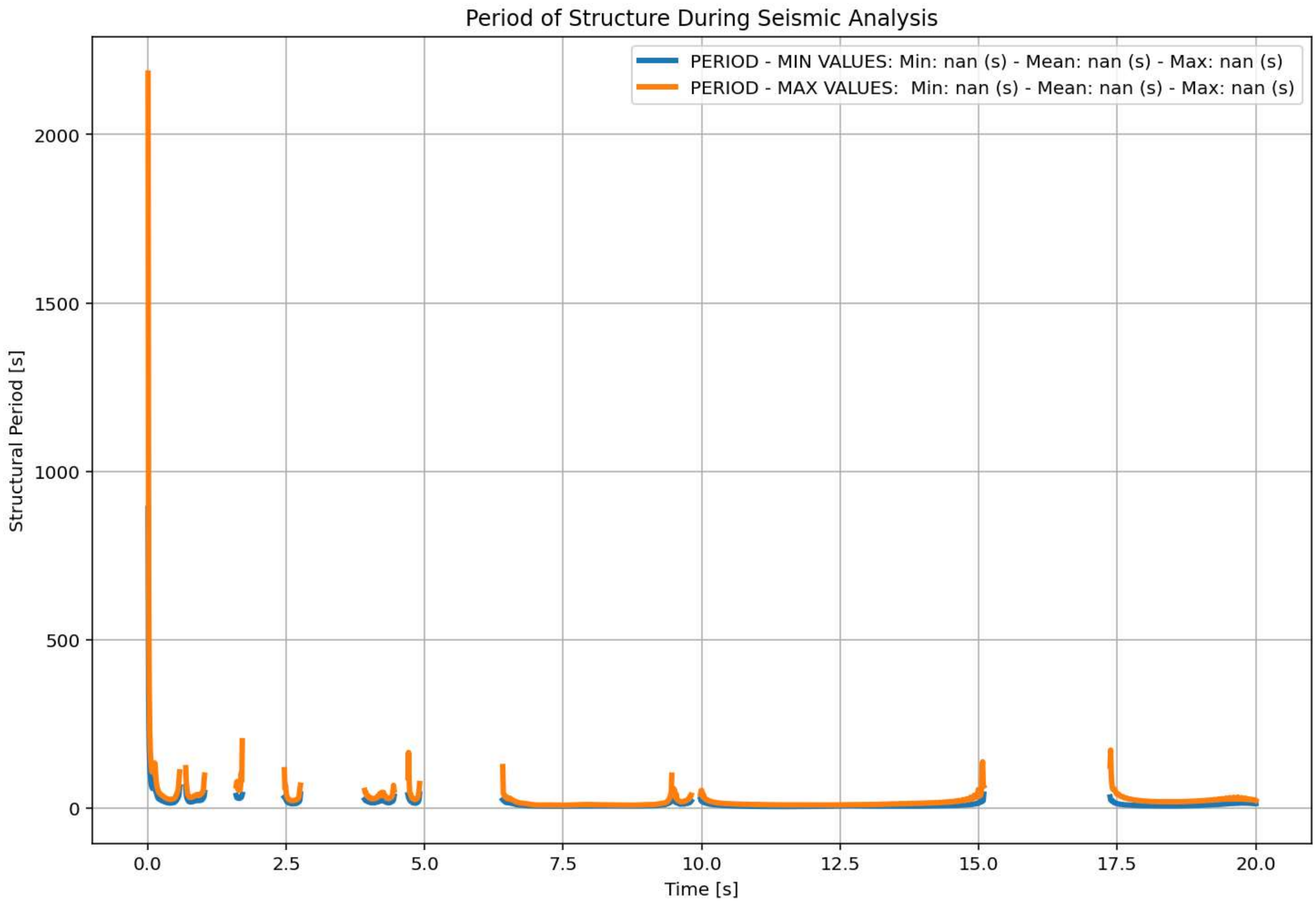

elements Strain in X Dir. vs Time for Cable Element During Seismic Analysis
1e−5
8
6
4
2
0
−2
Element Strain [mm]
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Time [s]
Ele. 1 - MAX. ABS. : 8.3161e-05
Ele. 2 - MAX. ABS. : 8.0280e-05
Ele. 3 - MAX. ABS. : 8.3077e-05
Ele. 4 - MAX. ABS. : 8.3077e-05
Ele. 5 - MAX. ABS. : 8.0280e-05
Ele. 6 - MAX. ABS. : 8.3161e-05

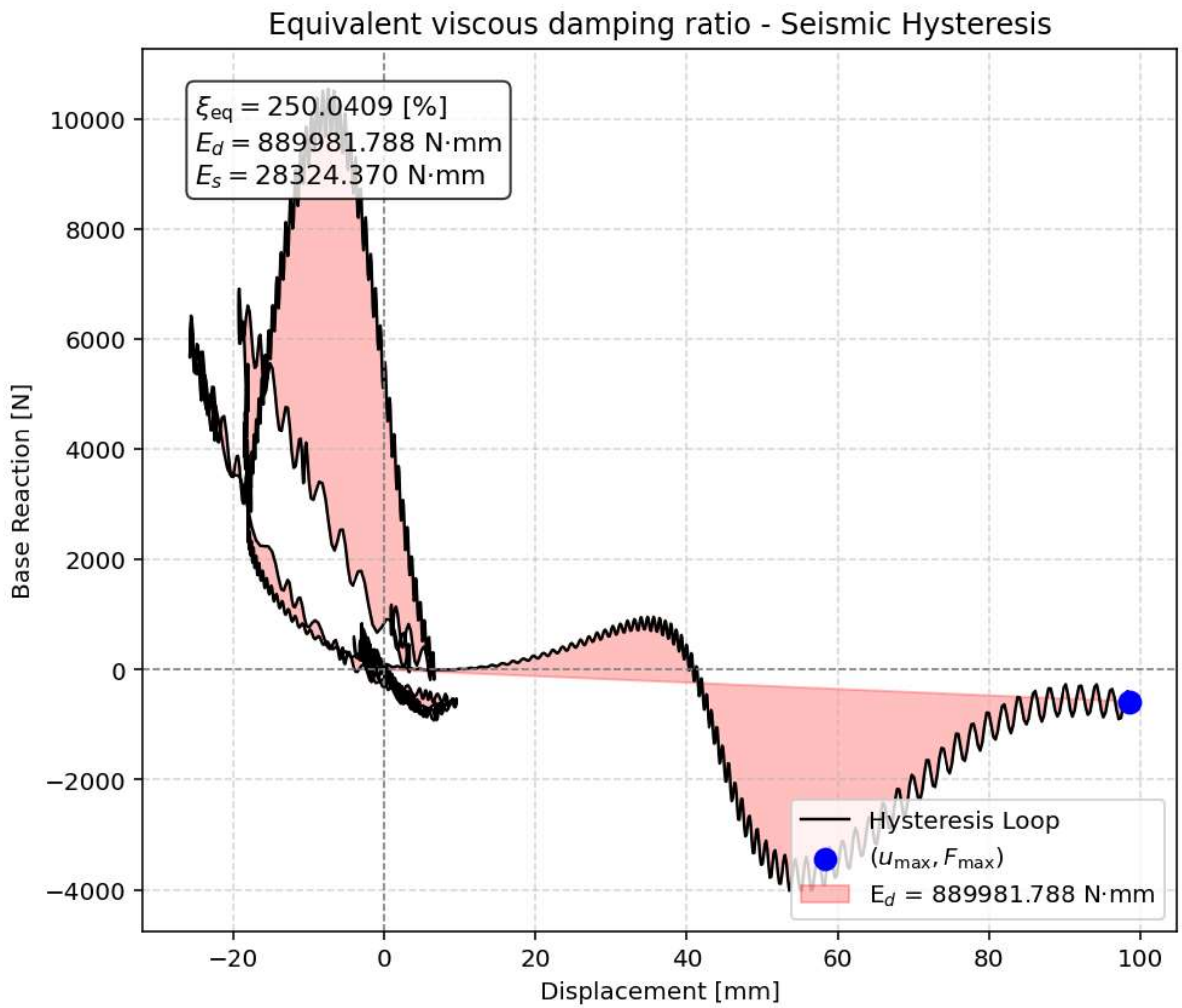

elements Damage-index [%] in X Dir. vs Time for Cable Element During Seismic Analysis
1e−5
Ele. 1 - MAX. ABS. : 8.3161e-05
Ele. 2 - MAX. ABS. : 8.0280e-05
Ele. 3 - MAX. ABS. : 8.3077e-05
Ele. 4 - MAX. ABS. : 8.3077e-05
Ele. 5 - MAX. ABS. : 8.0280e-05
Ele. 6 - MAX. ABS. : 8.3161e-05
Element Damage-index [%]
Time [s]
8
6
4
2
0
−2
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0



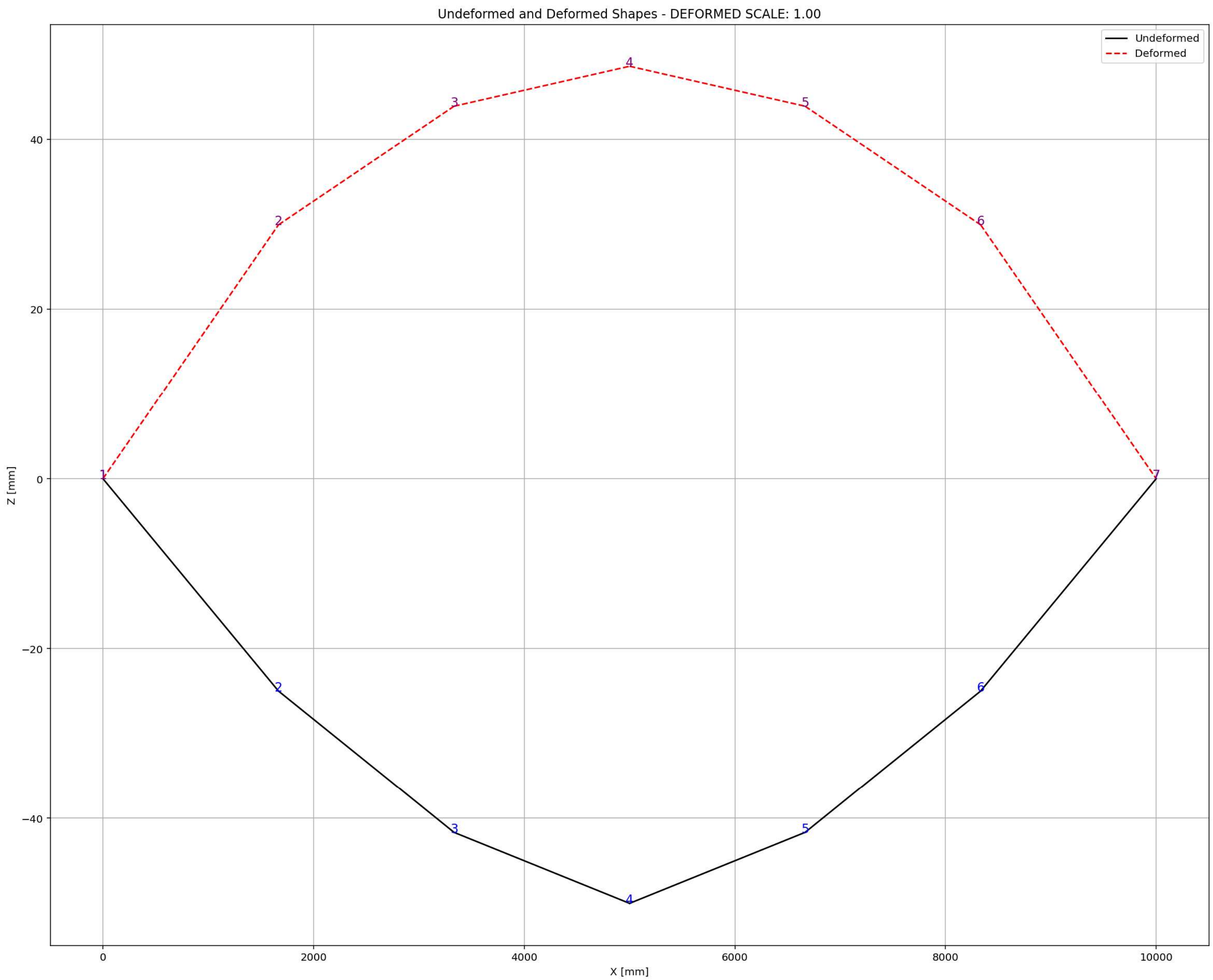
Node Displacements in X Dir. vs Time for Cable Element During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 3.4068e-01
Node 3 - MAX. ABS. : 1.5927e-01
Node 4 - MAX. ABS. : 5.2777e-12
Node 5 - MAX. ABS. : 1.5927e-01
Node 6 - MAX. ABS. : 3.4068e-01
Node 7 - MAX. ABS. : 0.0000e+00
Displacement [mm]
Time [s]


Node Displacements in Y Dir. vs Time for Cable Element During Seismic Analysis
Node 1 - MAX. ABS. : 0.0000e+00
Node 2 - MAX. ABS. : 5.9429e+01
Node 3 - MAX. ABS. : 8.5563e+01
Node 4 - MAX. ABS. : 9.8605e+01
Node 5 - MAX. ABS. : 8.5563e+01
Node 6 - MAX. ABS. : 5.9429e+01
Node 7 - MAX. ABS. : 0.0000e+00
Displacement [mm]
Time [s]
100
80
60
40
20
0
−20
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

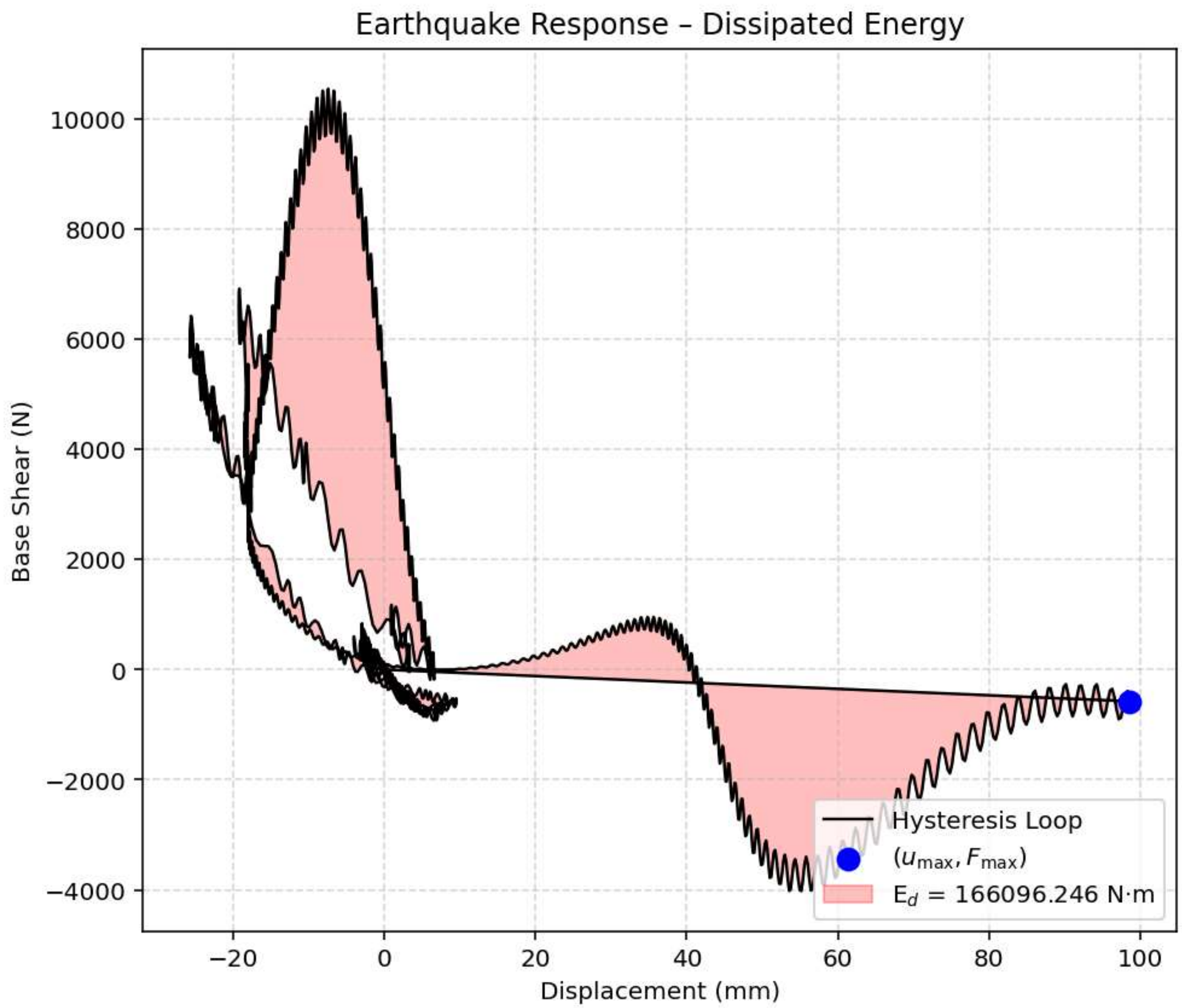

Period of Structure During Seismic Analysis
PERIOD - MIN VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
PERIOD - MAX VALUES: Min: nan (s) - Mean: nan (s) - Max: nan (s)
Structural Period [s]
Time [s]
0
500
1000
1500
2000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

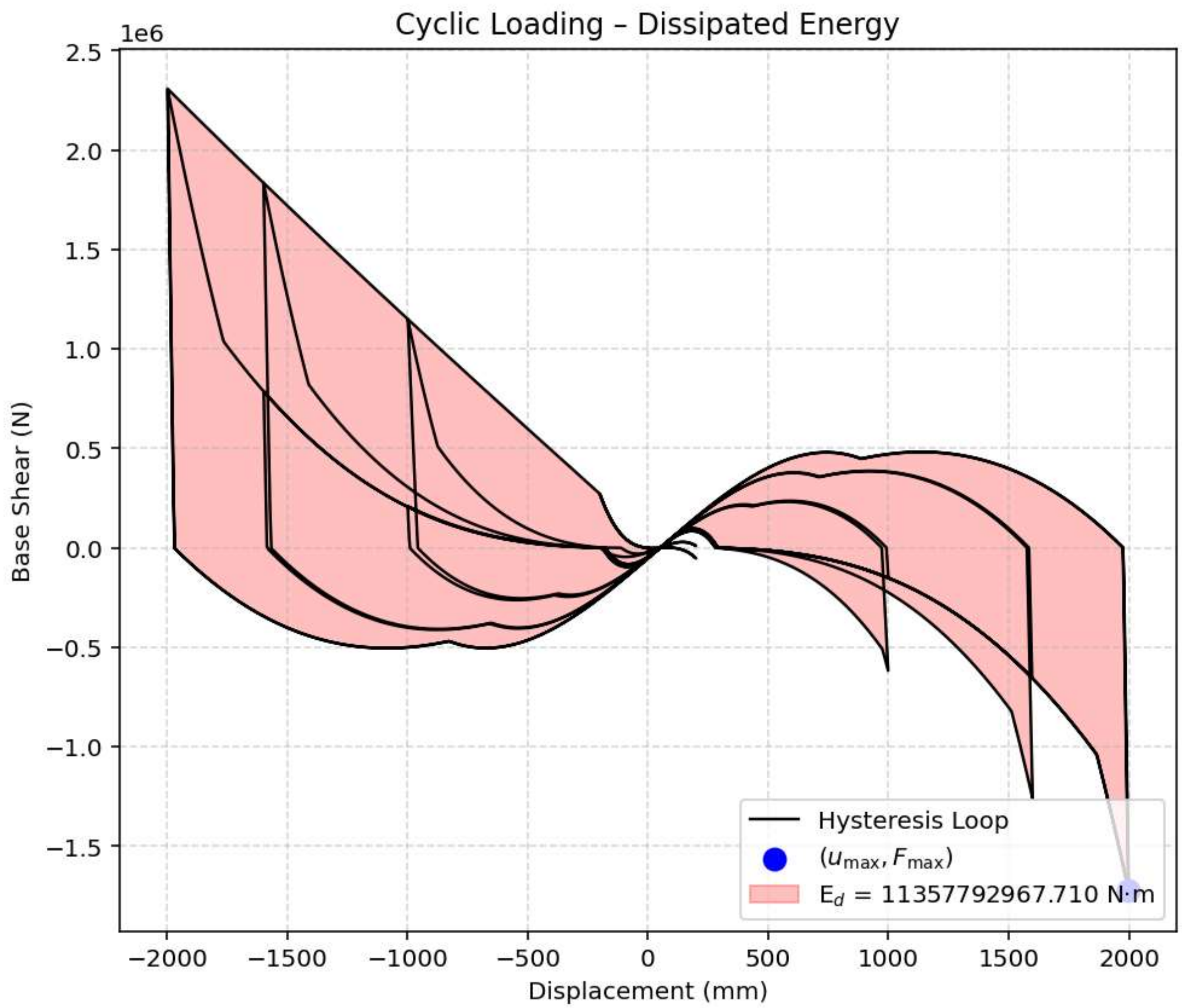


Equivalent viscous damping ratio - Seismic Hysteresis
$\xi_{eq} = 250.0409$ [%]
$E_d = 889981.788$ N·mm
$E_s = 28324.370$ N·mm
Base Reaction [N]
10000
8000
6000
4000
2000
0
−2000
−4000
−20
0
20
40
60
80
100
Displacement [mm]
Hysteresis Loop
(u_{max}, F_{max})
E_d = 889981.788 N·mm



Undeformed and Deformed Shapes - DEFORMED SCALE: 1.00
Undeformed
Deformed
Z [mm]
X [mm]
40
20
0
−20
−40
0
2000
4000
6000
8000
10000
1
2
3
4
5
6
7

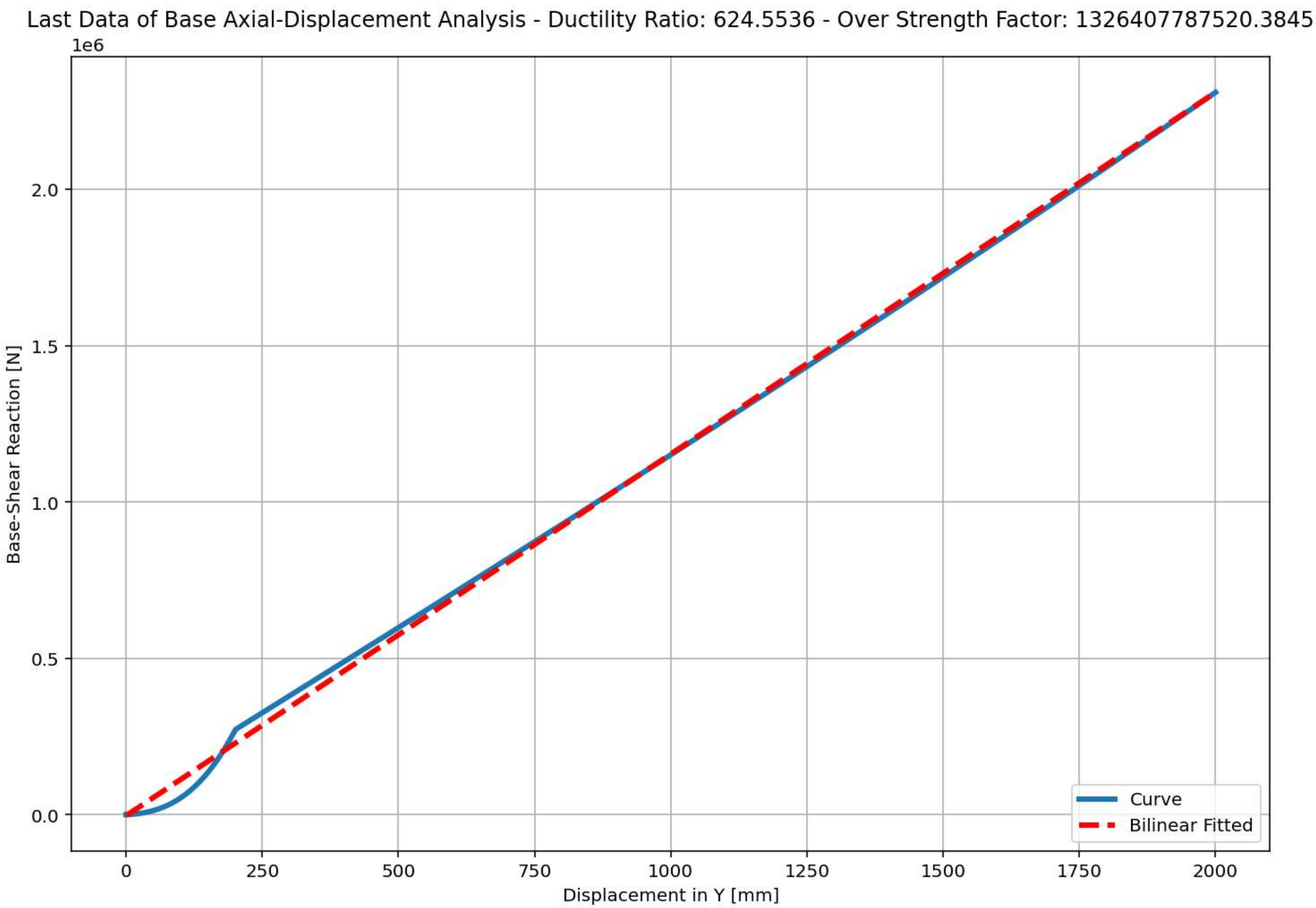

Earthquake Response – Dissipated Energy
Base Shear (N)
Displacement (mm)
10000
8000
6000
4000
2000
0
−2000
−4000
−20
0
20
40
60
80
100
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 166096.246 N·m



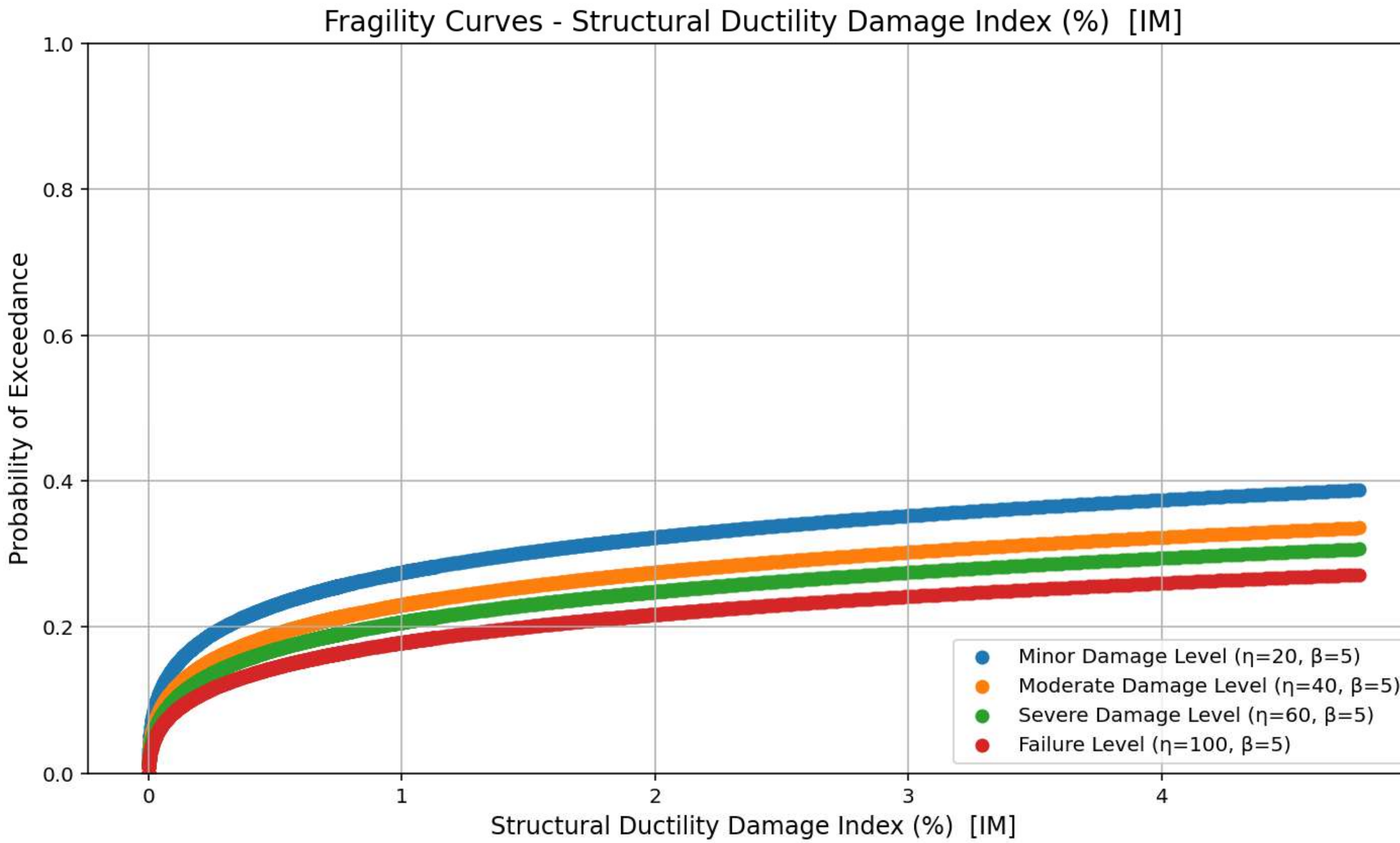
Cyclic Loading – Dissipated Energy
1e6
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
Base Shear (N)
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Displacement (mm)
Hysteresis Loop
$(u_{\max}, F_{\max})$
E_d = 11357792967.710 N·m

Dissipated Energy from Earthquake= 166096.25 N·mm
Dissipated Energy from Cyclic Displacement= 11357792967.71 N·mm

DISSIPATED ENERGY CAPACITY INDEX: 0.001 [%]
ZONE 1: VERY MINOR DAMAGE


Last Data of Base Axial-Displacement Analysis - Ductility Ratio: 624.5536 - Over Strength Factor: 1326407787520.3845
1e6
2.0
1.5
1.0
0.5
0.0
Base-Shear Reaction [N]
0
250
500
750
1000
1250
1500
1750
2000
Displacement in Y [mm]
Curve
Bilinear Fitted

```
+==========================+
=   Analysis curve fitted =
    Disp       Base Shear
----------------------------
[[0.00000000e+00 0.00000000e+00]
 [3.20228738e+00 1.74154749e-06]
 [2.00000000e+03 2.31000216e+06]]
+==========================+
+--------------------------------------------------+
 Structure Elastic Stiffness :     0.00
 Structure Plastic Stiffness :     1155.00
 Structure Tangent Stiffness :     1156.85
 Structure Ductility Ratio :      624.55
 Structure Over Strength Factor:   1326407787520.38
+--------------------------------------------------+
Over Strength Coefficient (Ω0):     1326407787520.3845
Displacement Ductility Ratio (μ):    624.5536
Ductility Coefficient (Rμ):        624.5536
Structural Behavior Coefficient (R): 828412712963939.6250
Structural Ductility Damage Index in Y Direction: 4.7778 (%)
```


Fragility Curves - Structural Ductility Damage Index (%) [IM]
Probability of Exceedance
Structural Ductility Damage Index (%) [IM]
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
Minor Damage Level (η=20, β=5)
Moderate Damage Level (η=40, β=5)
Severe Damage Level (η=60, β=5)
Failure Level (η=100, β=5)